

Honeywell

TotalPlant Solutions (TPS) System

**Planning, Installation, and Service for
TPN-Connected Nodes
(SVR2/WKS2)**

TP38-100

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About This Document

References

The following list identifies all documents that may be sources of reference for material discussed in this publication.

Document Title
Button Configuration Data Entry
Honeywell Icon Series Console Planning, Installation, and Service Guide
Keyboards
LCN Planning Guide
LCN System Installation Guide
TPS System Administration Guide
TPS System Administration Guide for Windows 2000
TPS System Implementation Guide
TPS System Implementation Guide for Windows 2000
TPS System Configuration Utility User's Guide
TPS System Configuration Utility User's Guide for Windows

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Symbol Definitions

The following table lists those symbols used in this document to denote certain conditions.

Symbol	Definition
	ATTENTION: Identifies information that requires special consideration.
	TIP: Identifies advice or hints for the user, often in terms of performing a task.
	REFERENCE -EXTERNAL: Identifies an additional source of information outside of the book set.
	REFERENCE - INTERNAL: Identifies an additional source of information within the book set.
CAUTION	Indicates a situation which, if not avoided, may result in equipment or work (data) on the system being damaged or lost, or may result in the inability to properly operate the process.
	CAUTION: Indicates a potentially hazardous situation, which if not avoided, may result in minor or moderate injury. It may also be used to alert against unsafe practices. CAUTION symbol on the equipment refers the user to the product manual for additional information. The symbol appears next to required information in the manual.
	WARNING: Indicates a potentially hazardous situation, which if not avoided, could result in serious injury or death. WARNING symbol on the equipment refers the user to the product manual for additional information. The symbol appears next to required information in the manual.
	WARNING, Risk of electrical shock: Potential shock hazard where HAZARDOUS LIVE voltages greater than 30 Vrms, 42.4 Vpeak, or 60 VDC may be accessible.

Symbol	Definition
	ESD HAZARD: Danger of an electro-static discharge to which equipment may be sensitive. Observe precautions for handling electrostatic sensitive devices.
	Protective Earth (PE) terminal: Provided for connection of the protective earth (green or green/yellow) supply system conductor.
	Functional earth terminal: Used for non-safety purposes such as noise immunity improvement. NOTE: This connection shall be bonded to Protective Earth at the source of supply in accordance with national local electrical code requirements.
	Earth Ground. Functional earth connection. NOTE: This connection shall be bonded to Protective Earth at the source of supply in accordance with national and local electrical code requirements.
	Chassis Ground: Identifies a connection to the chassis or frame of the equipment shall be bonded to Protective Earth at the source of supply in accordance with national and local electrical code requirements.

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1. System Planning

1.1 Overview

About the SVR2/WKS2 platform

The TPS Server (SVR2) and TPS Workstation (WKS2) provide PC-based functionality for the TPS platform including Global User Stations (GUSs) and APP Nodes. The SVR2 and WKS2 hardware connect to the TotalPlant Process Network (TPN) (formerly known as LCN) coax through the LCN Media Access Unit (MAU). The MAU has two coax connections to the LCN: Cable A coax cable and Cable B coax cable.

Additional references

The SVR2 and WKS2 platforms can be configured with a variety of options. The following table lists other Honeywell publications that contain additional information on options that can be used with the SVR2 and WKS2 platforms.

Table 1-1 Additional Publications

Number	Publication	Contains information on
HC05	Planning, Installation and Service for Honeywell Icon Series Console	Includes instructions for mounting the SVR2/WKS2 in the Icon console.
US13-600	Universal Station Service	Includes information on the Classic console furniture and peripheral devices.
US13-520	Universal Station (Ergonomic) Service	Includes information on the Z and EZ console furniture and peripheral devices.
RE01	Honeywell Remoting Options	Contains information for using a remote system with the computer platform.
ADP01	Honeywell Peripheral Adapters	Contains information for using the OEP/IKB and Carroll touchscreen adapters with computer platforms that do not have the ISA card.

SVR2/WKS2 Software Requirements

Refer to the latest SCN for software qualified on the SVR2 and WKS2 platforms.

1.2 SVR2 Description

Overview

The peripheral electronics assemblies in the SVR2 are based upon the Peripheral Component Interconnect (PCI) bus. All mass storage devices are connected via the IDE. The floppy drive and EIDE CDRW/DVDRW drive are connected via the floppy disk connector and the IDE connector on the motherboard. The RAID-1 EIDE controller card has two IDE internal connectors to connect the two hard drives, one hard drive on each of the internal IDE connectors.

The standard DDR SDRAM memory for this platform is 1.0 GB, expandable to 2.0 GB, with no cache memory options.

The standard mass storage for this platform is a 61 GB EIDE hard drive, which is mirrored with another 61 GB EIDE hard drive.

The model numbers for this platform are structured to include the QWERTY keyboard and mouse with the platform.

SVR2 common standard features

The SVR2 platform is configured with the standard hardware components listed below.

- Pentium 4 motherboard (single CPU)
- 512 KB Cache
- 1 AGP 2X/4X/8X Slot
- Ports: 1 serial, 1 parallel, 8 USB 2.0 (6 - back-panel; 2 – internal)
- Integrated Sound and LAN (Intel PRO/1000 CT Desktop)
- 5 PCI slots
- Dual bus Master IDE Ultra DMA 33/66/100
- 300W Dual Power Supply
- 2-512 MB DDR333, ECC SDRAM
- EIDE CDRW/DVDRW(+/-) Drive
- EIDE 61 GB or larger, 7200 RPM, Hard Drive
- 3.5" Floppy drive
- Dual Head AGP Video Graphics Card

1.3 WKS2 Description

Overview

The peripheral electronics assemblies in the WKS2 are based upon the Peripheral Component Interconnect (PCI) bus. All mass storage and removable media devices (except for the floppy drive) are connected via the EIDE or SCSI, depending on the WKS2 configuration. There are two separate EIDE interconnection cables: one for the mass storage (hard disk) drive and the second for the CDRW/DVDRW(+/-) drive.

The standard DDR SDRAM memory for this platform is 512 MB, expandable to 1.5 GB, with no cache memory options.

The standard mass storage for this platform is a 61 GB or larger hard drive, expandable with one additional 61 GB or larger hard drive.

The model numbers for this platform are structured to include the QWERTY keyboard and mouse with the platform.

WKS2 common standard features

The WKS2 platform is configured with the standard hardware components listed below.

- Pentium 4 motherboard (single CPU)
- 512 KB Cache
- 1 AGP 2X/4X/8X Slot
- Ports: 1 serial, 1 parallel, 8 USB 2.0 (6 - back-panel; 2 – internal)
- Integrated Sound and LAN (Intel PRO/1000 CT Desktop)
- 5 PCI slots
- Dual bus Master IDE Ultra DMA 33/66/100
- 250W Power Supply
- 2-256 MB DDR333, ECC SDRAM
- EIDE CDRW/DVDRW(+/-) Drive
- EIDE 61 GB or larger, 7200 RPM, Hard Drive (second one optional)
- 3.5" Floppy drive
- Dual Head AGP Video Graphics Card

1. System Planning

1.3. WKS2 Description

System architecture with GUS and APP nodes

The GUS and APP node relationship to the TPN and Plant Internet (PIN) is shown in the following TPS system architectural drawing.

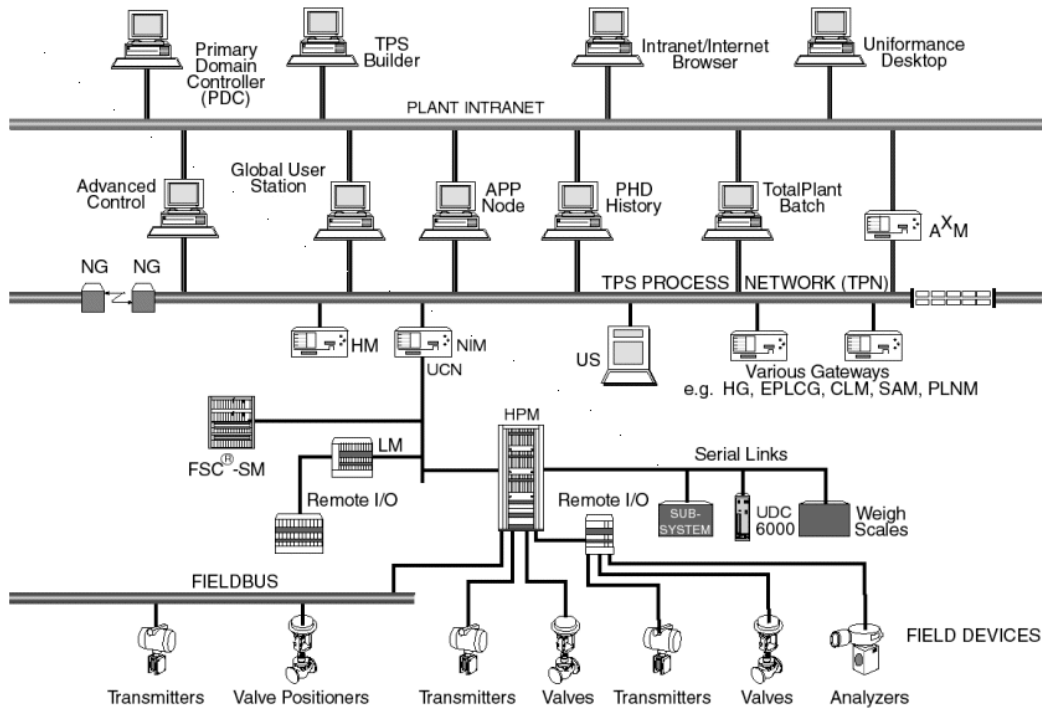


Figure 1-1 GUS and APP Node in a TPS System

Main memory configuration

The standard installed memory is 1.0 GB. You can, however, increase the memory up to 2.0 GB. Memory can be increased by installing 1 - 512 MB DDR SDRAM in DIMM socket 1 channel A and 1 - 512 MB DDR SDRAM in DIMM socket 1 channel B on the motherboard. See the tables below for memory configurations

Table 1-2 Standard Memory Configuration for SVR2

DIMM Socket	Channel A	Channel B	Total Memory
0	512 MB	512 MB	1.0 GB
1	-	-	

Table 1-3 Memory Upgrade Configuration for SVR2

DIMM Socket	Channel A	Channel B	Total Memory
0	512 MB	512 MB	
1	512 MB	512 MB	2.0 GB

Bios configuration

All SVR2 systems must have Release BZ87510A.86A.0030.P07 or later version of BIOS installed.

Single/Dual screen video configuration for App

The SVR2 uses an AGP dual head video graphics controller card that is inserted in the AGP slot on the motherboard.

Single screen video configuration for GUS

Standard video in the SVR2 is a dual screen AGP controller card with an add-in video memory card for a total of 16-MB of video memory. The dual screen AGP video controller requires an AGP slot. The video display signal cable is connected to the top video port of the video card.

1. System Planning

1.4. SVR2 Platform Configuration Options

Dual screen video configuration for GUS

An option of the TPS platform is dual screen capability. This option provides the ability to output to two video displays from a single TPS platform.

In the desktop configuration, the primary video display is to the left. In the console configuration the primary video display must be located in the lower tier of the furniture. The secondary video display may be vertically oriented directly overhead in the upper tier or it may be placed in the bay to the right of the bay containing the TPS electronics. This option consists of two signal cables and two video displays. The dual-screen controller card consumes the AGP slot. The primary video display signal cable is connected to the top video port of the video card and the secondary video display signal cable is connected to the lower video port of the video card.

NOTE: No touch-screen is supported for the secondary video display in the dual screen option.

Hard disk drive configuration

The SVR2 platform has two internal hard disk drive bays. The system has two internal 61 GB IDE hard drives for RAID-1. Drive 0 is located in the drive bay closest to the motherboard. Drive 1 is located in the drive bay furthest from the motherboard. Both RAID-1 hard drives are pinned as “Device 0 (Master)” (See *Note below).

***Note:** Use the 16 HEADS pin outs diagram on the hard drive.

1.5 WKS2 Platform Configuration Options

Main Memory Configuration

The standard installed memory is 512 MB. You can, however, increase the memory up to 1.5 GB. Memory can be increased by installing 1 - 512 MB DDR SDRAM in DIMM socket 1 channel A and 1 - 512 MB DDR SDRAM in DIMM socket 1 channel B on the motherboard. See the tables below for memory configurations:

Table 1-4 Standard Memory Configuration for WKS2

DIMM Socket	Channel A	Channel B	Total Memory
0	256 MB	256 MB	512 MB
1	-	-	

Table 1-5 Memory Upgrade Configuration for WKS2

DIMM Socket	Channel A	Channel B	Total Memory
0	256 MB	256 MB	
1	512 MB	512 MB	1.5 MB

BIOS configuration

All WKS2 systems must have Release BZ87510A.86A.0030.P07 or later version BIOS installed.

Single screen video configuration

Standard video in the WKS2 is a dual screen AGP controller card with an add-in video memory card for a total of 16-MB of video memory. The dual screen AGP video controller requires an AGP slot. The video display signal cable is connected to the top video port of the video card.

1. System Planning

1.6. Furniture and Device Configuration Options

Dual screen video configuration

An option of the TPS platform is dual screen capability. This option provides the ability to output to two video displays from a single TPS platform.

In the desktop configuration, the primary video display is to the left. In the console configuration the primary video display must be located in the lower tier of the furniture. The secondary video display may be vertically oriented directly overhead in the upper tier or it may be placed in the bay to the right of the bay containing the TPS electronics. This option consists of two signal cables and two video displays. The dual-screen controller card consumes the AGP slot. The primary video display signal cable is connected to the top video port of the video card and the secondary video display signal cable is connected to the lower video port of the video card.

NOTE: No touch-screen is supported for the secondary video display in the dual screen option.

Quad screen video configuration

The quad screen configuration provides the ability to output to three or four FPDs from a single WKS2 platform. This option consists of a PCI quad screen controller card located in PCI slot #1.

If using 3 FPDs, attach the FPDs to channel 1,2 and 4.

Hard disk drive configuration

The WKS2 platform has two internal hard disk drive bays. The base system has one internal 61 GB or larger EIDE hard drive.

The base 61 GB or larger hard drive is mounted in hard drive bay 1 and is pinned as "Device 0 (Master)" (See *Note below) and an optional expansion drive can be mounted in hard drive bay 0 and must be pinned "Device 1 (Slave)" (See *Note below).

***Note:** Use the 16 HEADS pin outs diagram on the hard drive.

All hard drives are daisy chained together on an IDE cable connected to IDE1 connector on the motherboard. Only one additional 61 GB or larger hard drive can be added.

1.6 Furniture and Device Configuration Options

Configuration descriptions

The SVR2/WKS2 platform is the building block for eight different configuration options. Table 1-6 briefly describes these options.

Table 1-6 SVR2/WKS2 Configuration Options

Deskside/Desktop APP Node	The platform is placed on a desk in a vertical or horizontal position, with a monitor, keyboard, and mouse.
Deskside/Desktop GUS	The platform is placed on a desk in a vertical or horizontal position with a monitor, keyboard, and mouse.
Cabinet-mounted App Node	The platform is mounted in a cabinet in a horizontal position. The user interface is placed in a cabinet or the Remote electronics option is used to connect the user interface.
Cabinet mounted GUS	The platform is mounted in a cabinet in a horizontal position. The Remote electronics is used to connect the user interface.
Icon Console GUS	The platform is mounted horizontally in an Icon Console. See the Honeywell Icon Series Console Planning, Installation, and Service Guide HC05 for additional information.
Z-Console GUS	The platform is mounted horizontally in Z-Console furniture with remote peripheral drives located on the console.
EZ-Console GUS	The platform is mounted horizontally in EZ-Console furniture with remote peripheral drives located on the console. EZ consoles can have single or dual desktop swivel monitors. The EZ-Console furniture option provides a platform for mounting standard desktop monitor(s) or a multipurpose mounting enclosure on top of a standard Z-Furniture base
Classic-Console GUS	The platform is mounted in Classic-Console furniture with remote peripheral drives located on the console.

1. System Planning

1.6. Furniture and Device Configuration Options

GUS Node options

GUS software can run on a WKS2 Workstation or a SVR2 Server. A description of the hardware options available for a GUS is in Table 1-7.. Available options for your GUS node depend on the furniture being used. The following table lists all of the options available for GUS nodes on the WKS2 and SVR2 platforms. The Parts Online section at the <http://www.acs.honeywell.com/> website contains information on specific part and model numbers. See also Section 12, “Spare Parts” in this document for ordering additional or replacement parts.

Table 1-7 GUS Options for SVR2/WKS2

<i>Video display options</i>
17-inch Desktop Monitor, SVGA, with cables
21-inch Desktop Monitor, SVGA, with cables (touchscreen not supported).
21-inch Desktop Monitor, SVGA with cables (touchscreen supported for only one video)
20.1-inch Desktop Flat Panel Display, with cables
17-inch EZ Console lower monitor with clamps and cables
17-inch EZ Console Upper monitor with clamps and cables
21-inch EZ Console lower monitor with clamps and cables
21-inch EZ Console Upper monitor with clamps and cables
19-inch lower monitor and Integrated Keyboard for Classic Console
19-inch lower monitor and Operator Keyboard for Classic Console
Upper 21-inch monitor (2 nd monitor), 4 MB dual channel drive board and cables
<i>Video display and keyboard options</i>
Lower Tier, 21-inch CRT and IKB for Z-Console
Lower Tier, 21-inch CRT and OEP for Z-Console
Upper Tier, second 21-inch CRT, 4 MB per channel, dual channel drive board and cables
<i>Touchscreen options</i>
Icon FPD USB touchscreen
Desktop flat panel monitor touchscreen
21-inch monitor with IR touchscreen
Z-console IR touchscreen for lower tier
Classic Console IR touchscreen for lower tier
<i>Honeywell keyboard options</i>
Desktop Integrated Keyboard without trackball
Desktop Integrated Keyboard with trackball
Desktop Operator Keyboard

Table 1-7 GUS Options for SVR2/WKS2

EZ Console IKB Package
EZ Console OEP Package
EZ-Console PC (QWERTY) Keyboard Package
<i>Remote (external) removable media options</i>
CD-ROM/DVD-ROM Drive (SCSI)
CD-RW/DVD-RW(+/-) Drive (EIDE to USB) for Windows 2000
100 MB Zip cartridge drive (SCSI to USB) for Windows 2000
1/4" 1.6/3.2 GB Tape Drive (SCSI to USB) for Windows 2000

Other options for WKS2

The following table lists additional options to be used with the WKS2 platform.

Table 1-8 Additional Options for WKS2

<i>Accessory options</i>
Industrial CE-Mark QWERTY keyboard
Remote Electronics System, 25 feet
Remote Electronics System, 50 feet
Remote Electronics System, 100 feet
Drive enclosure - work surface drive bay mounting
Drive enclosure - U/Tier (half-height) drive bay mounting
Industrial CE-Mark mouse
<i>Furniture options</i>
EZ Console Monitor Lower Mounting Platform
EZ Console Monitor Upper Mounting Platform
<i>Mounting Kits</i>
Mounting hardware
Rack mounting kit
Mounting hardware for console

1. System Planning

1.6. Furniture and Device Configuration Options

2. Compliance and Specifications

2.1 Regulatory Compliance

Desktop Regulatory Compliance

Table 2-1 Electromagnetic Compatibility (EMC)

European Community:

Emissions:	EN50081-1, 1992
Immunity:	EN50082-1, 1992

Australian Communications Authority:

Emissions:	AS/NZS 2064-1997 (CISPR 11, Class A) (SCN N314)
------------	---

United States, Federal Communications Commission:

Emissions:	FCC Part 15, Class B, 1996
------------	----------------------------

Attention: The following formula is a proximity guideline, for use of Portable Transceivers (walkie-talkies), in the frequency range of 80MHz to 1GHz:

$D > \sqrt{P}$ (D must be greater than the square root of P)
D = Distance from equipment, in meters.

P = Power Output of the Portable Transceivers (walkie-talkies), in Watts

Examples:

P = 10 Watts,	D > 3.162 meters
P = 5 Watts,	D > 2.236 meters
P = 1 Watt,	D > 1.000 meters

Note: Electrical cables, which are routed external to the equipment, must be fully shielded cables (360 degree metallic shielding) in order to comply with the above EMC standards.

Product Safety Compliance

CAN/CSA-C22.2 No. 950

EN60950

Note: Within these standards (above) is a 'Normative Reference' section citing additional standards which may apply as suited and required for product compliance.

2.1. Regulatory Compliance



Console/Cabinet Regulatory Compliance

Table 2-2 Electromagnetic Compatibility (EMC)

Note: Electrical cables, which are routed external to the equipment, must be fully shielded cables (360 degree metallic shielding) in order to comply with the above EMC standards.

Note: Within these standards (above) is a ‘Normative Reference’ section citing additional standards, which may apply as suited and may be required for product compliance.



WARNING

Honeywell does not claim Safety Compliance or Electromagnetic Compatibility (EMC) Compliance for system equipment configurations that have not been described in this manual as standard system configurations. Any equipment configuration other than that described in this publication decertifies the Safety and EMC compliance of this product.

2.2 Electronics Module Specifications

SVR2 platform specifications

Table 2-3 SVR2 Enclosure Specifications

Spec. Type	Description
DC Power	N/A
AC Voltage	120 (90-132) Vrms 240 (180-260) Vrms
AC RMS Current	2.06 Arms 1.13 Arms
AC Power	124 Watts 125 Watts
Height	177.8 mm
Width	412.8 mm
Depth	428.6 mm
Weight	13.9 Kg

2. Compliance and Specifications

2.3. Media Drive Specifications

WKS2 platform specifications

Table 2-4 WKS2 Enclosure Specifications

Spec. Type	Description
DC Power	N/A
AC Voltage	120 (90-135) Vrms 240 (180-265) Vrms
AC RMS Current	1.0 Arms 0.6 Arms
AC Power	250 Watts Max
Height	165 mm
Width	419 mm
Depth	445 mm
Weight	12.7 Kg depending on options

2.3 Media Drive Specifications

Hard disk drive specifications

Table 2-5 61 GB Hard Disk Specifications

Spec. Type	Description
DC 5 volt Power (typ)	5 Watts +/- 5%
DC 12 volt Power (typ)	12 Watts +10%, -8%
Other DC POWER	22 Watts Max, 10.0 Watts idle
AC Power	N/A
Height	25.4 mm
Width	101.6 mm
Depth	146.0 mm
Weight	0.6 kg

Removable media drive specifications

Table 2-6 Mini Cartridge Tape Drive Specifications

Spec. Type	Description
DC 5 volt power	5 Watts
DC 12 volt power	3 Watts
DC POWER (Other)	N/A
AC Voltage	N/A
AC RMS Current	N/A
AC Power	N/A
Height	42.5 mm
Width	148 mm
Depth	152 mm
Weight	0.46 kg

Table 2-7 Zip Disk Drive Specifications

Spec. Type	Description
DC 5 volt power	5.0 Watts
DC 12 volt power	N/A
DC POWER (Other)	N/A
AC Voltage	N/A
AC RMS Current	N/A
AC Power	N/A
Height	41.5 mm
Width	148 mm
Depth	163 mm
Weight	0.67 kg

2. Compliance and Specifications

2.3. Media Drive Specifications

Table 2-8 DAT Drive Specifications

Spec. Type	Description
DC 5 volt power	4.0 Watts
DC 12 volt power	5 Watts
DC POWER (Other)	N/A
AC Voltage	N/A
AC RMS Current	N/A
AC Power	N/A
Height	40.6 mm
Width	148 mm
Depth	1456.5 mm
Weight	1.09 kg

Table 2-9 CDRW/DVDRW Drive Specifications

Spec. Type	Description
DC 5 volt power	6.0 Watts
DC 12 volt power	21.6 Watts
DC POWER (Other)	N/A
AC Voltage	N/A
AC RMS Current	N/A
AC Power	N/A
Height	40.64 mm
Width	146.05 mm
Depth	14.22 mm
Weight	1.20 kg

2.4 Monitor Specifications

21-Inch monitor specifications

Table 2-10 21-Inch Desktop Monitor

Spec. Type	Description
Height	491 mm (19.3 inches)
Width	498 mm (19.6 inches)
Depth	478 mm (18.8 inches)
Weight	32 Kg (70.5 pounds)
AC Voltage Ranges	110 to 120 VAC, 50 to 60 Hz, 2.0 Amp 220 to 240 VAC, 50 to 60 Hz, 1.0 Amp
AC Load Normal Operation	≤ 145 Watts
Standby Mode	≤ 100 Watts
Suspended Mode	≤ 15 Watts
Active Off Mode	~ 1 Watt
Power Off	≤ 0 Watts
Max Inrush 120 VAC	50 Amp
240 VAC	80 Amp

2. Compliance and Specifications

2.4. Monitor Specifications

20.1-Inch Flat Panel Display specifications

Table 2-11 20.1-Inch Flat Panel Display

Spec. Type	Description
Height	Landscape: 501 mm (19.7 inches) Portrait: 541 mm (21.3 inches)
Width	Landscape: 498 mm (19.6 inches) Portrait: 418 mm (16.5 inches)
Depth	246 mm (9.6 inches)
Weight	17 Kg (16.1 pounds)
Height Adjustment	70 mm (2.8 inches)
Power Supply	AC 100 to 120 V, 50/60 Hz AC 220 – 240 V, 50/60 Hz
Current Rating	0.7A at 100 to 120 V 0.3A at 220 – 240 V

Sony 21-Inch Monitor Specifications

Table 2-12 Specifications for Sony 21-Inch Monitor (51197178-100)

Spec. Type	Description
CRT	0.24 mm aperture grille pitch, 90-degree deflection, FD Trinitron 21 inches measured diagonally.
Viewable image size	Approximately 403.8 x 302.2 mm (w/h) (16 x 12 inches) 19.8" viewing image
Resolution	<u>Maximum:</u> Horizontal: 2048 dots Vertical: 1536 lines <u>Recommended:</u> Horizontal: 1280 dots Vertical: 1024 lines
Deflection frequency	Horizontal: 30 to 110 kHz Vertical: 48 to 70 kHz
Input signal levels Video signal Sync signal	Analog RGB: 0.700 Vp-p (positive), 75 Ω Separate H/V, or composite sync: TTL 2 Ω , Polarity Free Sync on green: 0.3 Vp-p (negative)

2. Compliance and Specifications

2.4. Monitor Specifications

Sony 21-Inch Monitor Specifications

Table 2-12 Specifications for Sony 21-Inch Monitor (51197178-100)

Spec. Type	Description
Standard image area	Approximately 388 x 291 mm (15 3/8" x 11 1/2")
AC input voltage/current	100 to 240 V, 50-60 Hz, 2.0-1.0A
Power consumption	Approximately 135 W
Operating temperature	10 °C to 40°C
Dimensions	Approximately 495 x 395 x 565 mm (w/h/d) [19.50" x 15.54" x 22.25"] inches
Weight	Approximately 31.4 Kgs (69.2 lbs.)

21-inch Flat Screen Monitor Specifications

Spec. Type	Description
Resolution	up to 1600 x 1280 @ 90 Hz
Frequency	78 ± 2 KHz
Interface Video Sync Impedance	HD15, 0.7 Vp-p, positive Separate HD/VD, TTL Polarity Free 75 Ohm
Weight	33.6 Kgs (74 lb.)

2. Compliance and Specifications

2.5. Z-Console and EZ-Console General Specifications

19-inch Flat Screen Open-Frame Monitor Specifications

Table 2-13 19-inch Flat Screen Open-Frame Monitor Specifications

Spec. Type	Description
Resolution	Up to 1600 x 1280 @ 90 Hz
Frequency	78 ± 2 KHz
Interface Video Sync Video Impedance	HD15, 0.7 Vp-p, positive Separate HD/VD, TTL Polarity Free 75 ohm
AC Input Voltage Frequency Power	90 to 132 VAC 198 to 264 VAC 47 to 63 Hz ≤ 140 W
Weight	25.47 Kgs (56.15 lbs.)

2.5 Z-Console and EZ-Console General Specifications

Electrical specifications

Voltage	120 VAC +10%, -15% Single-Phase 2-wire (plus safety ground)
	240 VAC +10%, -15% Single-Phase 2-wire (plus safety ground)
Frequency	50 Hz or 60 Hz +3%, -6% with no more than 8% harmonic distortion
Power	1.4 kW maximum (per cabinet)
Grounding	Console must be grounded by a separate conductor (green or yellow-green wire) run with the power conductors.
Current	15 amperes maximum

2. Compliance and Specifications

2.5. Z-Console and EZ-Console General Specifications

Environmental requirements

Storage	Temperature: -35°C to 65°C Humidity: 5 to 95% Noncondensing Altitude: 10,668 meters (35,000 feet) Vibration: 0.5 G at 3 to 200 Hz at 1 octave/min
Operational	Temperature: 10°C to 35°C Humidity: 8 to 80% Noncondensing Altitude: 2,286 meters (7,500 feet)
Limitations	Certain peripheral devices have more critical requirements. The Global User Stations will be limited by the requirements of those devices. Refer to each peripheral Drive Specification published by its manufacturer.

Shipping requirements

Weight	Single Monitor Console: 149.6 kg (330 pounds) maximum Dual Monitor Console: 181.4 kg (400 pounds) maximum Single Monitor EZ-Furniture: 108.0 kg (238 pounds) maximum Dual Monitor EZ-Furniture: 177.8 kg (392 pounds) maximum Desktop Monitor: 32.7 Kg (72 lb.)
Bearing Load	A unit with a single monitor requires a floor capable of supporting an average loading of 32.2 kg/m ² (70.9 pounds/square foot) and point loading capability of 37.4 kg (82.5 pounds). A unit with double monitors requires a floor capable of supporting an average loading of 39 kg/m ² (86.0 pounds/square foot) and point loading capability of 45.4 kg (100 pounds).
Moving	When moving, the lifting force must be applied by a method that will bridge both edges of the base. Any accessory items used in lifting or moving and any moving devices must be prevented from bearing against end panels, doors, covers, keyboards, or table tops as these are nonstructural parts and may be damaged.

2. Compliance and Specifications

2.6. Classic Console General Specifications

Operating conditions

Weight	Single Monitor Console: 149.6 kg (330 pounds) maximum Dual Monitor Console: 181.4 kg (400 pounds) maximum
Bearing Load	A unit with single or double monitors requires a floor capable of supporting an average loading of 39 kg/m ² (86 pounds/square foot). A unit with a single monitor requires point-loading capability of 37.4 kg (82.5 pounds) and a unit with double monitors requires point-loading capability of 45.4 kg (100 pounds) when either unit is resting on the leveling feet.
Heat	1400 Watts (4778 BTU/hr.) maximum per bay. Exhaust temperature is 20°C maximum above inlet temperature. Required airflow is 95 liters/second (200 CFM) maximum per bay.
Vibration Operational	0.10 G input at base of console; 5 to 200 to 5 Hz at 1 octave/min, 6 sweeps for 60 minutes total per axis, all three mutually perpendicular axes.

2.6 Classic Console General Specifications

Electrical specifications

Voltage	120 VAC +10%, -15% Single Phase 2 wire (plus safety ground)
	240 VAC +10%, -15% Single Phase 2 wire (plus safety ground)
Frequency	50 Hz or 60 Hz +3%, -6% with no more than 8% harmonic distortion
Power	1.4 kW maximum (per cabinet)
Grounding	Console must be grounded by a separate conductor (green or yellow-green wire) run with the power conductors.
Current	15 amperes maximum

Environmental requirements

Storage	Temperature: -35°C to 70°C Humidity: 5 to 95% Non-condensing Altitude: 10,668 meters (35,000 feet) Vibration: 0.5 G at 3 to 200 Hz at 1 octave/min
Operational	Temperature: 10°C to 35°C Humidity: 8 to 80% Non-condensing Altitude: (10,000 feet)
Limitations	Certain peripheral devices have more critical requirements. The Global User Stations will be limited by the requirements of those devices.

Shipping requirements

Weight	Single Monitor: 204 kg (450 pounds) maximum Dual Monitor: 363 kg (800 pounds) maximum
Bearing Load	The unit requires a floor capable of supporting an average loading of 391 kg/m ² (80 pounds/square foot) for single and double bay consoles. Single bay unit requires point-loading capability of 51 kg (113 pounds) and the double bay unit requires point-loading capability of 91 kg (200 pounds) when either unit is resting on the leveling feet.
Moving	When moving, the lifting force must be applied by a method that will bridge both edges of the base. Any accessory items used in lifting or moving, and any moving devices must be prevented from bearing against end panels, doors, covers, keyboards, or table tops as these are nonstructural parts and may be damaged.

2. Compliance and Specifications

2.6. Classic Console General Specifications

Operating conditions

Weight	Single Bay: 204 kg (450 pounds) maximum Double Bay: 363 kg (800 pounds) maximum
Bearing Load	The unit requires a floor capable of supporting an average loading of 391 kg/m ² (80 pounds/square foot) for single and double bay consoles. Single bay unit requires point-loading capability of 51 kg (113 pounds), and the double bay unit requires point-loading capability of 91 kg (200 pounds) when either unit is resting on the leveling feet.
Heat	1400 watts (4778 BTU/hr.) maximum per bay. Exhaust temperature is 20°C maximum above inlet temperature. Required airflow is 60 CFM maximum per bay.
Vibration Operational	0.10 G input at base of console; 5 to 200 to 5 Hz at 1 octave/min, 6 sweeps for 60 minutes total per axis, all three mutually perpendicular axes.

3. Platform Installation

3.1 Pre-installation

Overview

The two cables and T-connectors (and terminators, if applicable) must be located underneath (the future location) of the GUS unit, with an additional loop, 1 meter in length (with a T-connector), to be routed into the rear of the platform where the connection is made to the LCN MAU. The cable between the LCNP4 board and the LCN MAU is 2 meters in length. See Figure 3-6.

Refer to *LCN System Installation*, Subsection 3.6 for the rules and techniques of installing an LCN cable system.

APP Node and GUS AC Power and Grounding

The ground connection is made through the third wire in the AC power cord.

Z-console and Classic console AC Power and Grounding

Ground the station as shown in *LCN System Installation* manual (see figure entitled, "Logic Ground (LG) Connection to Safety Ground Stake").

AC Power Warning



WARNING

The power supply circuit is connected to AC power when the power cable is connected. The power control switch on the front panel only enables the power supply circuit outputs.



ATTENTION

It is strongly recommended that the power cord be connected to a clean power source with backup such as an Uninterruptible Power Source (UPS). Refer to *TPS System Site Planning* and *LCN System Installation* manuals.

3.2 SVR2/WKS2 Enclosures

Overview

The enclosure used for the SVR2/WKS2 platform contains the PC platform used in the Global User Stations (GUSs), and fits all installed base cabinets and consoles. The enclosure provides for rapid repair, including swapping of fans, power supplies, and hard drives. It is easily upgradeable and supported in the field.

Description

The enclosure used for SVR2/WKS2 is shown in Figure 3-1. The chassis has provisions for rack mounting in a standard 19-inch RETMA rack. The chassis can be converted for desktop application. It is slide-mounted for easy access with a rear cable management system. To provide system cooling, filtered air is drawn in the right side of the enclosure and exhausted out the back.

*

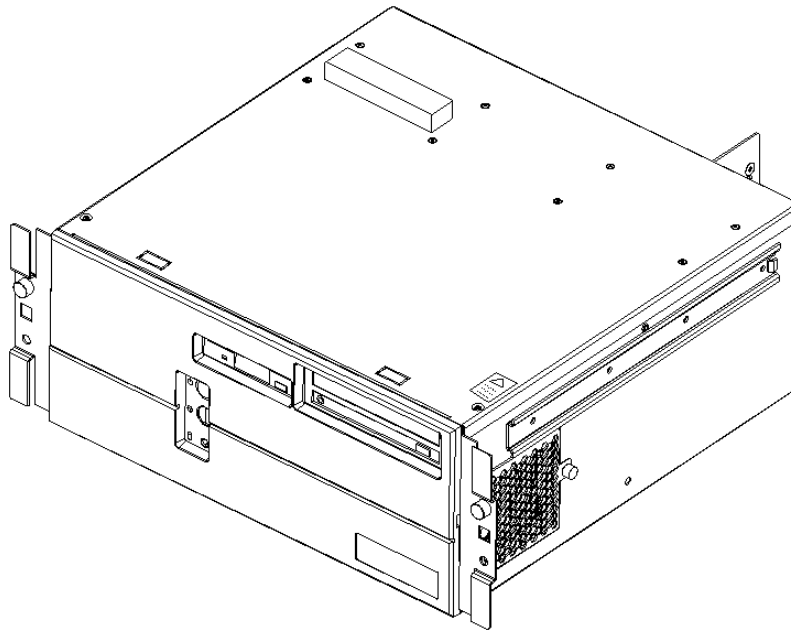


Figure 3-1 SVR2/WKS2 Enclosure for Cabinets and Consoles

Serviceability

The serviceability of the SVR2/WKS2 enclosure is based upon commercial desktop guidelines. Servicing requires common hand tools. It has an MTTR of one hour for hot swapping. Internal diagnostics rely on office desktop standards for troubleshooting common PC failures. Modular access to failing ORUs (Optimum Replacement Units) provides for ease of maintenance. The following figure follows shows a view of these modules.

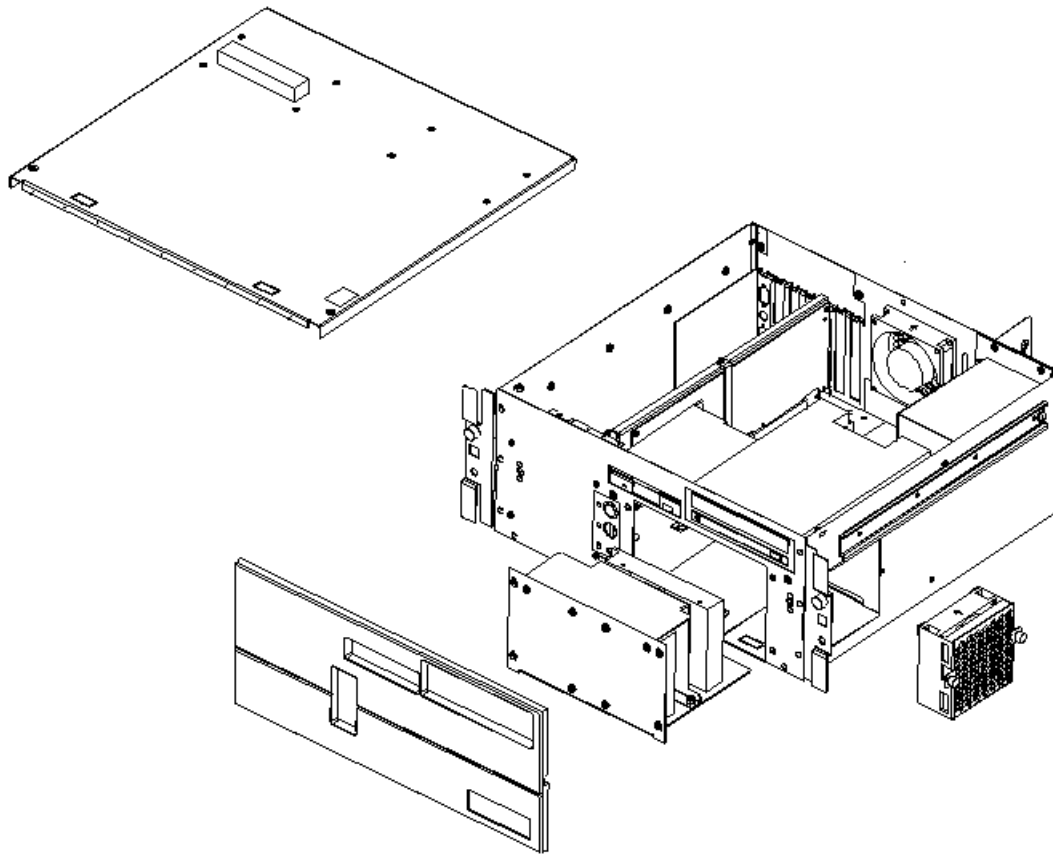


Figure 3-2 Exploded View of Enclosure ORUs

3. Platform Installation

3.2. SVR2/WKS2 Enclosures

Front view of enclosure

The following figure shows the front view of the enclosure.

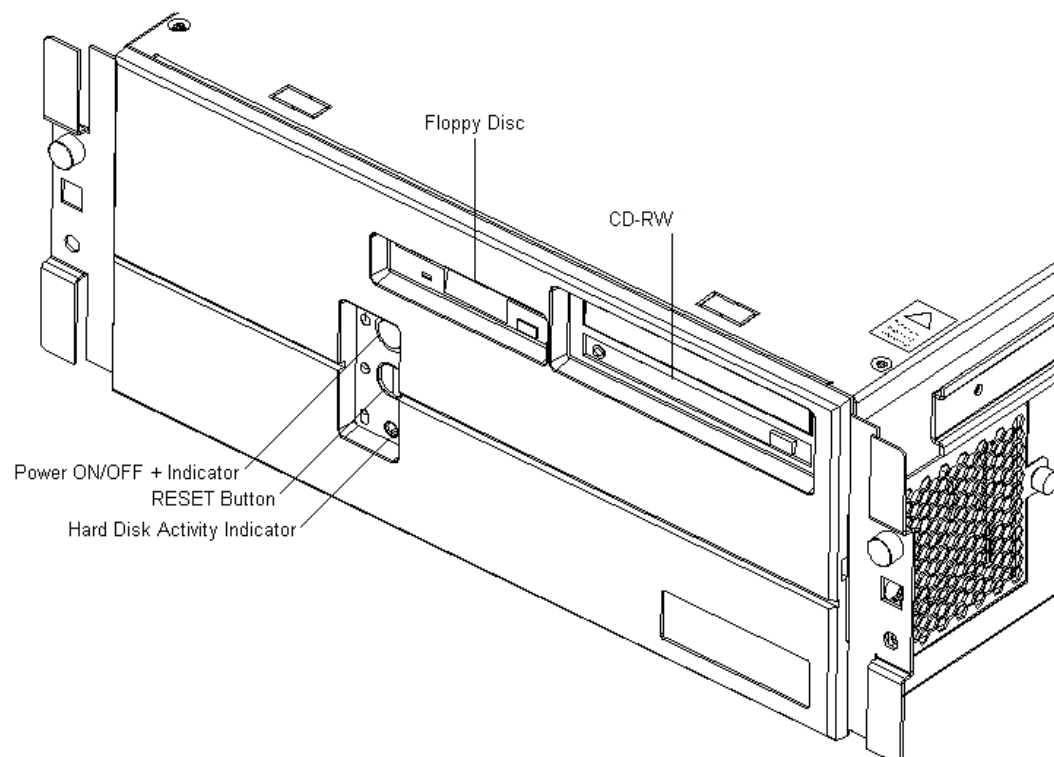


Figure 3-3 Front View of Enclosure

Top view of enclosure

The following figure shows the top view of the platform enclosure.

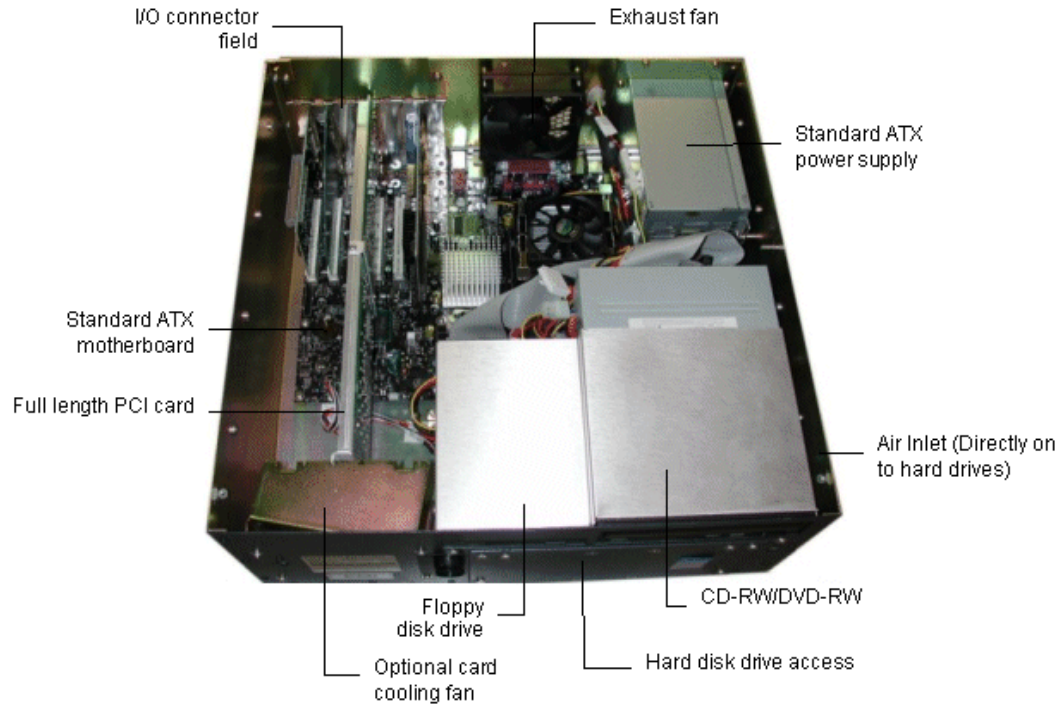


Figure 3-4 Top View Details of Enclosure

3.3 Cable Connections

Overview

The following picture illustrates the connectors for the platform.

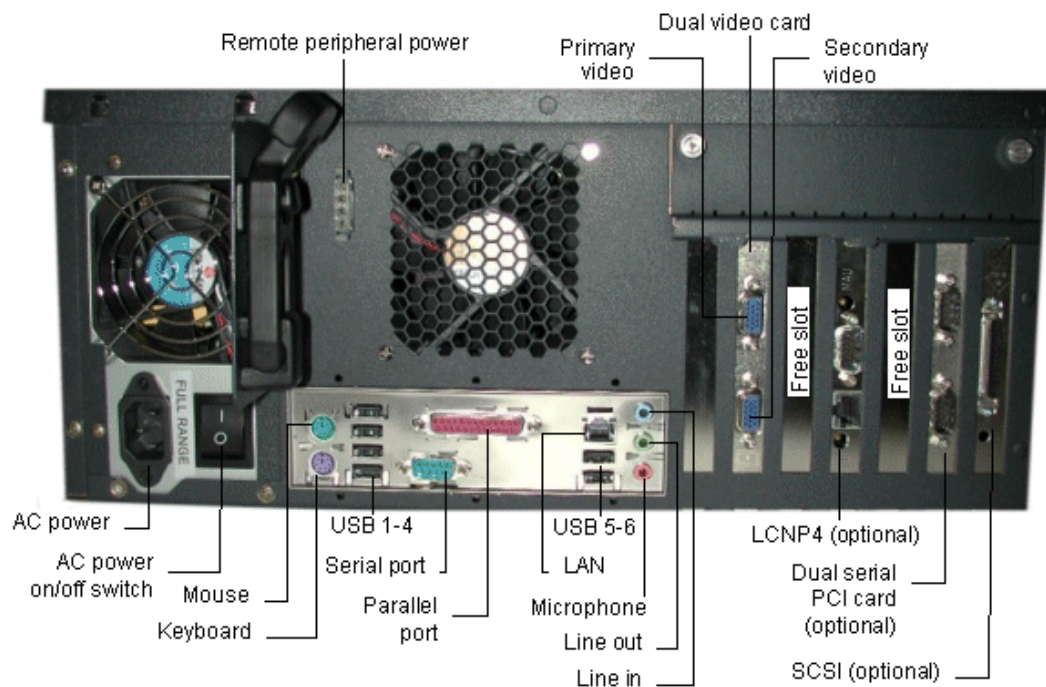


Figure 3-5 Cable Connections for WKS2 Node

GUS Cable connections

Use the following procedure to connect the GUS Node cables to the back of the WKS2 platform enclosure.

Step	Action
1	Connect the Ethernet cable to the RJ-45 connector on the Network Interface Card.
2	If you are not installing an Integrated Keyboard (IKB), connect the 104-keyboard cable to the Keyboard connector on the rear of the enclosure.
3	If you are installing an IKB, install the IKB according to the instructions in Section 8.4, "Using the Adapter to Connect the IKB."
4	Connect the mouse cable to the Mouse connector on the rear of the enclosure.
5	If you are installing a printer, connect the printer cable to the Parallel connector on the rear of the enclosure.
6	If you have a VGA monitor(s), configure the monitor(s) for VGA using the switch on the rear of the monitor.
7	For a single screen, connect the video cable to the primary video connector on the Dual Video card.
8	For two screens, connect the two video cables to the primary and secondary video connectors on the Dual Video card.
9	Connect the LCN MAU cable to the MAU connector on the LCNP4 board.
10	If you have a video with the Carroll touchscreen option, connect the touchscreen according to the instructions in Section 7.2, "Using the Touchscreen Adapters to Connect to the Computer."
11	If you have a video with the MicroTouch touchscreen option, connect the touchscreen cable to the Serial 1 connector.

3. Platform Installation

3.3. Cable Connections

Console cable connections

Use the following procedure to connect the cables when you are installing the platform in a console.

Step	Action
1	Connect the Ethernet cable to the RJ-45 connector on the NIC (Network Interface Card).
2	If you are installing an IKB, install the IKB according to the instructions in Section 8.4, "Using the Adapter to Connect the IKB."
3	Connect the mouse cable to the Mouse connector on the rear of the enclosure.
4	If you are installing a printer, connect the printer cable to the Parallel connector on the rear of the enclosure.
5	Configure the monitor(s) for VGA sync using the switch on the rear of the monitor.
6	For a single screen, connect the video cable to the primary video connector on the Dual Video card.
7	For two screens, connect the two video cables to the primary and secondary video connectors on the Dual Video card.
8	Connect the LCN MAU cable to the MAU connector on the LCNP4 board.

Icon Console cable connections

Use the following procedure to connect the cables when you are installing the platform in an Icon console.

Step	Action
1	Connect the Ethernet cable to the RJ-45 connector on the NIC (Network Interface Card).
2	Install the IKB according to the instructions in Section 8.4, "Using the Adapter to Connect the IKB."
3	Connect the mouse cable to the Mouse connector on the rear of the enclosure.
4	If you are installing a printer, connect the printer cable to the Parallel connector on the rear of the enclosure.
5	For a single screen, connect the video cable to the primary video connector on the Dual Video card.
6	For two screens, connect the two video cables to the primary and secondary video connectors on the Dual Video card.
7	For three screens (FPDs), connect the three video cables to the Channel 1, 2, and 4 video connectors on the Quad Video card.
8	For four screens (FPDs), connect the four video cables to the Channel 1, 2, 3, and 4 video connectors on the Quad Video card.
9	Connect the LCN MAU cable to the MAU connector on the LCNP4 board.

3.4 LCN Connection

Overview

The LCN Cable A and Cable B connections are made through a single cable from the LCNP4 board to the LCN Media Access Unit (MAU) contained in a metal

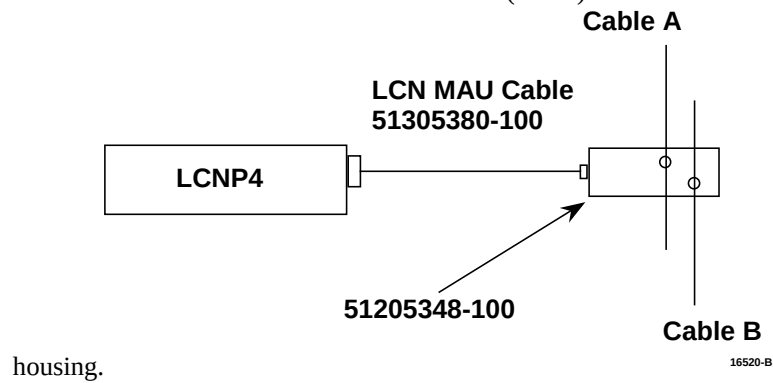


Figure 3-6 LCN MAU to LCN Cabling

MAU connection

Connect the MAU to both Cable A and Cable B coax T-connector as shown below.

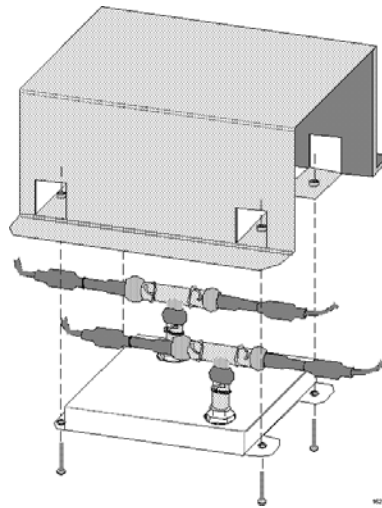


Figure 3-7 LCN MAU to LCN Cable T-Connections

3. Platform Installation

3.4. LCN Connection

Z-Console connection to the LCN coax cable

The mounting for the LCN MAU is located in the footrest area of the station.

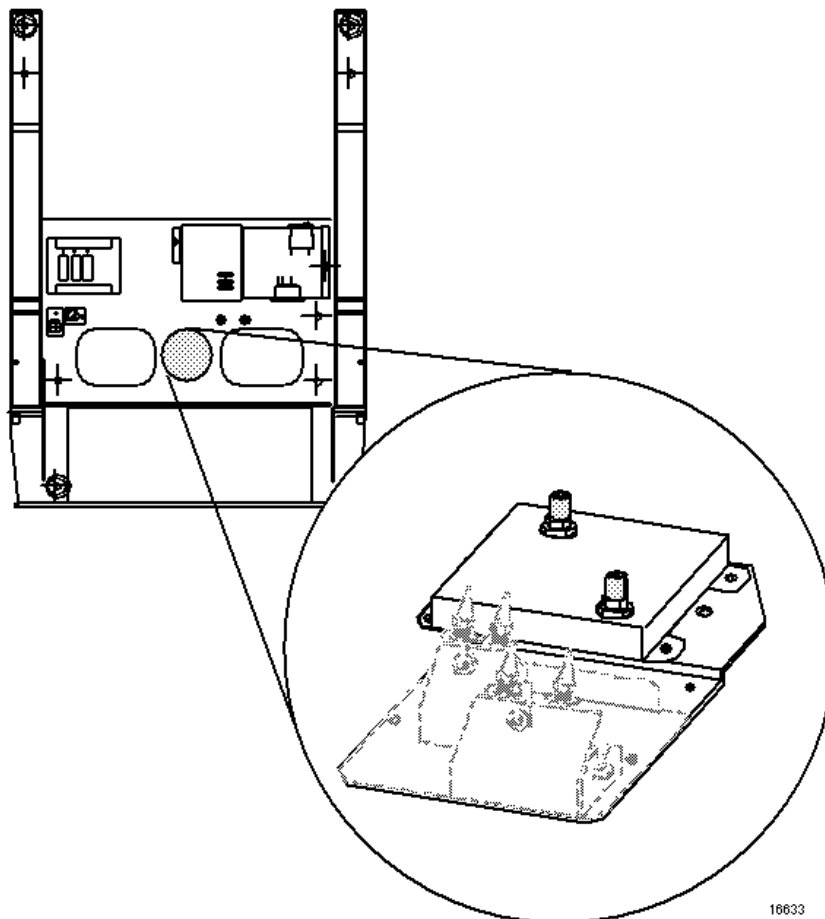


Figure 3-8 **Location of LCN MAU**

Connecting LCN coax to LCN MAU in Z-Console

Step	Action
1	Connect the color-coded A cables to the color-coded T-connector on the BNC connector marked A.
2	Connect the color-coded B cables to the color-coded T-connector on the BNC connector marked B.

Classic Console connection to the LCN Coax Cable

The LCN MAU is mounted to the EMC cover mounted to the rear of the platform enclosure.

Step	Action
1	Follow the color-coding while connecting the color-coded A cables to the color-coded T-connector on the BNC connector marked A.
2	Follow the color-coding while connecting the color-coded B cables to the color-coded T-connector on the BNC connector marked B.

3.5 Power-On Sequence

Turn on power

Step	Action
1	See Figure 3-5 for the location of the AC power switch.
2	Place the AC power switch, located on the rear panel, in the "1" position.
3	After turning the monitor and waiting for the raster to appear, press the Power On button on the front panel (top button in the middle of the front panel). The unit will boot itself from the hard drive and beep when the procedure is complete.

Check LCNP Status Display

The indicators in the LCNP Status Display discussed in the *TPS System Operations Guide* should be verified to ensure that the unit has passed the self-tests successfully.

3.6 Deskside GUS Operation

Switches and buttons

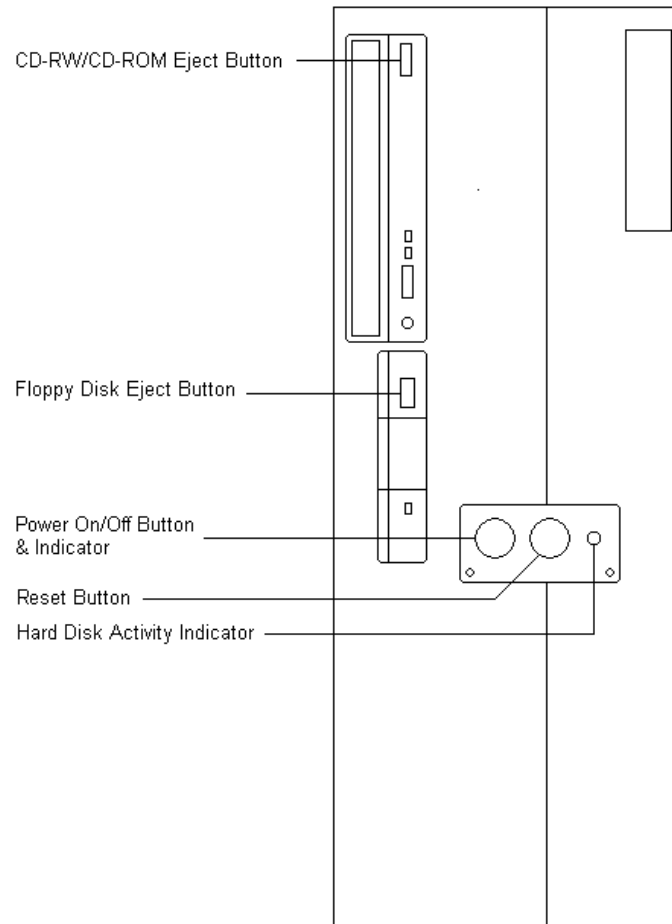


Figure 3-9 Front Panel Buttons and Indicators

3. Platform Installation

3.6. Deskside GUS Operation

4. Platform Operation and Service

4.1 Overview

Communication Port Settings

The following table summarizes the Communication Port I/O address and IRQ settings for a GUS Node on the SVR2/WKS2 platform.

Table 4-1 COM Port Settings

Port	I/O Address	IRQ
COM1	3F8	04
COM3	B800	19
COM4	BC00	19

4.2 Slot Configurations for SVR2

Overview

One AGP slot.

- AGP specification compliant.
- AGP 1.5V or 0.8V for 2x/4x/8x device support.

Five 32-bit Master PCI Bus slots.

Slot configuration options

The I/O interface boards are placed in the SVR2 enclosure. There are four I/O board slot configurations for the SVR2 GUS nodes:

- Board Configuration for Single/Dual Screen Systems
- Board Configuration for Quad Screen Systems

4. Platform Operation and Service

4.2. Slot Configurations for SVR2

Board Configuration for Single/Dual Screen Systems

Dual NIC	PCI5
Promise Tech RAID	PCI4
StarTech	PCI3
LCNP4	PCI2
SCSI (optional)	PCI1
Dual screen: G450 Dual Video Single screen: empty	AGP Slot

- **Note:** Single Screen – Connect video cable to top port of video card.

Board Configuration for Quad Screen Systems

Dual NIC	PCI5
Promise Tech RAID	PCI4
StarTech	PCI3
LCNP4	PCI2
G200 Quad Video	PCI1
Empty	AGP Slot

- **Note:** There is no internal SCSI controller option for the SVR2 with Quad video.

4.3 Slot Configurations for WKS2

Overview

One AGP slot.

- AGP specification compliant.
- AGP 1.5V or 0.8V for 2x/4x/8x device support.

Five 32-bit Master PCI Bus slots.

Slot configuration options

The I/O interface boards are placed in the WKS2 enclosure. There are six I/O board slot configurations for the WKS2:

- Board Configuration for Single/Dual Screen Systems
- Board Configuration for Quad Screen Systems

Board Configuration for Single/Dual Screen Systems

Dual NIC	PCI5
SCSI	PCI4
StarTech	PCI3
LCNP4	PCI2
Empty	PCI1
Dual screen: G450 Dual Video Single screen: empty	AGP Slot

- Note: Single Screen – Connect video cable to top port of video card.

Board Configuration for Quad Screen Systems

Dual NIC	PCI5
SCSI	PCI4
StarTech	PCI3
LCNP4	PCI2
G200 Quad Video	PCI1
Empty	AGP Slot

4. Platform Operation and Service

4.4. Servicing and Replacing an SVR2/WKS2 in a Console or Cabinet

4.4 Servicing and Replacing an SVR2/WKS2 in a Console or Cabinet

Accessing the Electronics

The platform enclosure provides access to all internal, hardware components without having to remove the enclosure from the Console or Cabinet in which it is mounted, which simplifies servicing.

Step	Action
1	Shut down the Operating System.
2	Press the Power On/Off Switch on the front of the enclosure to turn the power off.
3	Using a screwdriver, loosen the two captivated panel fasteners that secure the enclosure to its mounting rails in the console/cabinet.
4	Grasp the mounting ears and carefully slide out the enclosure until the detents in the rails click into place.
5	Remove the top cover. Refer to the "Remove the top cover" procedure.
6	Service the platform hardware components as required.
7	Replace the top cover. Refer to the "Replace the top cover" procedure.
8	Slide the enclosure back into the console/cabinet.
9	Tighten the two captivated panel fasteners that secure the enclosure to its mounting rails
10	Press the Power On/Off Switch on the front of the enclosure to turn the power back on.
11	Restart the Operating System.

If you need to replace the platform from the Console or Cabinet, execute the "Mounting platforms" procedure in the following section.

Mounting platforms

The SVR2/WKS2 enclosure is designed to rack mount in a rail system that conforms to the industry standard 19-inch RETMA criteria. If a non-Honeywell equipment rack is used, make sure that parallel side-to-aft cooling airflow is provided.

Figure 4-1 and Figure 4-2 show a platform mounted in a Z-Console and a Classic Console.



Figure 4-1 **Platform Mounted in a Z-Console**

4. Platform Operation and Service

4.4. Servicing and Replacing an SVR2/WKS2 in a Console or Cabinet



Figure 4-2 **Platform Mounted in a Classic Console**

4. Platform Operation and Service

4.4. Servicing and Replacing an SVR2/WKS2 in a Console or Cabinet

Replace the SVR2/WKS2 in a Console/Cabinet

Use the following procedure to install a new SVR2/WKS2 into a console.

Step	Action
<i>Remove existing Platform</i>	
1	Shut down the Operating System.
2	Press the Power On/Off Switch on the front of the enclosure to turn the power off.
3	Using a screwdriver, loosen the two captivated panel fasteners that secure the enclosure to its mounting rails in the console/cabinet.
4	From the rear of the cabinet, disconnect all cables from the connectors on the rear of the enclosure and then remove them from the cable restraint clamp.
5	Press the two release buttons located on the outside of each slide rail.
6	Slide the platform forward, off the rails and set it on your workbench.
7	If the replacement platform does not already have a chassis slide rail installed, remove it from the old unit and install it on the new one.
<i>Install the new Platform</i>	
8	Grasp the mounting ears and carefully slide the enclosure back in until the detents in the rails click into place.
9	Using a screwdriver, tighten the two captivated panel fasteners to secure the enclosure to its mounting rails in the console/cabinet.
10	Reconnect all cables to the connectors on the rear of the platform and reattach the cable restraint clamp.

4. Platform Operation and Service

4.5. Replacing the Power Supply

4.5 Replacing the Power Supply

Replace the power supply for SVR2/WKS2

Use the following procedure to replace the Power Supply.

Step	Action
1	Press the Power On/Off button on the front of the platform to turn the power off.
2	Remove the front cover from the platform enclosure. Refer to the "Remove the front cover" procedure in Section 4.10.
3	Remove the top cover from the platform enclosure. Refer to the "Remove the top cover" procedure in Section 4.10.
4	Remove the power cable to the power supply.
5	Remove all cables connecting the power supply to the peripherals and the motherboard.
6	Remove the four (4) Phillips screws that secure the power supply to the back of the SVR2/WKS2 enclosure.
7	Remove the Phillips screw that secures the power supply to the side of the SVR2/WKS2 enclosure.
8	Carefully lift out the power supply and replace it with the new one.
9	Replace the Phillips screw that secures the power supply to the side of the SVR2/WKS2 enclosure.
10	Replace the four (4) Phillips screws that secure the power supply to the back of the SVR2/WKS2 enclosure.
11	Replace all cables to the peripherals and the motherboard.
12	Replace the power cable to the power supply.
13	Replace the top cover from the platform enclosure. Refer to the "Replace the top cover" procedure in Section 4.10.
14	Replace the front cover from the platform enclosure.
15	Press the Power On/Off button on the front of the platform to turn the power back on.

4.6 Replacing the Cooling Fans

Overview of intake fan

The SVR2/WKS2 platform is equipped with a **hot-swappable** cooling fan. The intake fan provides side-to-aft cooling of the SVR2/WKS2 enclosure. The removal and replacement of the failed cooling fan can be performed in a “hot-swap” condition without having to take the platform or network offline. See Figure 4-3 for the location of the fan.

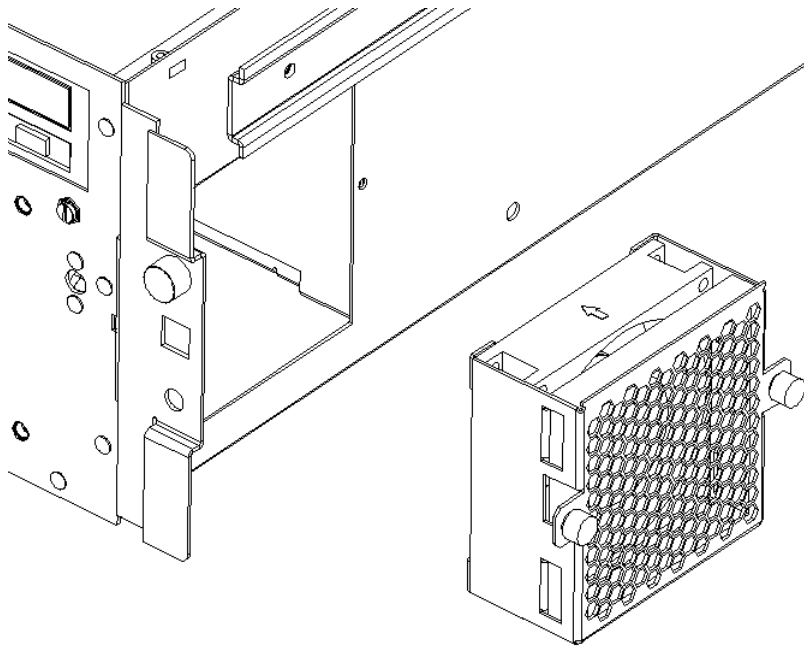


Figure 4-3 Intake Cooling Fan

4. Platform Operation and Service

4.6. Replacing the Cooling Fans

Replace the intake cooling fan

The following procedure will allow you to identify the failed cooling fan, remove it, and replace it with a new fan, with the platform still operating.

Step	Action
1	Verify that the intake cooling fan is not operational by placing a sheet of paper over the grillwork. If the fan is operational, the air intake will cause the paper to cling to the grillwork. If the paper does not cling to the grillwork, proceed to the next step.
2	Loosen the two (2) knurled-knob retaining screws with a slot-head screwdriver. Once loose, continue to loosen with thumb and fingers.
3	Carefully remove the fan assembly until the power plug is accessible. Unplug power from the fan assembly.
4	Remove the four (4) screws that secure the fan to the intake fan module. Slide out the failed fan. Remove the air filter located in front of the fan mounting location and make certain it is clean and unrestricted. Clean the filter if necessary and reinstall. Install the new replacement fan.
5	Position the replacement fan near the power plug and plug the power cable into the receptacle on the fan assembly. Verify proper fan operation.
6	Align the two (2) knurled-knob retaining screws with the mounting holes on the side of the enclosure. Finger-tighten the retaining screws. Further secure the two (2) knurled-knob retaining screws with a slot-head screwdriver.

Overview of exhaust fan

The SVR2/WKS2 platform is equipped with a **hot-swappable** exhaust cooling fan. See Figure 4-4 below for the location of the fan.

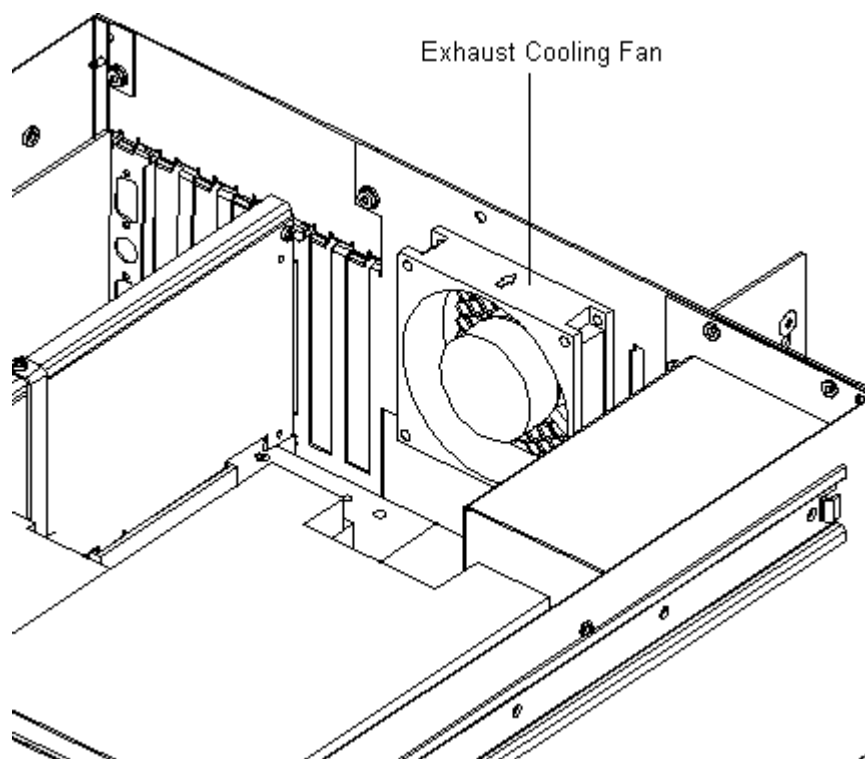


Figure 4-4 Exhaust Cooling Fan

4. Platform Operation and Service

4.6. Replacing the Cooling Fans

Replace the exhaust cooling fan

The following procedure will allow you to identify the failed cooling fan, remove it, and replace it with a new fan.

Step	Action
1	Verify that the exhaust fan is not operational by placing a sheet of paper in front of the grillwork. If the fan is operational, the airflow will blow the paper away from the grillwork. If the paper does not blow away from the grillwork, proceed to Step 2 if it is cabinet-mounted, or Step 4 if the enclosure is configured as a desktop/deskside.
2	If the SVR2/WKS2 enclosure is cabinet-mounted, go to Step 3. If the platform is configured as a desktop/deskside, go to Step 5.
3	If the SVR2/WKS2 enclosure is mounted in an equipment cabinet, loosen the black plastic panel fastener knobs located on the WKS2 enclosure mounting flanges that secure the enclosure to the cabinet rail mounting rails.
4	With the panel fasteners fully disengaged, pull the slide-mounted SVR2/WKS2 enclosure forward until dents in the sides lock into place.
5	Remove the top cover from the enclosure. Refer to the "Remove the top cover" procedure in Section 4.10.
6	Disconnect the fan power connector. Remove the four (4) screws that secure the fan to the rear wall of the enclosure.
7	Replace the failed fan with a new fan, securing it with the four screws.
8	Reconnect the fan power cable and verify proper fan operation.

4.7 Servicing and Replacing LCNP4 Boards

LCNP4 board description

The LCNP4 board has 16 MB of on-board memory that is soldered to the board.

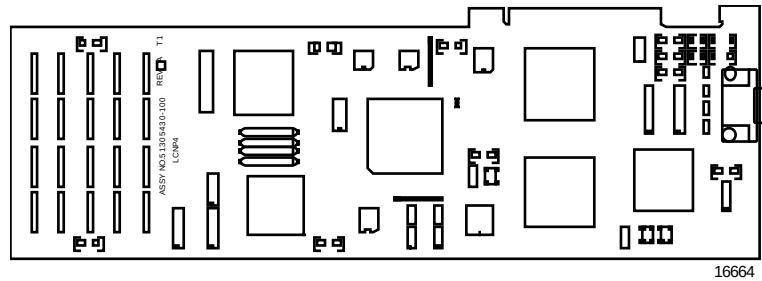


Figure 4-5 LCNP4 Board

4. Platform Operation and Service

4.7. Servicing and Replacing LCNP4 Boards

Replace the LCNP4 board

The LCNP4 board is located in the PCI2 slot, as shown in Figure 3-5. Use the following procedure to replace any of the PCI boards after the required board pinning has been verified.

Use ESD



Wrist Strap
16310

Step	Action
1	Shut down the Operating System.
2	Press the Power On/Off button on the front of the platform to turn the power off.
3	Remove the top cover from the SVR2/WKS2 enclosure. Refer to the "Remove the top cover" procedure in Section 4.10.
4	Disconnect the LCN MAU cable from the LCNP4 board.
5	Loosen the two screws and remove the Card Retainer Bracket.
6	Using a grounded ESD wrist strap, remove the LCNP4 board assembly from the riser connector and replace it with the new board.
7	Replace the Card Retainer Bracket and tighten the two screws.
8	Replace the top cover from the SVR2/WKS2 enclosure. Refer to the "Replace the top cover" procedure in Section 4.10.
9	Press the Power On/Off button on the front of the platform to turn the power back on.
10	Run the Configurator to re-establish the correct node number.
11	Reconnect the LCN MAU cable.
12	Restart the Operating System.

LCNP Status Display

The status display for the LCNP4 board is documented in the *TPS Operations Guide*.

4.8 Replacing the Video Board for WKS2

Overview

The Dual Video board is located in the AGP slot.



CAUTION

Be careful not to damage the EMI gasket fingers when removing/installing boards.

Replace the dual video board

Perform the following procedure to replace the Dual Video Board.

Step	Action
1	Shut down the Operating System.
2	Press the Power On/Off button on the front of the platform to turn the power off.
3	Remove the top cover from the SVR2/WKS2 enclosure. Refer to the "Remove the top cover" procedure.
4	Loosen the two screws and remove the Card Retainer bracket.
5	Disconnect the cables from the Dual Video board.
6	Remove the Dual Video board from the riser connector and replace it with the new board.
7	Reconnect the cables to the Dual Video board.
8	Replace the Card Retainer Bracket and tighten the two screws.
9	Replace the top cover from the SVR2/WKS2 enclosure. Refer to the "Replace the top cover" procedure in Section 4.10.
10	Press the Power On/Off button on the front of the platform to turn the power back on.
11	Restart the Operating System.

4.9 Troubleshooting

System interrupt assignments (with Carroll Touchscreen)

These PCI Bus devices are dynamically reconfigurable by the system. The actual IRQs assigned may change from one boot to the next. They may even show overlapping IRQ assignments among themselves. This is normal.

IRQ	Device	Notes
01	Standard 101/102-key or Microsoft Natural PS/2 Keyboard	PC Keyboard Port
04	Communication Port (COM1)	Serial port 1 on motherboard COM 1
11	Honeywell LCNP4GPS	LCNP4
18	Intel PRO/1000 CT	Ethernet
06	Standard Floppy disk controller	Floppy
19	Lava D serial PCI Port A (COM3)	COM Port 3
19	Lava D serial PCI Port B (COM4)	COM Port 4
18	Symbios Logic 875xs/ Dm 2280x PCI SCSI Adapter	LSI SCSI Controller
17	SondMAX Integrated Digital Audio	On board sound
12	Microsoft PS/2 Mouse	Mouse Port

Ethernet Troubleshooting

There are two indicators located in the RJ-45 connector on the back of the NIC card in the platform.

The left indicator is the Link Integrity Indicator and the right indicator is the Activity Indicator.

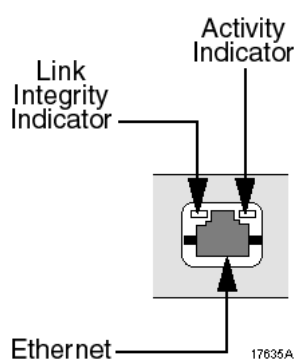


Figure 4-6 Ethernet Indicators on the RJ-45 Connector

4.10 Enclosure Service

Front Cover

The front cover can be mounted on an SVR2/WKS2 enclosure in either a rack-mounted (cabinet) or desktop configuration. This front cover will shield the hardware assembly components from the casual observer and improve the appearance of the enclosure. The panel is held in place by two studs mounted on the cover that fit into friction connections on the front of the enclosure

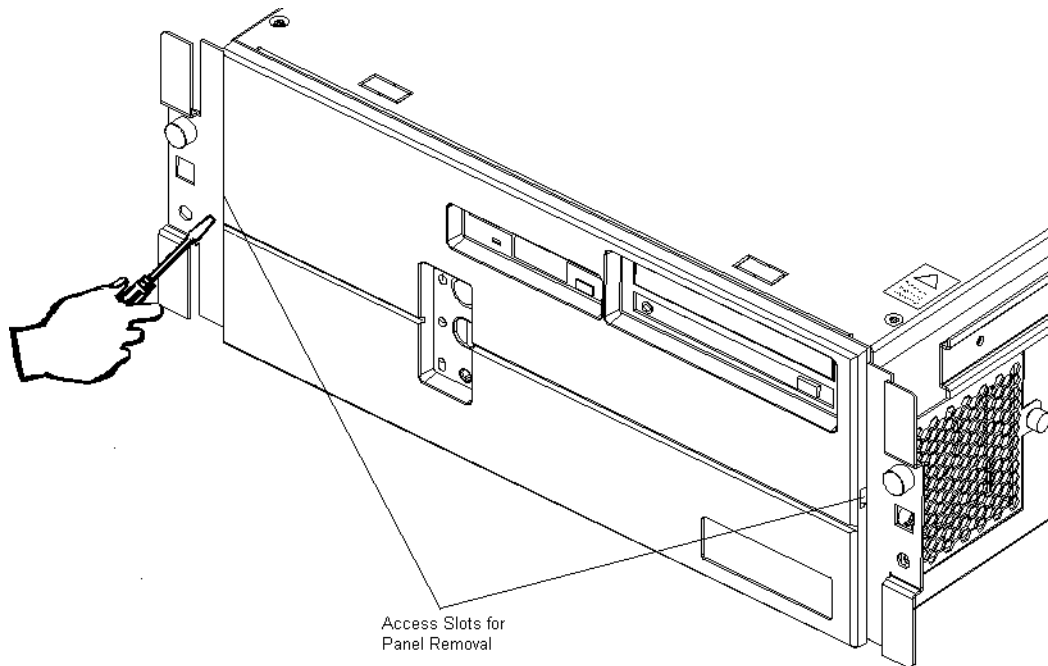


Figure 4-7 Front Cover for SVR2/WKS2 Enclosure

Install the front cover

Use the following procedure to install the front cover.

Step	Action
1	Align the two mounting studs to the corresponding mounting receptacles on the front panel. Apply pressure to both sides until the cover is friction seated into the mounting receptacles.

Remove the front cover

Perform the following procedure to remove the front cover.

Step	Action
1	Use a slot-head screwdriver to access the slotted hole on either side of the front cover. To pop loose the cover, twist or pry the screwdriver in the slot. Grasp both sides and remove it from the front panel of the SVR2/WKS2 enclosure.

Top cover

Servicing of the TPS SVR2/WKS2 enclosure is based on standard PC repair guidelines. Access to the common PC components is from the top of the enclosure, and requires the removal of the top cover. The top cover must also be removed to replace the exhaust fan.



CAUTION

A warning label on the cover of the enclosure states that you should consult this manual prior to removing the cover. There is a rubber block under the cover that provides cushioning to the adapter boards and connectors. Correct removal of the cover will prevent damage to the board and/or connectors.

4. Platform Operation and Service

4.10. Enclosure Service

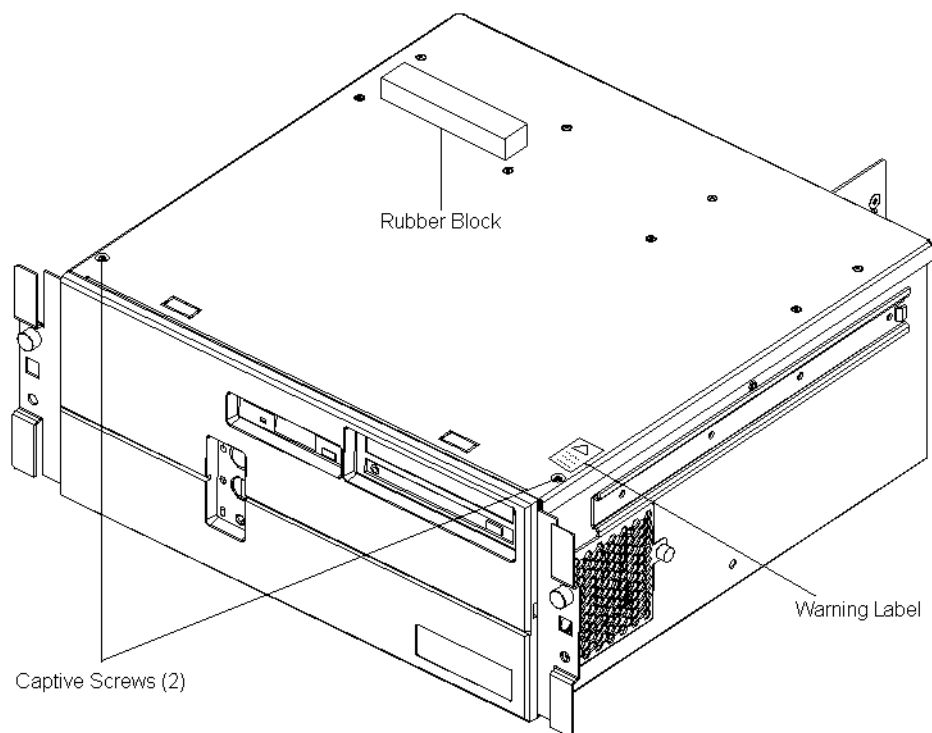


Figure 4-8 Top Cover Removal Details

Remove the top cover

Use the following procedure to remove the front cover.

Step	Action
1	Loosen the captive screws (2) located at the front of the cover.
2	Lift the cover slightly to clear the captive screws from their holes.
3	Slide the cover back slightly and then set the cover back down.
4	Push the cover back with a slight pressure to compress the rubber block that hits the connector.
5	Lift the cover straight up and away from the enclosure.

Replace the top cover

Perform the following procedure to replace the front cover.

Step	Action
1	Place the cover back on the enclosure.
2	Push the cover forward away from the rubber block that hits the connector.
3	Slide the cover forward slightly and then set the cover back down.
4	Move the cover so that the captive screws are exposed.
5	Tighten the captive screws (2) located at the front of the cover.

4.10. Enclosure Service

5. Media Drive Operation and Service

5.1 Overview

SVR2 hard drive

The SVR2 platform has two internal hard disk drive bays. The base system has two internal 61 GB or larger EIDE hard drives. The base 61 GB hard drive is mounted in hard drive bay 1 and pinned as “Device 0 (Master)” and the second drive is mounted in hard drive bay 0 and must be pinned as “Device 0 (Master)” (See *Note below).

***NOTE:** Use the 16 HEADS pin outs diagram on the hard drive.

Both hard drives are connected to the IDE RAID controller; one hard drive is connected to the IDE1 connector and the other is connected to the IDE2 connector using two separate IDE cables.

WKS2 hard drive

The WKS2 platform has two internal hard disk drive bays. The base system has one internal 61 GB or larger EIDE hard drives. The base 61 GB or larger hard drive is mounted in hard drive bay 1 and is pinned as “Device 0 (Master)” (See *Note below) and an optional expansion drive can be mounted in hard drive bay 0 and must be pinned “Device 1 (Slave)” (See *Note below).

***NOTE:** Use the 16 HEADS pin outs diagram on the hard drive.

All hard drives are daisy chained together on an IDE cable connected to IDE1 connector on the motherboard. Only one additional 61 GB or larger hard drive can be added.

SVR2/WKS2 Removable Media

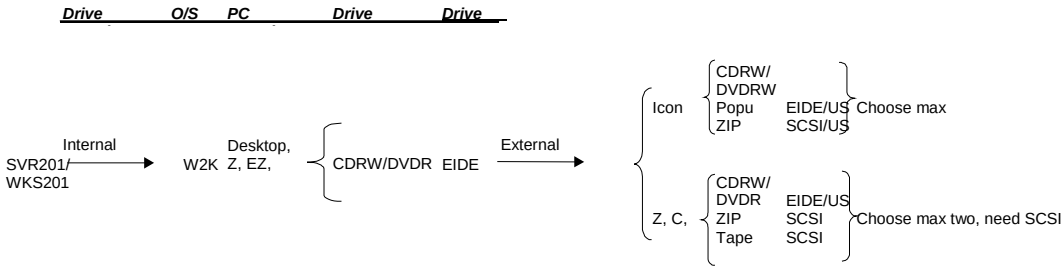
The SVR2/WKS2 platform has one 3.5-inch bay, dedicated to a 3.5-inch floppy drive. This floppy drive is to be left in place; it may not be remotely mounted in consoles.

The SVR2/WKS2 platform is configured with one (1) removable media drive (in addition to the floppy drive mounted in the electronics module); this drive is a CD-RW/DVD-RW (+/-) drive. The CD-RW/DVD-RW (+/-) drive is pinned “Master” and is connected to the IDE2 connector on the motherboard.

5. Media Drive Operation and Service

5.1. Overview

SVR2/WKS2 Removable media chart



CAN HAVE BOTH INTERNAL AND CONSOLE DRIVE SCENARIOS IN

Model	TP- DVDRW/CDRW Drive for Z, Classic, EZ (GUS 3xx and later only)
Rule	Total number of external CDRWDVDRW drives Total number of installable remote removable media drives

5.2 Servicing and Replacing the Hard Disk Drive for WKS2

Description

Drive 1 is the easiest to access for replacement purposes. The optional hard disk drive is designated as Drive 0 (drive closest to the motherboard). Drive 0 requires some additional time to remove the mounting bracket in order to gain access. See the following figure for details.

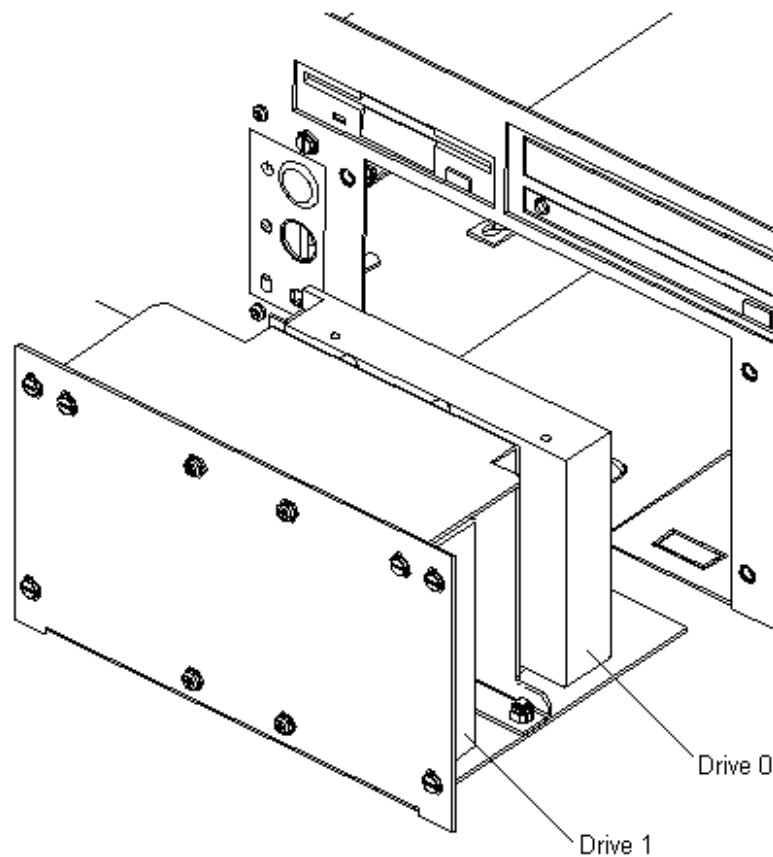


Figure 5-1 Hard Disk Drives

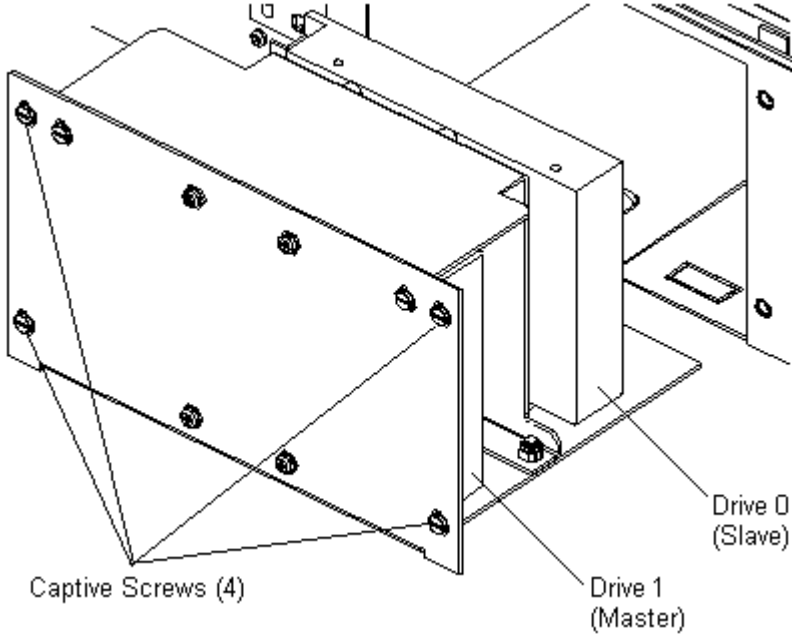
5. Media Drive Operation and Service

5.2. Servicing and Replacing the Hard Disk Drive for WKS2

Replace the Hard Disk Drive(s) for WKS2

Perform the following procedure to replace one or both hard disk drives.

Step	Action
Accessing the hard disk drives	
1	Shut down the Operating System.
2	Press the Power On/Off Switch on the front of the platform to turn the power off.
3	Remove the front cover from the WKS2 platform enclosure. Refer to the "Remove the front cover" procedure in Section 0 for details.
4	Loosen the four (4) Slot-head captive screws on the outside edges of the front panel on the hard drive assembly.



The diagram illustrates the internal hard drive assembly of the WKS2 platform. It shows a front panel held in place by four slot-head captive screws, which are highlighted with leader lines and labeled "Captive Screws (4)". Behind the panel, two hard disk drives are visible, labeled "Drive 0 (Slave)" and "Drive 1 (Master)". The drives are mounted on a metal frame within the enclosure.

5. Media Drive Operation and Service

5.2. Servicing and Replacing the Hard Disk Drive for WKS2

Step	Action
5	Slowly pull on the panel until the assembly panel starts moving toward you. The signal and power cables are located on the left side of the assembly. Make sure the cables do not bind up on the side, as this can prevent removal of the assembly. It may be necessary to remove the cable ties from the cables so you can slide the assembly out easily.
	ATTENTION You cannot slide out the hard disk drive assembly further than the service loops of the signal and power cable. Cable ties may have to be removed for better access.
	ATTENTION The hard disk drive assembly has a bottom plate that slides on retainer channels on the bottom plate of the enclosure. The bottom plate of the hard disk drive assembly extends itself under the motherboard. This channel slot prevents the assembly's bottom plate from tilting up into the bottom of the motherboard thereby preventing damage to the board.
6	If you are going to replace Drive 1, execute Steps 7 through 15, then Steps 31 through 33. If you are going to replace Drive 0, go to Step 16 and execute only Steps 16 through 31.
Replace Drive 1	
	ATTENTION Be sure you have performed Steps 1 through 6 in this section before you perform the following steps.
7	Remove the Power plug from its receptacle on Drive 1.
	TIP It is recommended that you place the thumb and index finger at the top and bottom of the plug. Apply pressure in a slight rocking motion while pulling on the plug to free it from the receptacle on Drive 1.
8	Remove the Signal plug from its receptacle on Drive 1. Refer to the TIP in the previous step.
9	Remove the four (4) Philips-head screws that secure device 1 to the front mounting plate.
	Then, carefully remove the hard disk drive out either side of the mounting assembly.

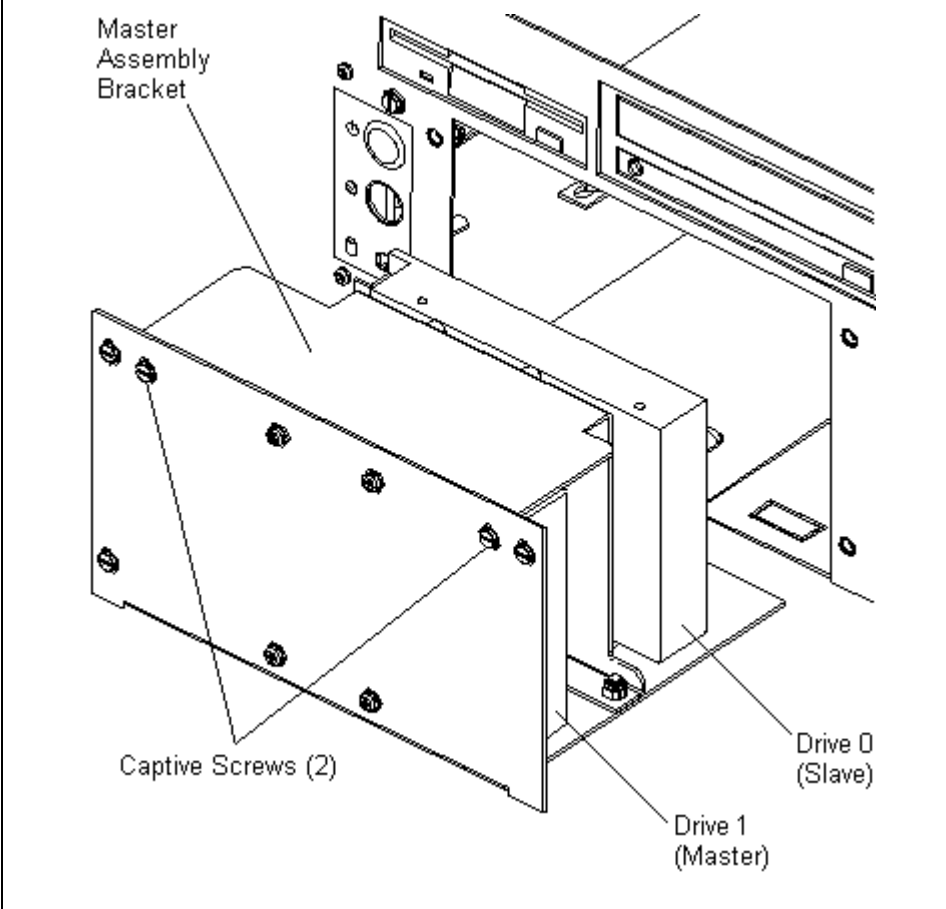
5. Media Drive Operation and Service

5.2. Servicing and Replacing the Hard Disk Drive for WKS2

Step	Action
10	Check the replacement hard disk drive to verify that the device address pinning of the replacement drive <u>matches</u> the pinning of the failed drive. Change pinning if necessary.
11	Place the new hard disk drive on the front mounting plate, align mounting holes, and secure in place with the four (4) Philips-head screws.
12	Plug the Signal cable into the receptacle on the replacement hard disk drive.
13	Plug the Power cable into the receptacle on the replacement hard disk drive.
	ATTENTION The power plug is keyed with yellow wire at the top
14	Verify the hard disk drive has power and is rotating by placing the hand on the surface of the housing and feel for drive vibration.
15	Go to Step 29 to place the hard drives back into the enclosure.
Replace Drive 0	
	ATTENTION Be sure you have performed Steps 1 through 6 in this section before you perform the following steps.
16	Remove the Power plug from its receptacle on Drive 0.
	TIP It is recommended that you place the thumb and index finger at the top and bottom of the plug. Apply pressure in a slight rocking motion while pulling on the plug to free it from the receptacle on Drive 0.
17	Remove the Signal plug from its receptacle on Drive 0. Refer to the TIP in the previous step.
18	Locate and remove the two (2) star nuts mounted on base plate studs. A socket drive will be required.
19	Loosen the two (2) Slot-head captive screws on the front mounting bracket assembly that secures Drive 0. Remove Drive 0 mounting bracket assembly out the top and place on a work surface.

5. Media Drive Operation and Service

5.2. Servicing and Replacing the Hard Disk Drive for WKS2

Step	Action
 The diagram illustrates the internal components of a server chassis. A Master Assembly Bracket is shown at the top, with two Captive Screws (2) indicated. Below the bracket, two hard disk drives are visible: Drive 0 (Slave) and Drive 1 (Master). The drives are mounted on a common rail system.	
20 	Remove the four (4) Philips-head screws that secure Drive 0 to its mounting plate. Then, carefully remove the hard disk drive.
21	Check the replacement hard disk drive to verify that the device address pinning of the replacement drive <u>matches</u> the pinning of the failed drive. Change pinning if necessary.

5. Media Drive Operation and Service

5.2. Servicing and Replacing the Hard Disk Drive for WKS2

Step	Action
22	Place the new hard disk drive on the Drive 0 mounting plate, align the mounting holes and secure in place with the four (4) Philips-head screws.
23	Retrieve the mounting bracket with the drive assembly mounted from the work surface and place it onto the dual drive assembly. Align the drive assembly onto the two (2) base mounting studs. Align the two (2) Slot-head captive screws on the front mounting plate and secure them.
24	Place the two (2) star nuts onto the base plate studs. Secure them in place with a socket driver.
25	Plug the Signal cable into the receptacle on the replacement hard disk drive.
26	Plug the Power cable into the receptacle on the replacement hard disk drive
	ATTENTION The power plug is keyed with yellow wire at the top
27	Verify the hard disk drive has power and is rotating by placing the hand on the surface of the housing and feel for drive vibration.
28	Continue on with Step 29 to place both hard drives back into the enclosure.
Place the drives back into the enclosure	
	ATTENTION When sliding or pushing the hard disk drive assembly back into the enclosure be careful not to accidentally press the Power On/Off or Reset button on the front panel. This is very easy to do, so be careful when pushing the assembly back into the enclosure. Remember that the platform is still "hot" and operating in a network environment.
29	Slowly push on the assembly front panel until the hard disk drive assembly starts moving into the enclosure. The signal and power cables are located on the left side of the assembly. Care should be given to the cables so they do not bind up on the side preventing the installation of the assembly. It may be necessary to dress the cables to ease sliding in the assembly.
30	Tighten the four (4) Slot-head captive screws on the outside edges of the front panel on the dual hard disk drive assembly.

5. Media Drive Operation and Service

5.2. Servicing and Replacing the Hard Disk Drive for WKS2

Step	Action
31	Replace the front cover from the WKS2 platform enclosure.
32	Press the Power On/Off Switch on the front of the platform to turn the power back on.
33	Restart the Operating System.

5. Media Drive Operation and Service

5.3. Servicing and Replacing the Hard Disk Drive for SVR2

5.3 Servicing and Replacing the Hard Disk Drive for SVR2

Overview

The SVR2 Server is equipped with dual, hot-swappable hard disk drives, together with a disk shadowing (RAID 1) SCSI RAID controller. The drive closest to the front access plate is designated as Drive 1. Drive 1 is the easiest to access for replacement purposes. The drive closest to the motherboard is designated as Drive 0. Drive 0 requires some additional time to remove the mounting bracket in order to gain access. See Figure 5-1 for details.

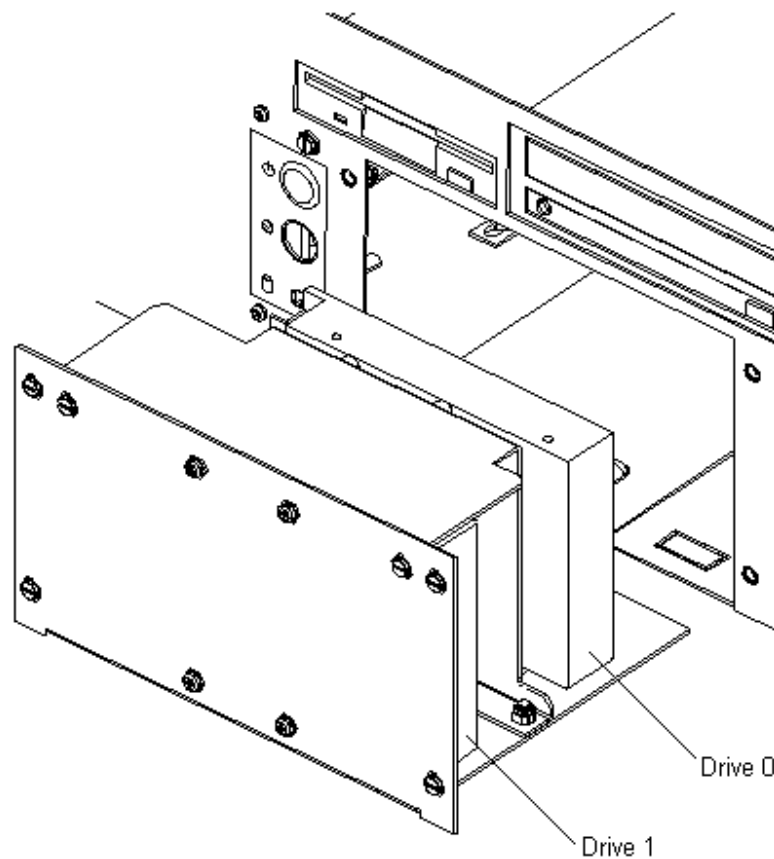


Figure 5-2 Hard Drives

The two hard disk drives are supplied with separate sources of power. The following illustration shows the power source connections for Drive 0 and Drive 1.

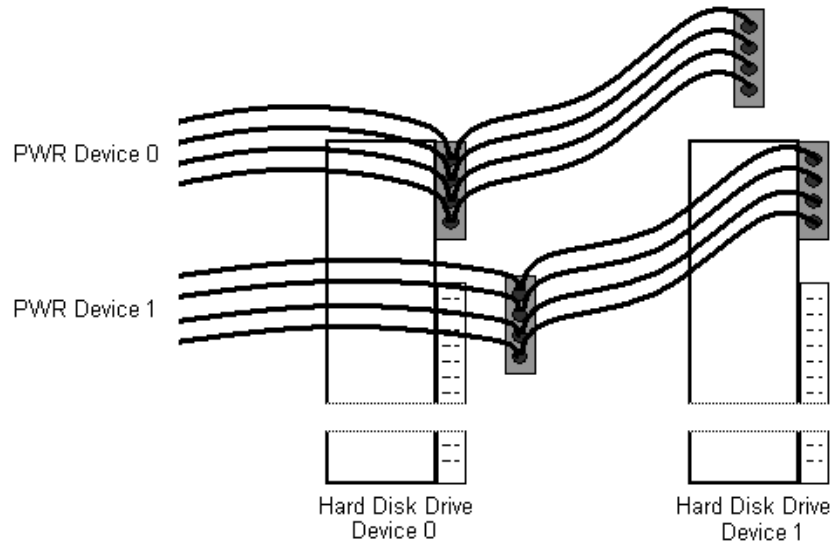


Figure 5-3 Power Cabling for Hard Disk Drives

Prerequisite

Prior to a hard disk drive failure, it is assumed that the Server (SVR2) is running error free and that the Promise Array Management Utility has been installed and used to configure redundancy and disk failure notification.

About the Promise Array Management (PAM) utility

The Promise Array Management (PAM) utility is the main user interface for monitoring and managing Promise Technology RAID products. See the *Promise Array Management Utility User's Manual* for complete instructions on performing all of the following tasks:

- Installing the PAM utility
- Setting up the utility, including creating arrays, setting up a message server and configuring alert notification events
- Using the interface
- Monitoring and maintaining RAID with the utility, including rebuilding an array

5. Media Drive Operation and Service

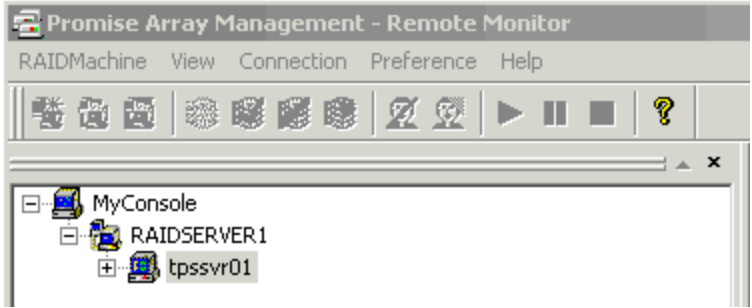
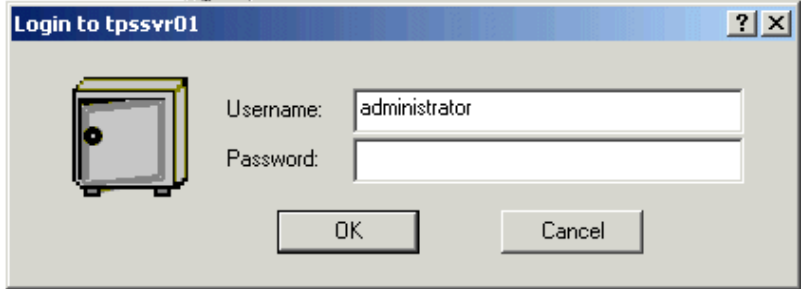
5.3. Servicing and Replacing the Hard Disk Drive for SVR2

Detecting a failure

A fault-tolerant feature of the Server is the user notification when one or both of the dual hard drives fail. The type of notification you receive when a failure occurs depends on how you have the alert notification events configured. See the *Promise Array Management Utility User's Manual* for complete instructions on setting up alert notifications.

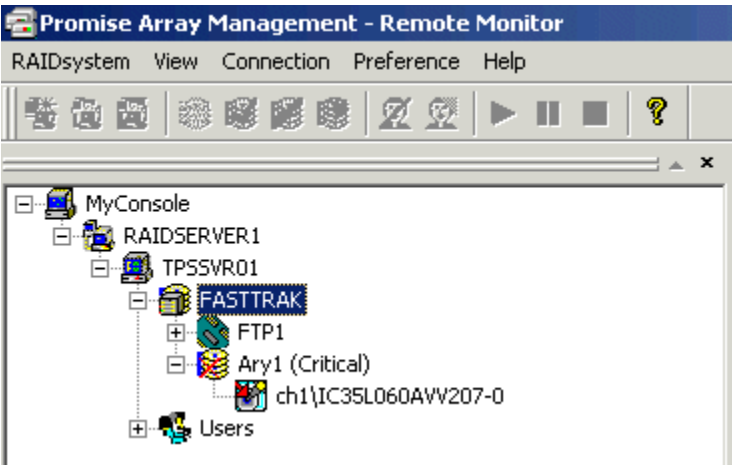
Identifying the failed device

Perform the following procedure to identify the failed device. See also the *Promise Array Management Utility User's Manual* for more details.

Step	Action
1	Start the PAM utility by selecting Start > Programs > Promise Array Management .
2	Select the machine that has the hard disk drive to be replaced: 
3	Enter the username and password in the login window. 

5. Media Drive Operation and Service

5.3. Servicing and Replacing the Hard Disk Drive for SVR2

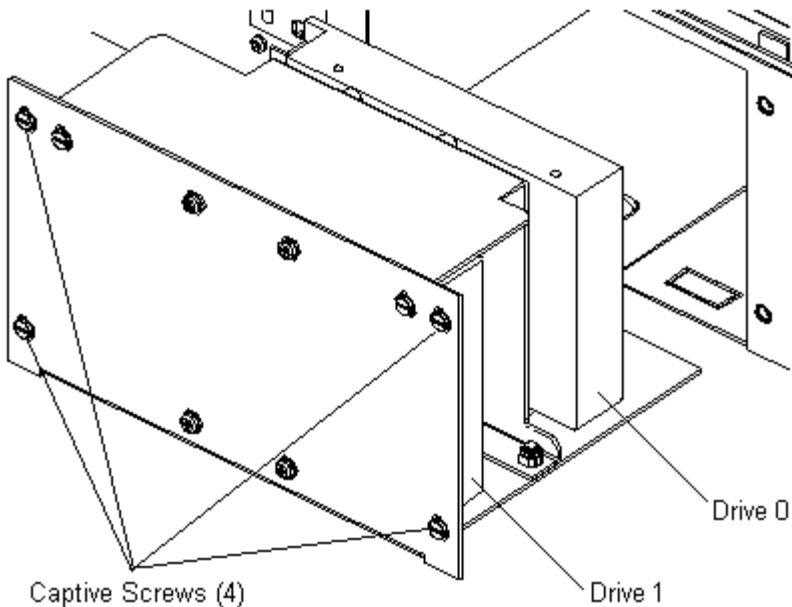
Step	Action
4	Expand the treeview to see the disk status and identify the failed drive.  Note: See also the <i>Promise Array Management Utility User's Manual</i> for further information on the user interface.
	TIP You may also view the status of the hard disks using MS Windows Event Viewer. See the operating system online help for details. 

5. Media Drive Operation and Service

5.3. Servicing and Replacing the Hard Disk Drive for SVR2

Accessing the hard disk drives

Step	Action
1	Remove the front cover from the SVR2 Server enclosure. Refer to the topic <i>Remove the front cover</i> for details.
2	Loosen the four (4) Slot-head captive screws on the outside edges of the front panel on the hard drive assembly.



Captive Screws (4)

Drive 0

Drive 1

3	Slowly pull on the panel until the assembly panel starts moving toward you. The signal and power cables are located on the left side of the assembly. Make sure the cables do not bind up on the side, as this can prevent removal of the assembly. It may be necessary to remove the cable ties from the cables so you can slide the assembly out easily.
	ATTENTION You cannot slide out the hard disk drive assembly further than the service loops of the signal and power cable. Cable ties may have to be removed for better access.

5. Media Drive Operation and Service

5.3. Servicing and Replacing the Hard Disk Drive for SVR2

Step	Action
	ATTENTION The hard disk drive assembly has a bottom plate that slides on retainer channels on the bottom plate of the enclosure. The bottom plate of the hard disk drive assembly extends itself under the ATX motherboard. This channel slot prevents the assembly's bottom plate from tilting up into the bottom of the ATX motherboard thereby preventing damage to the board.
	WARNING Removal and connection of cable plugs must be performed in a prescribed order. When performing the removal and replacement steps on the next few pages, you must follow the sequence shown below to avoid damage to the controller board or replacement hard disk drive (ribbon cable from controller provides ground reference signal). Here is the order you must follow: - Disconnect the Signal connector plug. - Disconnect the Power connector plug. - Remove and replace the failing hard disk drive. - Connect the Signal connect plug. - Connect the Power connector plug.
	ATTENTION The two hard disk drives are supplied with separate sources of power. See Figure 5-3 for a description of the cabling.
4	If you are going to replace Drive 1, execute Steps 10 through 18, then Steps 32 through 50. If you are going to replace Drive 0, go to Step 19 and execute only Steps 19 through 50.
Hot Swap Drive 1	
	ATTENTION Be sure you have performed Steps 1 through 4 in this procedure before you perform the following steps.
5	Remove the Power plug from the drive being replaced.
	TIP It is recommended that you place the thumb and index finger at the top and bottom of the plug. Apply pressure in a slight rocking motion while pulling on the plug to free it from the receptacle on the drive.

5. Media Drive Operation and Service

5.3. Servicing and Replacing the Hard Disk Drive for SVR2

Step	Action
6	Remove the Signal plug from the drive being replaced. Refer to the TIP in the previous step.
7	If you are replacing Drive 1, go to step 10. Otherwise, perform Steps 8 and 9 before going to Step 10.
8	<u>Drive 0 only</u> Locate and remove the two (2) star nuts mounted on base plate studs. A socket drive will be required.
9	<u>Drive 0 only</u> Loosen the two (2) Slot-head captive screws on the front mounting bracket assembly that secures Drive 0. Remove Drive 0 mounting bracket assembly out the top and place on a work surface.
10	Remove the four (4) Philips-head screws that secure the device to the front mounting plate.  Then, carefully remove the hard disk drive out either side of the mounting assembly.
11	Check the replacement hard disk drive to verify that the device address pinning of the replacement drive <u>matches</u> the pinning of the failed drive. Change pinning if necessary.
12	Place the new hard disk drive on the mounting plate, align mounting holes, and secure in place with the four (4) Philips-head screws.
13	If you are replacing Drive 1, go to step 16. Otherwise, perform Steps 14 and 15 before going to Step 16.
14	Retrieve the mounting bracket with the drive assembly mounted from the work surface and place it onto the dual drive assembly. Align the drive assembly onto the two (2) base mounting studs. Align the two (2) Slot-head captive screws on the front mounting plate and secure them.
15	Place the two (2) star nuts onto the base plate studs. Secure them in place with a socket driver.
16	Plug the Signal cable into the replacement hard disk drive.
17	Plug the Power cable into the replacement hard disk drive.  ATTENTION The power plug is keyed with yellow wire at the top.

5. Media Drive Operation and Service

5.3. Servicing and Replacing the Hard Disk Drive for SVR2

Step	Action
18	Verify the hard disk drive has power and is rotating by placing the hand on the surface of the housing and feel for drive vibration.
Place the hard drives back into the enclosure	
	ATTENTION When sliding or pushing the hard disk drive assembly back into the enclosure, be careful not to accidentally press the Power On/Off button or Reset button on the front of the workstation. This is very easy to do, so be careful when pushing the assembly back into the enclosure. Remember that the Server is still "hot" and operating in a network environment.
19	Slowly push on the assembly front panel until the hard disk drive assembly starts moving into the enclosure. The signal and power cables are located on the left side of the assembly. Make sure the cables do not bind up on the side, as it may prevent installation of the assembly. It may be necessary to dress the cables to slide in the assembly easily.
20	Tighten the four (4) slot-head captive screws on the outside edges of the front panel on the dual hard disk drive assembly.
21	Replace the front cover from the SVR2 Server enclosure.

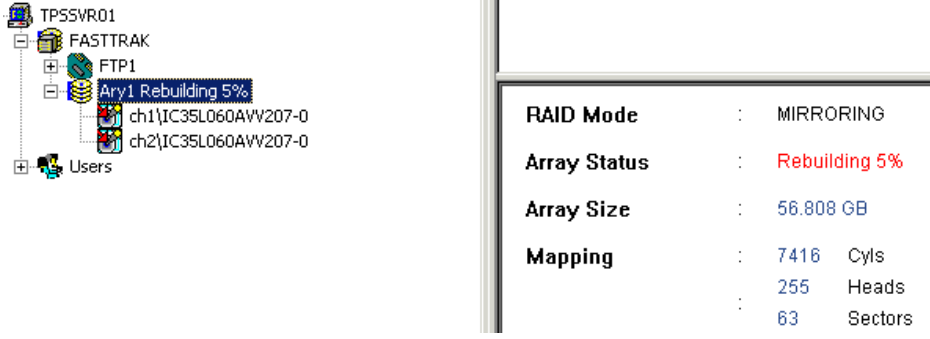
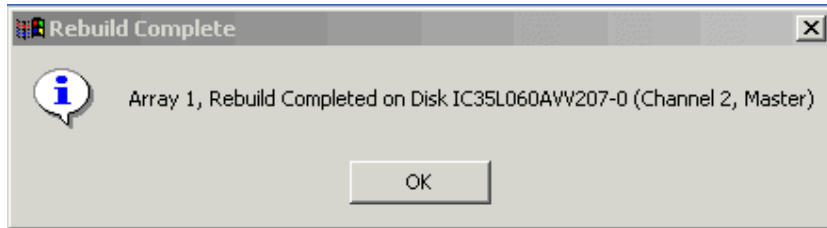
Rebuilding an array

Normally, the rebuild process begins automatically when you repair or replace a faulty disk. Use the following procedure to manually rebuild an array. See also the *Promise Array Management Utility User's Manual* for further information on rebuilding arrays.

Step	Action
1	If necessary, start the PAM utility.
2	Select the machine that has the replacement hard disk and log on.
3	Expand the treeview and select the Array icon of the array you want to rebuild.
4	Right click on the icon and select Rebuild from the popup menu. OR Click the Rebuild Array button in the Toolbar.
5	Follow the Rebuild Wizard to complete the rebuild process.
6	Allow a period of time (about forty-five minutes) for the new hard disk drive to be rebuilt with data from the non-failed hard disk.

5. Media Drive Operation and Service

5.3. Servicing and Replacing the Hard Disk Drive for SVR2

Step	Action
7	During the Rebuild, you can monitor the status of the mirroring process from the PAM utility window. 
8	You will receive the following message when the rebuild is complete. Click OK. 
9	Use the PAM utility window to verify the disk mirroring was successful. The Information view displays Functional. OR Open Windows Explorer and verify that the Local Disk (C:) hierarchy tree and the Local Disk (D:) hierarchy tree are identical.
The hard disk drive hot swap replacement is now complete.	

Troubleshooting an inoperative drive

In general, if only one drive fails, the drive is most likely at fault. If both drives are inoperative, check the power supply in the Server/Workstation and the power extension cable from the Server/Workstation to the drive enclosure.

Step	Action
1	If both drives fail, disconnect the power cable coming out of the rear of the Server/Workstation. +5 Vdc and +12 Vdc must be present. See Figure 3-5 for the location of the peripheral power connector.
2	Check for power at the input to the drive enclosure and at the input to the drive.
3	If no problem is found in Step 1, check that all cable connections are seated.
4	Replace the drive with a spare.
5	If all of the above failed to locate the problem, check the SCSI cable inside the Server/Workstation.
6	The last resort is to replace the motherboard inside the Server/Workstation.

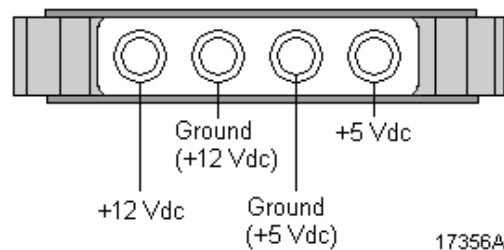


Figure 5-4 Remote Peripheral Power Connector

5. Media Drive Operation and Service

5.4. Backing Up the Hard Disk Drive

5.4 Backing Up the Hard Disk Drive

Overview

The contents of the hard disk drive for the platform can be backed up to a CD. This is accomplished using Roxio CD Recording software. The following procedures detail the installation and use of this utility to perform hard disk backups.

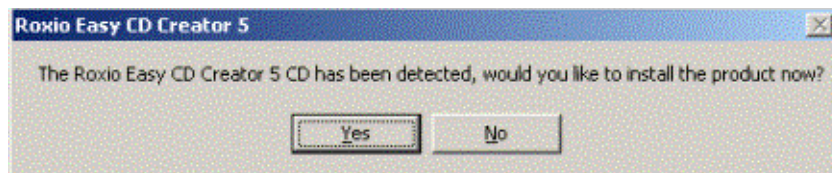
Install Roxio CD Recording Software

Perform the following procedure to install the ROXIO CD Recording software and to ensure the CD-RW drivers are installed and running.



ATTENTION

The following dialog box is displayed when the Roxio CD Recording Software CD is inserted in the CD-RW drive:



Clicking **Yes** will not start the ROXIO software installation because the CD-RW Drivers are not yet installed and running.

Step	Action
1	Insert the Roxio CD Recording Software CD-ROM, Easy CD Creator 6.1 or greater into the CD-RW drive.
2	Using Windows 2000, navigate to the CD and double-click setup.exe .
3	Choose the appropriate setup language and click OK .
4	Click Next .
5	Select Complete installation , and then click Next .
6	Click Install .
7	Click Finish .
8	When the prompt "Do you want to restart your computer?" appears, click No .
9	Shut down the node and cycle the power.

5. Media Drive Operation and Service

5.4. Backing Up the Hard Disk Drive

Step	Action
	ATTENTION Cycling power on the machine and the CD/RW device ensures the CD/RW drivers are loaded properly.

Using the Roxio Recording Software



ATTENTION

The CD Recording software included in this package will work on systems on which Windows 2000-Professional has been fully installed. The following operating systems are not officially supported and Roxio technical support will not be available when used on a Windows 2000 Server

However, most users do not encounter problems when using the Roxio CD Recording Software on these operating systems.

Potential CD-RW Reading and Copying Problems

For additional software information, visit the Roxio web site at <http://www.roxio.com/en/support/dcdwin/>.



WARNING

There are potential problems reading and copying files from the CD-RW device. You may encounter problems reading and writing files from and to a CD in certain situations. The following table details these situations and provides a solution.

If you do this, it may not read all directories and files . . .	
1	Format a disc on a CD-RW device operating in Windows 2000 Professional.
2	Read data from a different type of CD-RW device operating in Windows 2000 Professional – it may not read all directories and files.

To avoid potential disk read/write problems, do this . . .	
1	Format the CD on a CD-RW device operating in Windows 2000 Professional.
2	Read data from a CD/RW device also operating in Windows 2000 Professional.

5. Media Drive Operation and Service

5.5. Servicing and Replacing Removable Media Drives

5.5 Servicing and Replacing Removable Media Drives

Overview

The CD-ROM and Zip drives have a SCSI or EIDE interface, depending on the SVR2/WKS2 configuration, and are located in the peripheral slots for the platform. The optional second hard disk for the desktide platform also has SCSI interface and is located in a peripheral slot.

Supported remote removable media

The Windows 2000 operating system removable media drives can be mounted using the external “narrow” SCSI controller or with IDE to USB adapter or SCSI to USB adapter depending on which remote media drive(s) is chosen.

- Z/EZ/Classic Console - TP-DTAP02
- Streaming tape drive, TP-CNDRW1
- CD-RW/DVD-RW with USB to IDE adapter and TP-DZIP00, ZIP disk drive.
- ICON Console – See HCO5, *Planning, Installation and Service for Honeywell Icon Series Console* for information on removable media for Icon consoles.

SCSI Pin Assignments

Table 5-1 **SCSI Address Assignments**

Peripheral Drive	Address Assignment
Tape drive type should be set	SCSI address 6
ZIP disk drive must be pinned	SCSI address 4
Secondary dual tape drive	SCSI address 4
Secondary dual ZIP disk drive	SCSI address 6
CD/DVD-ROM	SCSI address 5.
Note: If these unit numbers conflict, use address 3, then 2.	

Replace the CD-RW/DVD-RW(+/-) Drive

Perform the following procedure to replace the CD-RW/DVD-RW(+/-) drive.



ESD HAZARD

Before you open your computer or remove the drive from its antistatic bag, ground yourself by touching the metal chassis of your computer or some other grounded object. For optimal protection, use an electrostatic grounding strap.

Step	Action
1	Press the Power On/Off button on the front of the platform to turn the power off.
2	Remove the front cover from the SVR2/WKS2 enclosure: Use a slot-head screwdriver to access the slotted hole on either side of the front cover. To pop loose the cover, twist or pry the screwdriver in the slot. Grasp both sides and remove it from the front panel of the SVR2/WKS2 enclosure.
3	Remove the top cover from the SVR2/WKS2 enclosure. Refer to "Remove the top cover" procedure in Section 4.10.
4	Disconnect the two Floppy Drive cables.
5	Disconnect the two CD-RW/DVD-RW (+/-) cables.
6	Loosen (do not remove) the two bolts securing the drives to the bottom of the enclosure.
7	Loosen (do not remove) the two screws securing the drives to the front of the enclosure.
8	Carefully lift the drives out of the enclosure.
9	Disconnect the power cable to the Intake Cooling Fan.
10	Remove the EMC drive shields, two on each end.
11	Remove the final four bolts from the CD-RW/DVD-RW (+/-) Drive.
12	Replace the CD-RW/DVD-RW(+/-) Drive with the new one and then reverse the previous 11 steps.

5. Media Drive Operation and Service

5.5. Servicing and Replacing Removable Media Drives

Replace the floppy drive

Perform the following procedure to replace the Floppy Drive.



ESD HAZARD

Before you open your computer or remove the drive from its antistatic bag, ground yourself by touching the metal chassis of your computer or some other grounded object. For optimal protection, use an electrostatic grounding strap.

Step	Action
1	Press the Power On/Off button on the front of the platform to turn the power off.
2	Remove the front cover from the SVR2/WKS2 enclosure: Use a slot-head screwdriver to access the slotted hole on either side of the front cover. To pop loose the cover, twist or pry the screwdriver in the slot. Grasp both sides and remove it from the front panel of the SVR2/WKS2 enclosure.
3	Remove the top cover from the SVR2/WKS2 enclosure. Refer to the "Remove the top cover" procedure in Section 4.10.
4	Disconnect the two Floppy Drive cables.
5	Disconnect the two CD-RW/DVD-RW(+/-) cables.
6	Loosen (do not remove) the two bolts securing the drives to the bottom of the SVR2/WKS2 enclosure.
7	Loosen (do not remove) the two screws securing the drives to the front of the SVR2/WKS2 enclosure.
8	Carefully lift the drives out of the enclosure.
9	Disconnect the power cable to the Intake Cooling Fan.
10	Remove the EMC drive shields, two on each end.
11	Remove the final four bolts from the Floppy Drive.
12	Replace the Floppy Drive with the new one and then reverse the previous 11 steps.

Cleaning the tape drive

The manufacturer recommends cleaning the tape drive as specified below.

Conditions	Heavy Daily Backup/Access	Periodic Daily Backup/Access
Office Conditions	Weekly	Monthly
Uncontrolled Conditions	Daily	Weekly

In situations where the environment is moderate to severe and the drive is used sporadically, we recommend cleaning the drive just prior to using it.

Cleaning kit description

The cleaning kit has a plastic cartridge that acts as an applicator/spreading device for the tubes of cleaning solution that are injected into it while it is inserted into a running drive. The part number for the cleaning kit is in the spare parts list at the end of this publication.

The process uses two tubes of cleaning solution for each cleaning and the kit contains 20 tubes and a plastic applicator. Follow the procedure printed on the back of the cleaning kit package.

5. Media Drive Operation and Service

5.5. Servicing and Replacing Removable Media Drives

Guidelines for best tape performance

Perform the following common error checks before replacing a tape drive.

- Perform cleaning and other maintenance as specified in this section.
 - Has the recommended cleaning, care, and maintenance been done as defined?
 - Did the problem occur as a result of following such maintenance?
- Use the tape media recommended by Honeywell (supplied by Honeywell).
 - Ensure that the tape used meets the same specifications as the tape supplied by Honeywell. Using the proper high-quality media will preserve the data on the tape for longer periods of time and prolong the life of the drive.
- Care for and store the tape media properly.
 - Follow the tape manufacturer's recommendations for the proper care and storage of tape media. See the general instructions in the *TPS System Site Planning* manual.
- Ensure the drive door is closed and locked.
 - It's easy to forget to push the tape in completely, leave the tape door open, or not lock the tape in place.
- Make sure you are not trying to write on a write-protected tape. You may get the following message:

ATTENTION

Is this a tape that is protected to preserve its contents?

- If you experience difficulty writing to a tape, check to make sure the write protect mechanism is not set.
 - Try another tape. The tape you are trying to use may be corrupted, defective, or broken.
- Make sure you are not trying to read a high-density tape in a low-density drive.
 - You can read a low-density tape in a high-density drive, but you cannot read a high-density tape in a low-density drive.
- Check to see if a drive is active.
 - Check to see if an access light illuminates or listen for the spinning noises. Make sure you are reading or writing to the correct drive.

Replace the Tape Drive

Perform the following procedure to replace the mini-cartridge tape drive. If you are replacing your tape drive with a newer model, the replacement tape drive can read both TR-4 and TR-5 tape media.



ESD HAZARD

Before you open your computer or remove the drive from its antistatic bag, ground yourself by touching the metal chassis of your computer or some other grounded object. For optimal protection, use an electrostatic grounding strap.

Step	Action
1	Press the Power On/Off button on the front of the platform to turn the power off.
2	Remove the front cover from the SVR2/WKS2 enclosure: Use a slot-head screwdriver to access the slotted hole on either side of the front cover. To pop loose the cover, twist or pry the screwdriver in the slot. Grasp both sides and remove it from the front panel of the SVR2/WKS2 enclosure.
3	Remove the top cover from the SVR2/WKS2 enclosure. Refer to "Remove the top cover" procedure in Section 4.10.
4	Disconnect the two Floppy Drive cables.
5	Disconnect the tape drive cable.
6	Loosen (do not remove) the two bolts securing the drives to the bottom of the enclosure.
7	Loosen (do not remove) the two screws securing the drives to the front of the enclosure.
8	Carefully lift the drives out of the enclosure.
9	Disconnect the power cable to the Intake Cooling Fan.
10	Remove the EMC drive shields, two on each end.
11	Remove the final four bolts from the tape drive.
12	Replace the tape drive with the new one and then reverse the previous 11 steps.

5. Media Drive Operation and Service

5.5. Servicing and Replacing Removable Media Drives

Remove a Zip cartridge with power off

The Zip cartridge can only be removed when power is off by accessing the rear of the drive and firmly inserting a small object, like a straightened paper clip, into the small hole on the rear of the drive, then extracting the cartridge.

6. Monitor Operation and Service

6.1 Overview

About this section

This section contains the following monitor sections:

- Operating and Servicing the 21-Inch Sony Monitor
- Operating and Servicing the 20.1 NEC Flat Panel Display
- Operating and Servicing the Sony 21-Inch Flat-Screen CRT
- Operating and Servicing the Sony 19-Inch Open-Frame CRT

Additional information in this document

For additional information on installing, or removing and replacing monitors in a specific console type, or for monitor specifications refer to the sections listed in the following table.

For further information on . . .	See this section . . .
Specifications	Section 2.4, "Monitor Specifications"
Servicing monitors in a Classic Console	Section 10, "Classic Console Operation and Service" • Removing and Replacing the Flat Screen Open-Frame CRT in a Classic Console
Servicing monitors in a Z console	Section 11, "Z Console Operation and Service" • Accessing and Removing a Monitor in a Z Console • Installing and Replacing a Monitor in a Z Console • Removing and Replacing the 21-inch Flat Screen CRT in Z Console

6. Monitor Operation and Service

6.2. Using the On-Screen Menu to Operate and Adjust the Monitor

6.2 Using the On-Screen Menu to Operate and Adjust the Monitor

Scope

If you do not see your specific monitor type in this section, use the procedures in this section to make adjustments to your monitor. New monitors may use a different on-screen menu than the examples in this section.

Using the control panel to access the on-screen menu

You can make numerous adjustments to your monitor using the on-screen menu. These adjustments are made using the membrane switches on the Monitor Control Panel.



Navigating the menu

Press the  button to display the main MENU on your screen.

6. Monitor Operation and Service

6.2. Using the On-Screen Menu to Operate and Adjust the Monitor

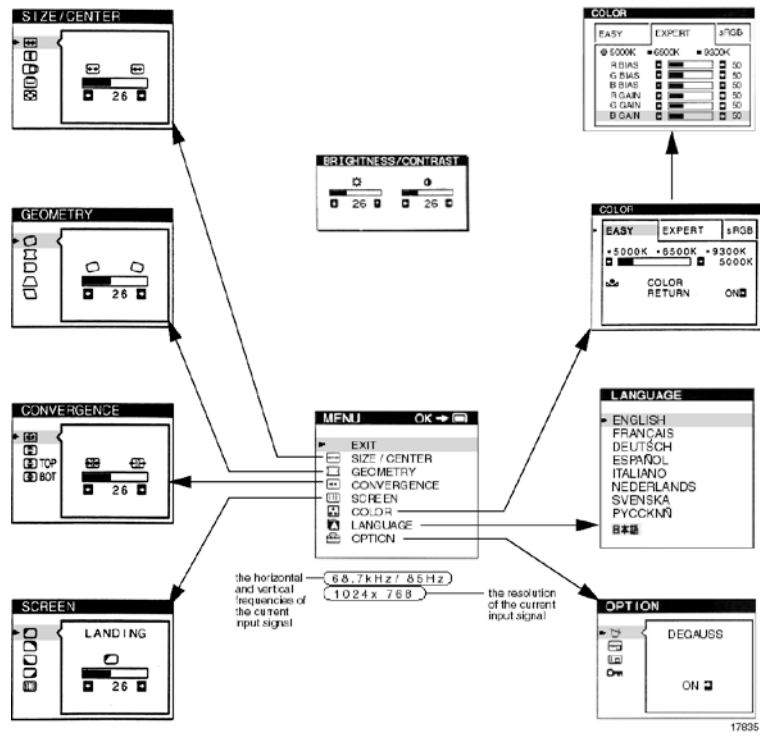


Figure 6-1 On-Screen Menu Structure

6. Monitor Operation and Service

6.2. Using the On-Screen Menu to Operate and Adjust the Monitor

Using the monitor control panel buttons

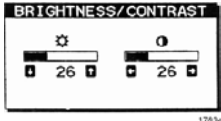
The following procedure will familiarize you with the Control Panel buttons.

Step	Action
1	Displaying the main MENU: Press the  button to display the main MENU on your screen.
2	Selecting the MENU containing the item you want to adjust: Press the  /  buttons to highlight the desired menu. Press the  button to select the menu item.
3	Selecting the adjustment item from the MENU: Press the  /  buttons to select the desired adjustment item. Press the  /  buttons to make the adjustment.
4	Closing the MENU: Press the  button once to return to the main MENU, and twice to return to normal viewing. If no buttons are pressed, the menu closes automatically after about 30 seconds.

Adjust the brightness and contrast

Brightness and contrast adjustments are made using a separate BRIGHTNESS/CONTRAST menu.

These settings are stored in memory for the signals from the currently selected input connector (default connection for video signal cable is HD15-1).

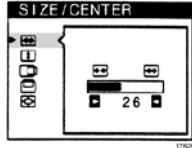
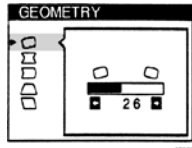
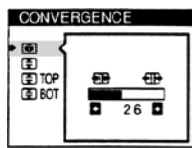
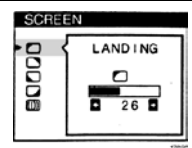
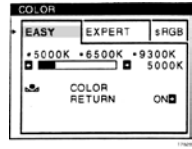
Step	Action
1	Press one of the  (brightness) / or  (contrast) / buttons. The BRIGHTNESS/CONTRAST menu appears on the screen. 
2	Press the  (brightness) / buttons to adjust the brightness () , and  (contrast) / buttons to adjust the contrast (). NOTE: The menu automatically disappears after about 3 seconds.

6. Monitor Operation and Service

6.2. Using the On-Screen Menu to Operate and Adjust the Monitor

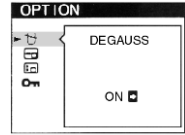
Adjusting other menu items

Other menu items can be adjusted using the various menus available. The following table highlights the basic operations of the main menu items.

No	Option	
1	EXIT Select EXIT to close the menu.	
2	SIZE/CENTER Select the SIZE/CENTER menu to adjust the picture's size, centering or zoom.	
3	GEOMETRY Select the GEOMETRY menu to adjust the picture's rotation and shape.	
4	CONVERGENCE Select the CONVERGENCE menu to adjust the picture's horizontal and vertical convergence.	
5	SCREEN Select the SCREEN menu to adjust the picture's quality. You can adjust the landing and moiré cancellation effect. Use LANDING to reduce or remove color impurities or irregularities primarily in the corners of the screen.	
6	COLOR Select the COLOR menu to adjust the picture's color temperature. You can use this to match the monitor's colors to a printed picture's colors.	

6. Monitor Operation and Service

6.2. Using the On-Screen Menu to Operate and Adjust the Monitor

No	Option	
7	LANGUAGE Select LANGUAGE to choose the on-screen menu's language.	
8	OPTION Select OPTION to adjust the monitor's options. The options include: – degaussing the screen – changing the on-screen menu position – locking the controls	

6. Monitor Operation and Service

6.2. Using the On-Screen Menu to Operate and Adjust the Monitor

Adjusting the Picture Quality (Screen)

The SCREEN settings allow you to adjust the quality of the picture by controlling the moiré and landing. See the “Adjust the color impurity” procedure in this section.

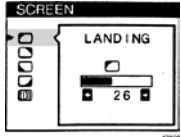
- If the color is irregular (impurity) at the corners of the screen, adjust the landing
- If elliptical or wavy patterns appear on the screen, cancel the moiré.

The CANCEL MOIRE and MOIRE ADJUST settings are stored in memory for the current input signal. All other settings are stored in memory for all input signals.

Monitors can pick up additional magnetic flux when left on for long periods of time. Whenever color purity is in question, degauss the CRT tube by performing the steps in the “Degaussing the monitor” procedure in Section 6.6. If a second degauss cycle is needed, allow a minimum interval of 20 minutes between attempts to degauss the monitor.

Adjust the color impurity

The following procedure provides a method for adjusting the color impurity of the monitor.

Step	Action
1	Press the  button. The main MENU appears on the screen.
2	Press the / to highlight the SCREEN menu entry and press the  button again. The SCREEN menu appears on the screen. 

6. Monitor Operation and Service

6.2. Using the On-Screen Menu to Operate and Adjust the Monitor

Step	Action	
3	Press the  /  to select the desired adjustment item. Then press the  /  buttons to make the adjustment.	
4	Use the action sequence below to adjust color impurity.	
	Select	To
	 LANDING	Reduce any color irregularities in the screen's top left corner to a minimum.
	 LANDING	Reduce any color irregularities in the screen's top right corner to a minimum.
	 LANDING	Reduce any color irregularities in the screen's bottom left corner to a minimum.
	 LANDING	Reduce any color irregularities in the screen's bottom right corner to a minimum.
	 CANCEL MOIRE*	Turn the moiré cancellation function ON or OFF.  (MOIRE ADJUST) appears in the menu when you select ON.
	 MOIRE ADJUST	Adjust the degree of moiré cancellation until the moiré is at a minimum.
	*Moiré is a type of natural interference that produces soft, wavy lines on your screen. It may appear due to interference between the pattern of the picture on the screen and the phosphor pitch pattern of the monitor.	

Adjustments for Touchscreen Optimization

Reference the *TPS System Implementation Guide* and perform the steps in the "Installing the Carroll Touch Screen" section.

6. Monitor Operation and Service

6.2. Using the On-Screen Menu to Operate and Adjust the Monitor

Displaying the Current Input Signal

The horizontal and vertical frequencies of the current input signal are displayed in the main MENU. If the signal matches one of this monitor's factory preset modes, the resolution is also displayed (see example in Figure 6-2).

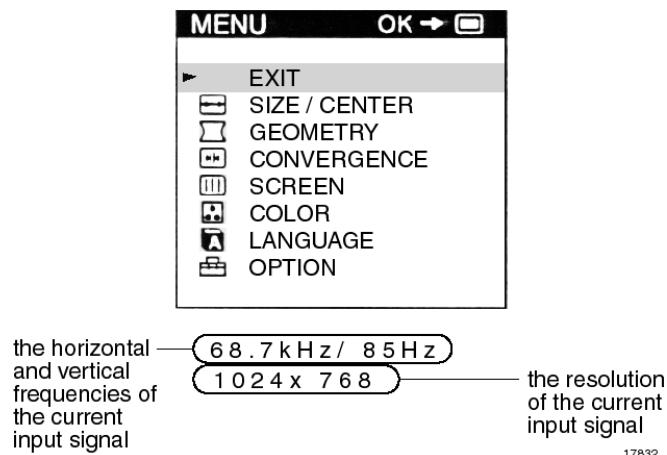


Figure 6-2 Main Menu Display of Input Signal

6.3 Operating and Servicing the 21-Inch Sony Monitor

Description

The 21-inch Sony flat screen high-resolution color display monitor with digital multi-scan capability is used as a desktop monitor for desktide TPN-Connected nodes. A tilt/swivel-mounting base allows for 180° horizontal and 20° vertical swivel.

This monitor is available with an optional touchscreen for EZ-Console (lower monitor only) and Desktide GUS (primary monitor only). The touchscreen is integrally bonded to the CRT face. The monitor must be procured either with or without the touchscreen since the touchscreen addition is an operation performed by the supplier.

References

For additional information, see the specific vendor documentation for your monitor.

6. Monitor Operation and Service

6.4. Operating and Servicing the 20.1 NEC Flat Panel Display

6.4 Operating and Servicing the 20.1 NEC Flat Panel Display

Description

The 20-inch high resolution, color TFT LCD MultiSync flat panel display monitor can be used as a desktop monitor for desktide TPN-Connected nodes. Keep in mind that the operation and behavior of FPD monitors is different than CRT monitors.

Attaching the monitor stand

Refer to the vendor user manual for specifics on attaching the monitor to the stand.

Using the correct monitor cable and video port

The monitor is packaged with an analog and a digital video signal cable. You will be using the DVI to Analog cable.

<i>WKS2 has</i>	<i>Use Video Signal Cable</i>	<i>Connect Video display end</i>	<i>To Video Display connector</i>	<i>Connect FPD end</i>	<i>To FPD Connector</i>
Standard analog video display	DVI to Analog	D-SUB	Analog	DVI-A	Input 1 DVI-I

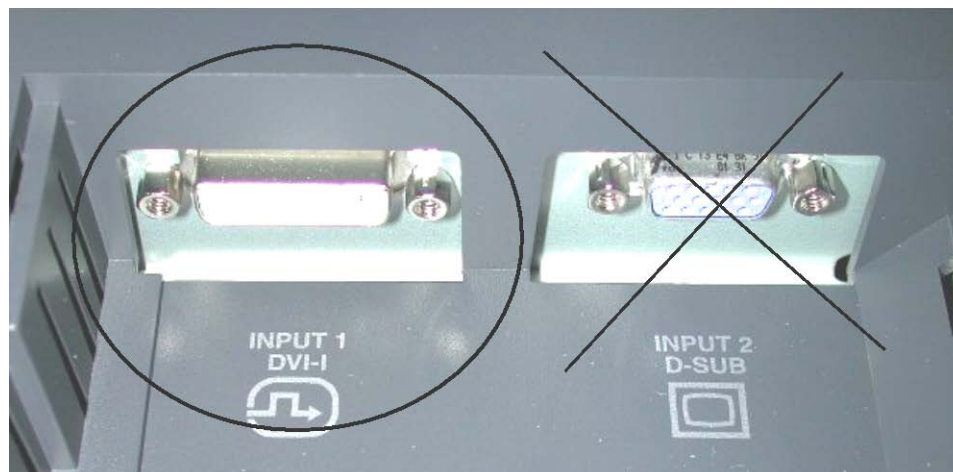


Figure 6-3 **Correct video port for Flat Panel Display Monitor**

Initial monitor settings



ATTENTION

Resolution: Although other resolutions are supported, the native mode for the LCDs is 1280 x 1024 @ 60Hz. NEC-Mitsubishi Electronics Display cites recommended resolution at 60 Hz for optimal display and color performance.

Color: Standard color setting is High Color (16 bit). Higher settings are supported, but may cause overall system performance problems.

You may need to perform the following tasks in order to properly configure your monitor. Detailed steps for these tasks are in this section.

- Correct color mismatch
- Set display bit rate
- Set monitor resolution
- Set monitor refresh frequency

If necessary, see also “Troubleshooting the Flat Panel Display Monitor”.

Clean the monitor

Use a lint-free, non-abrasive cloth to clean the FPD. Avoid using any cleaning solution or glass cleaner.

6. Monitor Operation and Service

6.4. Operating and Servicing the 20.1 NEC Flat Panel Display

Monitor screen adjustments

The monitor screen adjustments are all controlled using an on-screen menu. Access the on-screen menu using the control buttons on the front of the monitor.

Use the On-Screen Manager

The following table provides basic information about the On-Screen Manager. See the user manual for more details on the controls adjustments for your monitor.

Table 6-1 20.1 Inch FPD Monitor Adjustments		
To access OSM menu, press any of the control buttons (◀, ▶, ▲, ▼).		
To mute sound function, press the EXIT button.		
To rotate OSM between Landscape and Portrait modes, press the RESET button.		
To change DVI/D-SUB signal input, press the PROCEED button.		
NOTE: OSM must be closed in order to rotate, mute and change DVI/D-SUB signal input.		
Control Button	Main Menu	Sub-Menu
EXIT	Exits the OSM controls.	Exits to the OSM main menu.
CONTROL ▲ / ▼).	Moves the highlighted area up/down to select one of the controls.	Moves the highlighted area up/down to select one of the controls.
CONTROL ◀ / ▶).	Moves the highlighted area right/left to select control menus.	Moves the bar left/right to increase or decrease the adjustment.
PROCEED	Has no function Push PROCEED to activate Auto Adjust.	Activates Auto Adjust feature. In Tools and Information Mode, opens additional window.
RESET	Resets the highlighted control menu to the factory setting.	Resets the highlighted control to the factory setting.

Correct color mismatch

Step	Action
	TIP Perform these steps if there appears to be a color mismatch on adjacent console monitors.
1	Push the PROCEED button to activate Auto Adjust.
2	Highlight AUTO by using the arrow control buttons.
3	Press PROCEED again to execute the Auto Adjust.

Set display bit rate

Step	Action
	TIP Recommended Color Setting: High Color (16 bit)
1	Select Start > Settings > Control Panel .
2	Double-click the Display icon and select the Settings tab.
3	Under Colors area, select High Color (16 bit).
4	Click Apply .

Set monitor resolution

Step	Action
	TIP Recommended Resolution: Landscape: 1280 x 1024 @ 60 Hz
1	Select Start > Settings > Control Panel .
2	Double-click the Display icon and select the Settings tab.
3	Under Screen area, drag the slider to the appropriate resolution.
4	Click Apply .

6. Monitor Operation and Service

6.4. Operating and Servicing the 20.1 NEC Flat Panel Display

Set monitor refresh frequency

Step	Action
1	Select Start > Settings > Control Panel .
2	Double-click the Display icon and select the Settings tab.
3	Click the Advanced button and select the Monitor tab.
4	In the Monitor Settings area, select the Refresh Frequency from the drop down menu.
5	Click OK to apply changes.

Troubleshooting the Flat Panel Display Monitor

Under certain circumstances your monitor may appear to not be operating correctly. The following table lists known monitor performance deviations and solutions.

Table 6-2 Potential monitor abnormalities

Abnormality	May occur . . .	Resolution
In Windows 2000, if you previously disabled a monitor, the monitor may become enabled.	When you change the settings on any of the enabled monitors, and then cancel those settings.	Make all necessary settings before disabling one of the monitors. If necessary, go back and disable the monitor again.
Display pattern is not stable.	When the remote analog video signal cable is used.	Verify that the Horizontal Synchronous Adjustment cable is used and connected properly.
Display objects are shadowed or blurred on the right side.	When the 100-foot remote analog video signal cable is used.	Use the Image Adjust Control to access the Fine adjustment.
Color inaccuracies during warm-up.	When the short digital video signal cable is used.	Let the monitor completely warm-up.
Display text is off screen during boot (Safe mode video).	During start up when booted in safe mode.	Let the monitor boot up completely in order to view the text.
Adjoining displays may show a color mismatch.	When adjacent monitors have not been adjusted.	See the "Correct color mismatch" procedure in this section.
Small portion of a picture or text does not display properly.	If a dot or pixel on the monitor is defective. See the note below for details on defective dots and pixels.	All display objects should be at least 5 x 5 pixels. Move the object so it displays on a fully operational portion of the screen.
	FPD Specification Notes: - Each monitor pixel contains 3 dots (red + blue +green). - A dot is considered defective if more than 1/3 of its area does not display properly. - Industry standards specify that an FPD may have up to 15 defective dots. - No more than one of the defective dots should occur in any one 5 x 5 pixel area.	

6. Monitor Operation and Service

6.5. Operating and Servicing the Sony 21-Inch Flat-Screen CRT

6.5 Operating and Servicing the Sony 21-Inch Flat-Screen CRT

Description

The 21-inch flat screen high-resolution color display monitor (Part Number 51197178-100) with digital multi-scan capability is an Optimum Replaceable Unit (ORU). This monitor replaces the following predecessor monitors:

- Part Number 51197097-100
- Part Number 51197018-100.

This section includes information about configuring and operating the Sony 21-inch flat screen CRT.

Figure 6-4 Sony 21-inch Open-Frame CRT



Description of monitor assembly

The monitor assembly is comprised of a 21" monitor with a Flat Square Tube (FST) enclosed in an EMI shield. The monitor assembly includes the following features:

- A flat screen that minimizes image distortion and reflective glare
- A monitor control panel (part number 51197181-100) is available separately
- Digital panel control and a full suite of On-Screen Display (OSD) programming/calibration menus.
- A super fine pitch FD Trinitron[®] picture tube using magnetic deflection, electrostatic focus-precision in-line gun, and aperture grille technologies.

Adjust the Sony 21-Inch Flat-Screen CRT

The Sony monitor screen adjustments are all controlled by on-screen menus. This document applies to Sony monitors mounted in Classic (19 inch) and Ergonomic (21 inch) furniture.

In previously released monitors, the center button on the monitors was just a switch for the degauss. With the Sony monitors, the center button is a "menu callup" and "select" (enter) button. Use the up and down arrows move through the choices on the menus.

Step	Action
1	Once you select a parameter to change, the right and left arrows (or + and -) allow you to toggle it ON and OFF , or adjust it between its limits of 0 to 100 percent.
2	When done, push the center button to return to the Main Menu, and then select EXIT to terminate making adjustments. If no buttons are pushed for about 30 seconds the menu will clear by itself keeping the last changes you made.



CAUTION

A monitor's **DISPLAY** quality and purity can be affected whenever it is moved. When first installing and whenever moving a monitor, you must degauss it and check its screen quality.

The Sony does an automatic degauss every time the power is applied, but check the **Landing** and **Moiré** adjustments whenever the monitor is moved. There are many more adjustments that can be made to the monitor, which you can view from the on-screen menu.

6. Monitor Operation and Service

6.5. Operating and Servicing the Sony 21-Inch Flat-Screen CRT

Adjust Brightness & Contrast

Pressing any of the arrow buttons will bring up the Brightness/Contrast menu on the screen. Use the + and – buttons to change settings.

- Press either of the vertical arrows to change the brightness setting.  
- Press either of the horizontal arrows to change the contrast setting.  

The menu will automatically disappear after three seconds of inaction.

Degauss the monitor

If you ever need to manually degauss the monitor, it is located under the Main Menu **OPTION** selection.

- Press the center button  to bring up the main menu on the screen.
- Press the down arrow to get to the **OPTION** (or **SCREEN**) choice.
- Press the center button again to go to the Degauss menu.

DEGAUSS should be the top choice and automatically selected. If not selected, use the up or down arrows to select it.

Press the right arrow to activate the **DEGAUSS** option. The screen is degaussed for approximately two seconds. If a second degauss cycle is needed, allow a minimum interval of 20 minutes for the best results.

Landing (21-inch only)

The **LANDING** adjustment has four settings, one for each corner of the monitor. This is used to reduce any color irregularities in the corners of the monitor. These settings are found under the **SCREEN** selection from the Main Menu. Once at the **SCREEN** menu, you select which corner to adjust with the up or down arrows. The arrows on the right side then set the amount of adjustment between 0 and 100%.

Make Moiré adjustments

Moiré is a wavy pattern that may appear on the screen. The Moiré function is to help cancel any wavy pattern that may exist. The Moiré correction can be set **ON** or **OFF**. To make any Moiré correction, the **MOIRE** correction must be set to **ON** and then the **MOIRE ADJUST** option will appear. It can be adjusted from 0 to 100 percent.

Note: Use of the Moiré adjustment may make the picture fuzzy.

Make Other Adjustments

Other adjustments available from the **MAIN** menu include:

- Size & Centering
- Geometry
- Convergence
- Color
- Language.

Under the **OPTION** menu is a **LOCK** function. If the **LOCK** option is selected and the right arrow pressed,  all other menu functions will be locked from being changed.

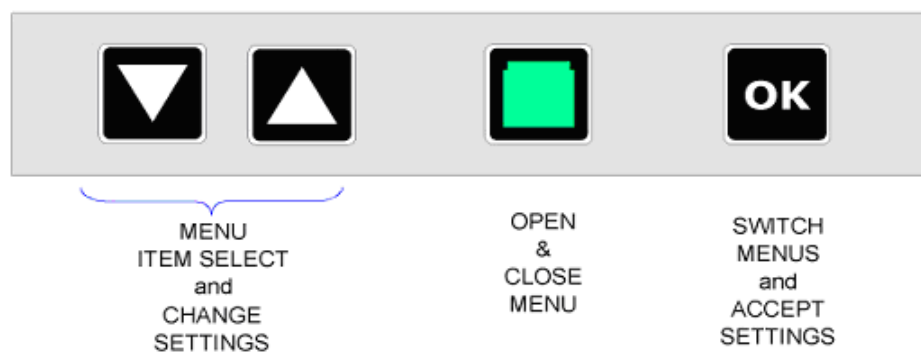
If the **LOCK** is active, selection of any other functions will make a key mark appear on the screen. To unlock the functions, select **LOCK** and press the right arrow again.

6. Monitor Operation and Service

6.5. Operating and Servicing the Sony 21-Inch Flat-Screen CRT

Using the On-Screen menu

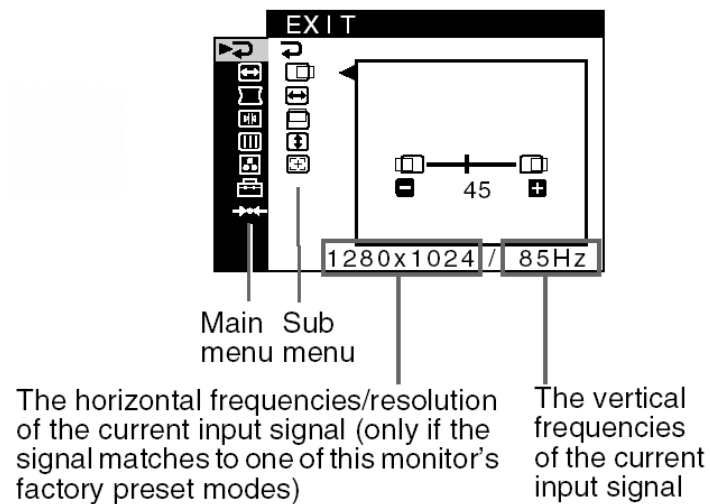
The monitor screen adjustments are all controlled by an on-screen menu. Access to these on-screen menus are made using the membrane switches on the monitor Control Panel as shown.



Using the On-Screen display

The following is the OSD (On-Screen Display) main menu and Brightness/Contrast menu for the second 19" monitor.

Figure 6-5 On-Screen Controls for Adjusting the Display



6. Monitor Operation and Service

6.5. Operating and Servicing the Sony 21-Inch Flat-Screen CRT

List of On-Screen menu adjustments

Main menu icons and adjustment items	Sub menu icons and adjustment items
Adjusting the contrast and brightness* ¹	Contrast Brightness
Adjusting the size or centering of the picture* ¹	Horizontal position Horizontal size Vertical position Vertical size Auto Size Center* ³
Adjusting the shape of the picture	Rotating the picture* ² Expanding or contracting the picture sides* ¹ Shifting the picture sides to the left or right* ¹ Adjusting the picture width at the top of the screen* ¹ Shifting the picture to the left or right at the top of the screen* ¹ RESET: Returns all settings to their factory default settings.
Adjusting the convergence* ²	Horizontally shifts red or blue shadows Vertically shifts red or blue shadows Vertically shifts red or blue shadows at the top of the screen Vertically shifts red or blue shadows at the bottom of the screen RESET: Returns all settings to their factory default settings.

Main menu icons and adjustment items	Sub menu icons and adjustment items
Adjusting the picture quality Example of Moire 	DEGAUSS: demagnetizes the monitor. CANCEL MOIRE: adjusts the degree of moire cancellation until the moire is at a minimum.* ¹ LANDING: reduces any color irregularities in the screen's top left corner to a minimum.* ² LANDING: reduces any color irregularities in the screen's top right corner to a minimum.* ² LANDING: reduces any color irregularities in the screen's bottom left corner to a minimum.* ² LANDING: reduces any color irregularities in the screen's bottom right corner to a minimum.* ² RESET: Returns all settings to their factory default settings.
Adjusting the color of the picture	See "": To adjust the color of the picture".
Additional settings	Protecting adjustment data (CONTROL LOCK)* ⁴ Selecting the on-screen menu language/Confirming the monitor's information LANGUAGE/INFORMATION* ⁵ Changing the menu's position for horizontal adjustment Changing the menu's position for vertical adjustment Selecting the color adjustment mode (See "": To adjust the color of the picture.")
Resetting the adjustments	* ¹ Resetting all the adjustment data for the current input signal.* ⁶ Select "OK". * ² Resetting all of the adjustment data for all input signals. Select "OK".

*¹ This adjustment is effective for the current input signal.

*² This adjustment is effective for all input signals.

*³ You may not be able to adjust the rotation by the earth magnetic field with the Auto Size Center function. If you use the Auto Size Center function, all other size or centering adjustments of the picture are reset.

*⁴ Only the (power) switch, MENU button, INPUT switch, and (CONTROL LOCK) menu will operate.

*⁵ Language Menu

• ENGLISH • NEDERLANDS: Dutch
 • FRANÇAIS: French • SVENSKA: Swedish
 • DEUTSCH: German • РУССКИЙ: Russian
 • ESPAÑOL: Spanish • 日本語: Japanese
 • ITALIANO: Italian

*⁶ The menu items , , and are not reset by this method.

Adjusting the picture color

The COLOR settings allow you to adjust the picture's color temperature by changing the color level of the white color field. Colors appear reddish if the temperature is low, and bluish if the temperature is high. This adjustment is useful for matching the monitor's color to a printed picture's colors.

Select one of the four-color, temperature-setting modes from the OPTION menu:

- EASY • EXPERT
- PRESET • SRBG

The following paragraphs describe each of these four temperature-setting modes.

EASY—(Default setting)—Use this setting to adjust the color temperature from 500K to 11000K.

PRESET—Use the preset color temperature from 5000K, 6500K or 9300K. The default setting is 9300K.

EXPERT—Select this mode to make additional fine adjustments to the color by selecting this mode. The **GAIN** sub mode adjusts the bright areas of the screen, while the **BIAS**  sub mode adjusts the dark areas of the screen.

Select	For
R 	R (Red) BIAS
G 	G (Green) BIAS
B 	B (Blue) BIAS
	RESET
R 	R (Red GAIN)
G 	G (Green GAIN)
B 	B (Blue GAIN)

sRGB—The sRGB color setting is an industry standard color space protocol designed to correlate the colors displayed on the monitor and those printed. In order to display the sRGB colors correctly, select the sRGB mode and your connected computer to the RGB sRGB profiles. If you select sRGB, you cannot operate the CONTRAST/BRIGHT menu adjustments.

6. Monitor Operation and Service

6.5. Operating and Servicing the Sony 21-Inch Flat-Screen CRT

Restoring Color from EASY, PRESET or SRGB Modes

Use the Image Restoration control () to restore colors to their original factory quality levels. Before using this feature, the monitor must have been in normal operating mode (green power indicator) for at least thirty minutes. You may need to adjust your computer's power saving settings.

If the monitor has not been on for at least thirty minutes, the **AVAILABLE AFTER WARMUP** message appears.

Note: This function may gradually lose its effectiveness due to the natural aging of the Trinitron picture tube.

Cabling Connections for the 21-inch Flat Screen CRT monitor

Table 6-3 lists the pin connections for the Sony 21" monitor.

Step	Action
1	Connect the Video signal cable of the monitor to the 15-pin connection port located at the rear of your computer. The pin assignment of the video signal cable is as shown in Table 6-3.

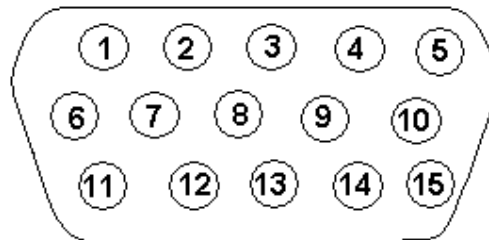


Table 6-3 Connecting and Modifying the Cables

Pin Number	Signal
1	Red
2	Green (Sync on Green)
3	Blue
4	ID (Ground)
5	DDC Ground*
6	Red Ground
7	Green Ground
8	Blue Ground
9	DDC + 5V*
10	Sync Return
11	ID (Ground)
12	Bi-Directional Data (SDA)
13	H. Sync
14	V. Sync
15	Data Clock (SCL)

DDC (Display Data Channel) is a standard of VESA.

6. Monitor Operation and Service

6.6. Operating and Servicing the Sony 19-Inch Open-Frame CRT

6.6 Operating and Servicing the Sony 19-Inch Open-Frame CRT

Monitor screen

An anti-reflection panel is provided on the CRT screen to reduce eyestrain. Avoid touching the CRT screen because soil and fingerprints reduce legibility and the effectiveness of the anti-reflection treatment. To remove soil or fingerprints from the screen, do the following:

- Use a cleaner (such as window cleaning solution) to clean the screen face. Be careful not to scratch or mar the anti-reflective treatment. Before using an untested cleaner, spot test an area of the screen to determine if the results are satisfactory.
- Remove the cleaner with a damp cloth and wipe the surface with a soft, dry cloth.

Cleaning the monitor screen

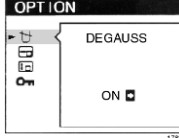
An anti-reflection panel is provided on the CRT screen to reduce eyestrain. Avoid touching the CRT screen because soil and fingerprints reduce legibility and the effectiveness of the anti-reflection treatment. To remove soil or fingerprints from the screen, do the following:

- Clean the screen with a soft cloth. If you use a cleaner (such as window cleaning solution), do not use any type of cleaner containing an anti-static solution or similar additive as this may scratch the screen's coating. Before using an untested cleaner, spot test an area of the screen to determine if the results are satisfactory.
- Remove the cleaner with a damp cloth and wipe the surface with a soft, dry cloth.

Degaussing the monitor

The monitor is automatically demagnetized (degaussed) each time the monitor is powered on. The degaussing eliminates color impurities and other distortions by neutralizing the effects of magnetic fields in the surrounding environment.

Monitors can pick up additional magnetic flux when left on for long periods of time. Whenever color purity is in question, degauss the CRT tube by performing the steps in the following procedure. If a second degauss cycle is needed, allow a minimum interval of 20 minutes between attempts to degauss the monitor.

Step	Action
1	Press the  button. The main MENU appears on the screen.
2	Press the / to highlight the OPTION menu entry and press the  button again. The OPTION menu appears on the screen. 
3	Press the / to select the DEGAUSS menu entry and then press the  button. The screen is degaussed for about 2 seconds. If a second degauss cycle is needed, allow a minimum interval of 20 minutes for the best result.

6. Monitor Operation and Service

6.6. Operating and Servicing the Sony 19-Inch Open-Frame CRT

Adjusting the Monitor

The monitor screen adjustments are all controlled by an on-screen menu. Access to these on-screen menus are made using the membrane switches on the monitor Control Panel as shown in Figure 6-6.

Monitor Screen Adjustments

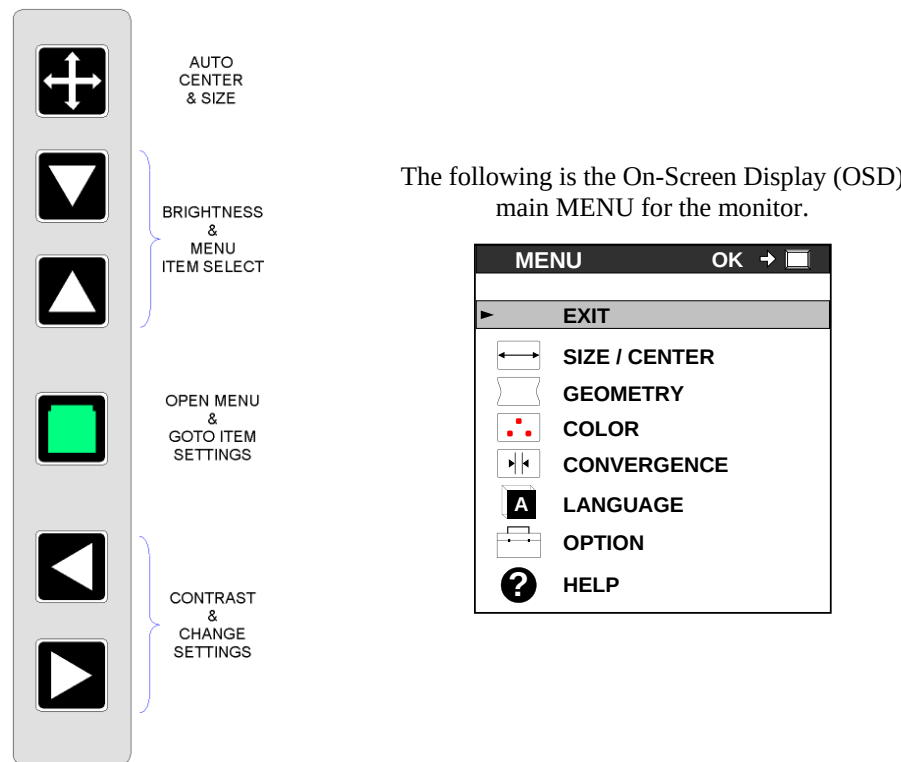


Figure 6-6 Monitor Screen Adjustments

Using the monitor control panel buttons

Step	Action
1	Displaying the main MENU: Press the  button to display the main MENU on your screen.
2	Selecting the MENU containing the item you want to adjust: Press the  /  buttons to highlight the desired menu. Press the  button to select the menu item.
3	Selecting the adjustment item from the MENU: Press the  /  buttons to select the desired adjustment item. Press the  /  buttons to make the adjustment.
4	Closing the MENU: Press the  button once to return to the main MENU, and twice to return to normal viewing. If no buttons are pressed, the menu closes automatically after about 30 seconds.
5	Auto sizing the picture on the screen: Press the  button to automatically reset the picture size to full screen and centered.

6. Monitor Operation and Service

6.6. Operating and Servicing the Sony 19-Inch Open-Frame CRT

Other Adjustments

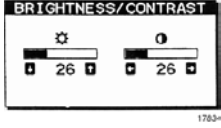
There are other adjustments available from the MAIN menu. The OPTION menu contains the following:

- Changing the OSD menu position (self explanatory)
- LOCK function.

If the LOCK option is selected and the  button is pressed, all other menu functions will be locked from being changed. If the LOCK is active, selection of any other functions will make a *key mark* appear on the screen. To unlock the functions, select LOCK and press the  again.

Adjusting the brightness and contrast

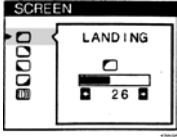
Brightness and contrast adjustments are made using a separate BRIGHTNESS/CONTRAST menu.

Step	Action
1	Press either one of the  (brightness) / or  (contrast) / buttons. The BRIGHTNESS/CONTRAST menu appears on the screen. 
2	Press the  (brightness) / buttons to adjust the brightness (), and  (contrast) / buttons to adjust the contrast (. NOTE: The menu automatically disappears after about 3 seconds.

Adjusting the Moiré (Picture Quality)

Moiré is a type of natural interference, which produces soft, wavy lines on your screen. It may appear due to interference between the pattern of the picture on the screen and the phosphor pitch pattern of the monitor. If elliptical or wavy patterns appear on the screen, cancel the moiré (CANCEL MOIRE).

The SCREEN settings allow you to adjust the quality of the picture by controlling the moiré. The CANCEL MOIRE and MOIRE ADJUST settings are stored in memory for the current input signal. All other settings are stored in memory for all input signals.

Step	Action
1	Press the  button. The main MENU appears on the screen.
2	Press / to highlight the SCREEN menu entry and press the  button again. The SCREEN menu appears on the screen. 
3	Press the / to select the desired adjustment item. Then press the / buttons to make the adjustment.

6. Monitor Operation and Service

6.6. Operating and Servicing the Sony 19-Inch Open-Frame CRT

Step	Action	
4	Use the action sequence below to adjust the Moiré.	
	Select	To
	 CANCEL MOIRE*	Turn the moiré cancellation function ON or OFF.  (MOIRE ADJUST) appears in the menu when you select ON.
	 MOIRE ADJUST	Adjust the degree of moiré cancellation until the moiré is at a minimum.
	NOTE: Use of the moiré adjustment may make the picture fuzzy.	

7. Touchscreen Operation and Service

7.1 19-Inch Touchscreen

Description of touchscreen assembly

The touchscreen assembly consists of a printed wiring assembly (PWA) and infrared (IR) windows. It is assembled to the inside of the plastic front bezel. The touchscreen provides cursor position information signals based on the position of an operator's finger or other stylus being placed in proximity to the monitor screen surface.

The touchscreen uses a matrix of opto-electronic devices operating in the infrared spectrum and controller electronics to provide cursor position information to the computer. The touchscreen assembly is mounted inside the plastic front bezel assembly for mounting to the screen of a 19-inch flat screen monitor. An interconnecting cable is provided for connection to the computer.

19-inch Touchscreen Assembly

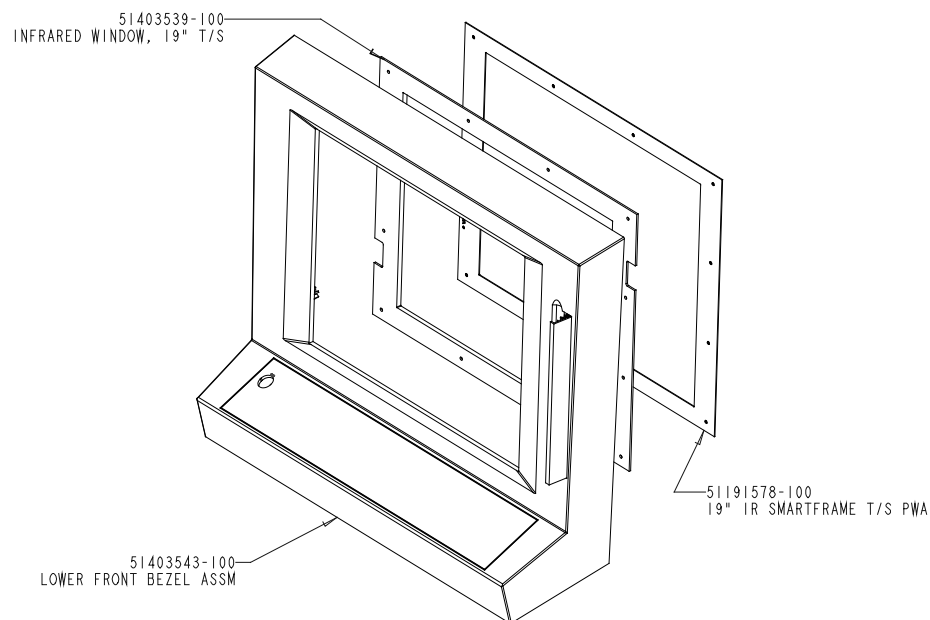


Figure 7-1 Exploded View of 19-inch Touchscreen Assembly

7. Touchscreen Operation and Service

7.1. 19-Inch Touchscreen

Clean the touchscreen window

Perform these steps to clean the touchscreen window.

Step	Action
1	Gently wipe the four inside edges of the touchscreen window with a clean, slightly damp, soft cloth.
2	Ensure that the surface of the touchscreen window is dry by wiping all four sides with a clean, soft, dry cloth.
3	If the touchscreen still does not function correctly, disassemble the touchscreen assembly and clean the individual LEDs and sensors on the touchscreen frame before replacing the touchscreen assembly.

Removing and Replacing the Touchscreen

The touchscreen assembly is attached to the inside of the monitor bezel. The monitor must be removed from the station before the bezel assembly can be removed. See Section 3 for instructions for access and retraction of the monitor.

Remove touchscreen

To remove a touchscreen assembly from the monitor, follow the procedure below and reference the illustration that follows.

Step	Action
1	Place the breaker switch inside the station in the OFF position.
2	Disconnect all cables from the monitor and remove the monitor assembly from the station. If you do not remove it now, you will have to later. After the monitor is removed, slide the monitor rails out of the way.
3	Disconnect the interface cable from the touchscreen.
4	Disconnect the power cable from the touchscreen.
5	Disconnect the two ground from the touchscreen.
6	After removing the four screws that fasten the four corners of the touchscreen to the bezel, remove the touchscreen.
7	Slide the touchscreen out of the bezel.

Replacing the touchscreen assembly

The touchscreen frame can be installed by reversing the previous procedure.

Calibrate the Touchscreen

After the node completes the Autoboot process, perform the following procedure to calibrate the cursor to the monitor screen.

Step	Action
1	After Autoboot completes, select CT PANEL from the Honeywell GUS (Common) window. This opens the Carroll Touch control panel.
2	Choose Calibration from the drop-down menu.
3	A large “+” mark appears in the upper-left corner of the screen. Follow the instructions that are displayed in the center of the screen to complete the process.

7. Touchscreen Operation and Service

7.2. Using the Touchscreen Adapters to Connect to the Computer

7.2 Using the Touchscreen Adapters to Connect to the Computer

Connecting the Touchscreen to the computer platform

Use the procedure in this section to connect the touchscreen to the SVR2/WKS2 platform. The touchscreen is connected to the COM4 serial port on the SVR2/WKS2 platform. The following diagrams show the Carroll touchscreen adapter kit connected on both a local and a remote system.

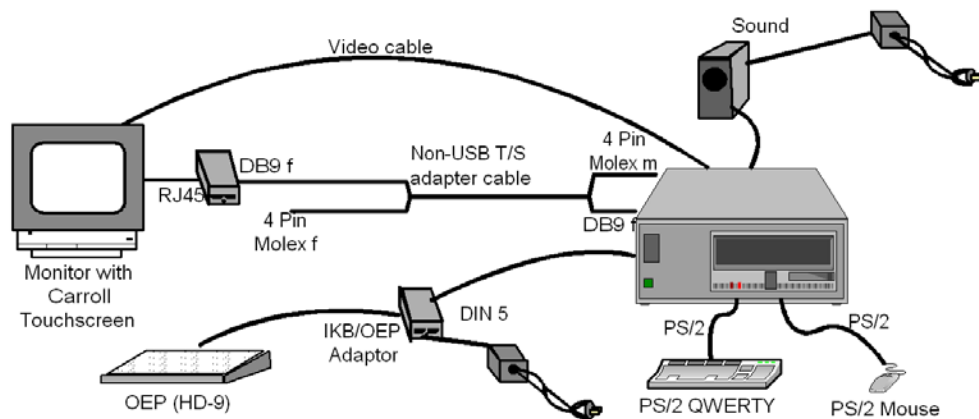


Figure 7-2 Touchscreen Adapter on SVR2/WKS2

7. Touchscreen Operation and Service

7.2. Using the Touchscreen Adapters to Connect to the Computer

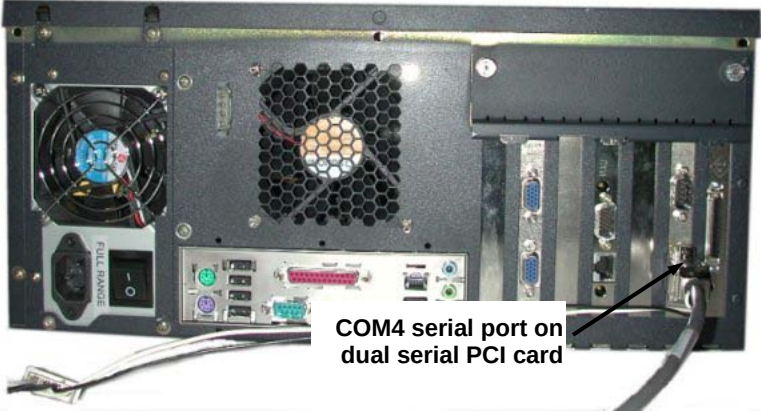
Connect the touchscreen adapter cable to the touchscreen and the computer

Perform the following procedure to connect the touchscreen adapter to the touchscreen and the computer.

Step	Action
1	Power down the computer and other peripheral devices.
2	Connect the male end of the RJ45 touchscreen cable to the touchscreen box assembly as shown. 

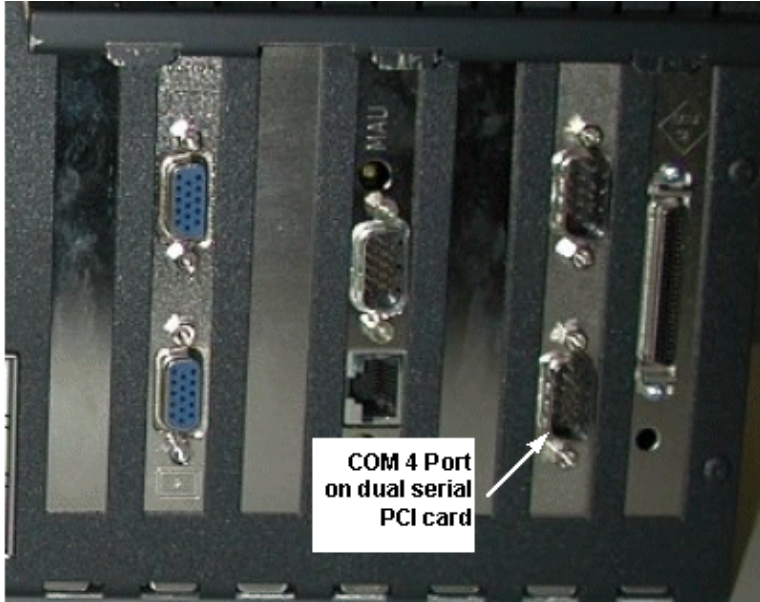
7. Touchscreen Operation and Service

7.2. Using the Touchscreen Adapters to Connect to the Computer

Step	Action
3	Connect the female end of the DB9 touchscreen adapter cable to the touchscreen box assembly as shown. Tighten the thumbscrews to secure the cable. 
4	Connect the other female end of the DB9 touchscreen adapter cable to the COM4 serial port on the computer as shown. 

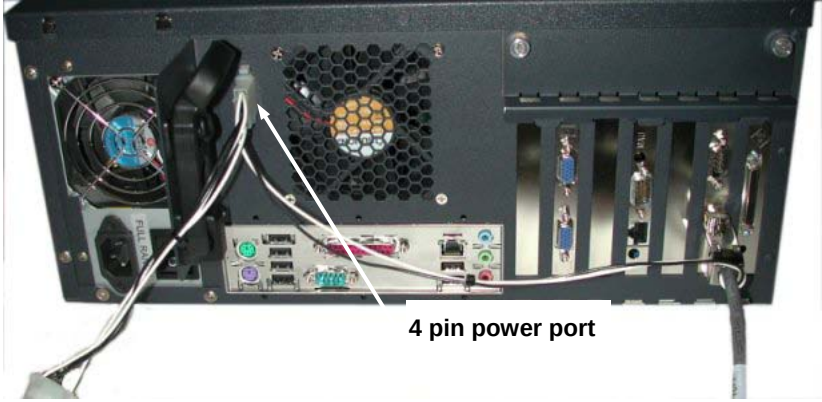
7. Touchscreen Operation and Service

7.2. Using the Touchscreen Adapters to Connect to the Computer

Step	Action
	ATTENTION The touchscreen adapter MUST be plugged into the COM4 Port on the dual serial PCI card as illustrated in the picture. 

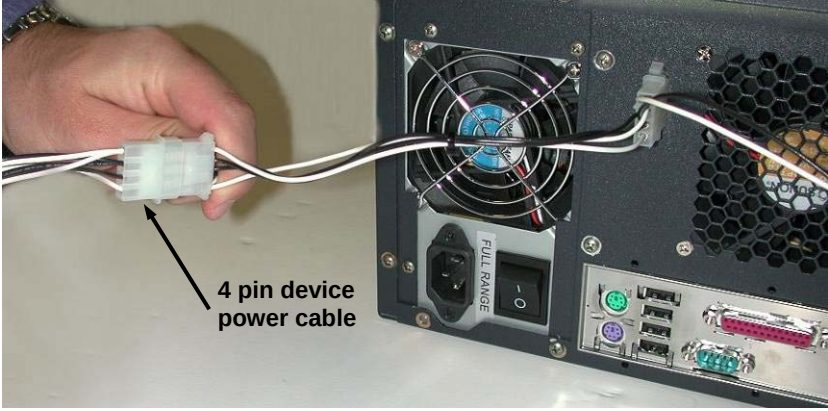
7. Touchscreen Operation and Service

7.2. Using the Touchscreen Adapters to Connect to the Computer

Step	Action
5	If there is currently a device connected to the computer's 4-pin power port, disconnect the power cable as shown. Otherwise, continue with the next step. 
6	Connect the male end of the touchscreen adapter cable's 4-pin connector to the computer's 4-pin power port as shown. 

7. Touchscreen Operation and Service

7.2. Using the Touchscreen Adapters to Connect to the Computer

Step	Action
7	If you disconnected a device in step 5, reconnect the power cable to the touchscreen adapter cable's 4 pin female connector. 
8	If you are connecting an IKB, connect the other end of the IKB12 PS/2 cable (51305381) to the Keyboard PS/2 Port on the computer.
9	Power up the system.

7. Touchscreen Operation and Service

7.2. Using the Touchscreen Adapters to Connect to the Computer

8. Integrated Keyboard (IKB) Operation and Service

8.1 Overview

Functional partition of the Integrated Keyboard

The integrated keyboard integrates the operator and engineer functions into one keyboard and has the following functional parts:

- Relegendable Function Key Panel (RFKP)
- Full-Travel Keyboard
- Trackball
- Operator Control Panel (OCP)



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Figure 8-1 Integrated Keyboard

8. Integrated Keyboard (IKB) Operation and Service

8.1. Overview

Relegendable function key panel

Relegendable function keys (buttons) can be defined or configured by the user to perform certain functions, such as the call-up of certain critical or frequently used displays or cross-screen displays within a console, or to activate a particular Control Language program. Refer to *Button Configuration Form Instructions* and *Button Configuration Data Entry* for button configuration. Refer to Sections 7 and 8 of the *Process Operations Manual* for Alarm Annunciation.



ATTENTION

The paper strip that can be labeled for the key legends can be removed and reinserted behind the panel surface.

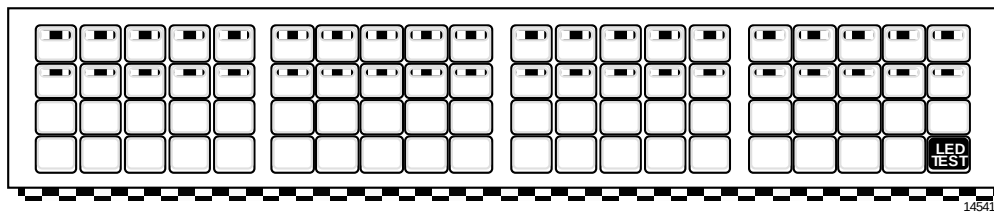


Figure 8-2 Relegendable Function Key Panel

Full-travel keyboard

This QWERTY keyboard functions as both operator and engineer's keyboard with multifunctional keys labeled in black, red, and blue lettering like the U^XS Auxiliary keyboard.

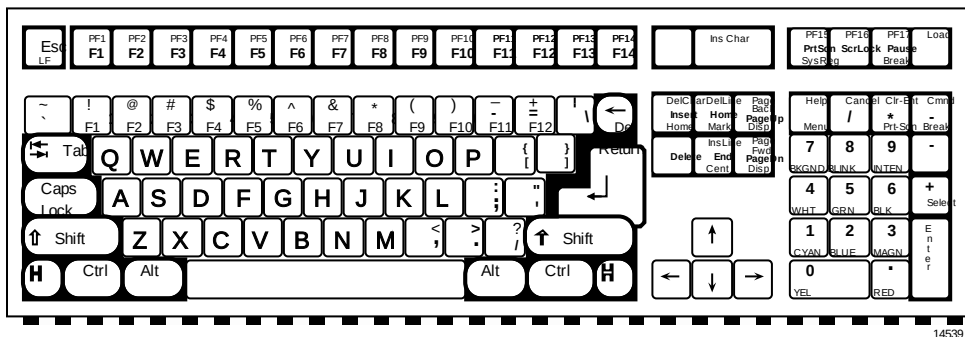
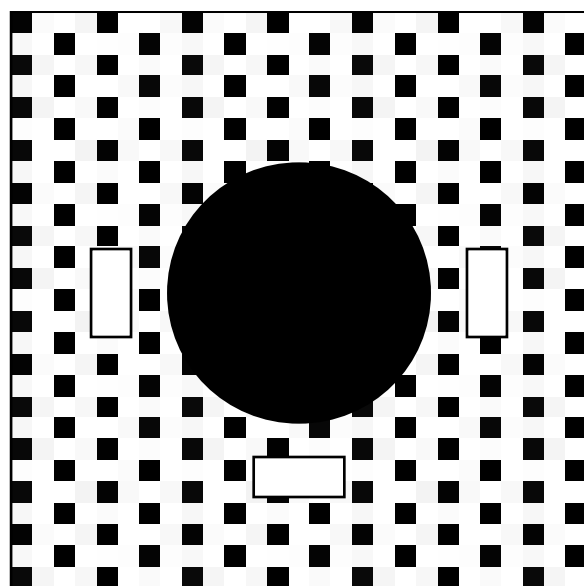


Figure 8-3 Full-Travel Keyboard

IKB Trackball

The trackball is an option to the IKB. It controls the movement of the cursor on the monitor screen. The functions of the trackball buttons depend upon the application the cursor is in when the button is used.



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Figure 8-4 **Integrated Keyboard Trackball**

8. Integrated Keyboard (IKB) Operation and Service

8.1. Overview

Operator Control Panel

The operator control panel contains the RESET switch and power Good/Fail LEDs, as well as the operator function keys. The following table lists the indicator lights.

Indicator	Description
Reset	Resets only the keyboard
Power Good	POWER GOOD FAIL   14659 Lit if the keyboard is properly powered.
Fail Indicator	POWER GOOD FAIL   14660 Lit if there is a hardware failure within the keyboard
Releghable keys	There are six releghable keys located on the far left side of the Operator Control Panel. To print the legend on these keys, raise the keyboard to access the paper insert, pull the paper insert out, label it, and replace the paper insert.

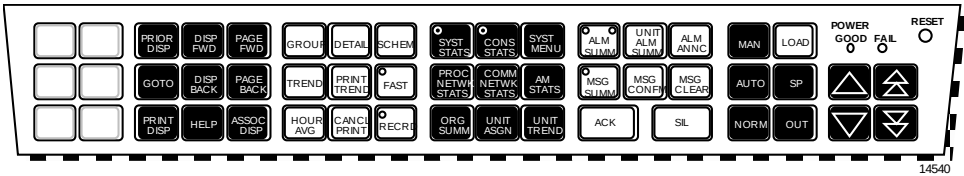


Figure 8-5 Operator's Control Panel

8.2 Servicing the Integrated Keyboard

Clean the keyboard

The IKB can be cleaned with a damp (not saturated) soft cloth with a light washing soap.

Step	Action
1	Press the Power On/Off button on the front of the platform to turn the power off.
2	Using a soft cloth slightly dampened with a dishwashing soap in water, wash the surface of the keyboard ensuring that no excess solution (there should not be any excess) runs into the keyboard assembly.
3	Rub the surfaces of the keyboard with a soft cloth to remove any remaining soap residue.
4	Press the Power On/Off button on the front of the platform to turn the power back on.

Clean the trackball

Step	Action
1	Press the Power On/Off button on the front of the platform to turn the power off.
2	Using a soft cloth slightly dampened with a dishwashing soap in water, wash the surface of the ball ensuring that it is dry and clean. Rub the surfaces of the trackball with a soft cloth to remove any remaining soap residue.
3	Press the Power On/Off button on the front of the platform to turn the power back on.

Troubleshoot the keyboard

Step	Action
1	Check the power LED on the Operator Control Panel. If it indicates a GOOD condition, try more than one function on each of the keyboard panels. If just the trackball fails, the problem could be in the cursor interface cable. If either the QWERTY keyboard or the button panel fails, the problem may be in the keyboard interface cable.
2	Try to reset the keyboard by pressing the keyboard RESET button with a pointed, but not sharp, object, such as a pen cap.

8. Integrated Keyboard (IKB) Operation and Service

8.3. Removing and Replacing Integrated Keyboard

8.3 Removing and Replacing Integrated Keyboard

Remove the Integrated Keyboard assembly from a console

The complete integrated keyboard, including the cables, can be removed as a complete assembly.

Step	Action
1	Shut down the Operating System for the node and go to another station and do a SHUTDOWN on this node before turning off the AC power. This prevents the LCN from ISOLATING the station.
2	Turn off the AC power to the Global User Station node. • If the GUS is the only node in the station, the main AC power switch on the rear of the footrest can be used. • If there is another node in the station, remove the upper-front cover panel and unplug the AC power cord from the GUS module.
3	If present, remove the two Phillips-head screws from each side of the IKB.
4	Using an 8 mm socket, remove the eight screws from under the front edge of the keyboard tray, as shown in Figure 8-6.
5	Using an 8 mm socket, loosen the eight screws from the panel located under the keyboard tray. See Figure 8-6.
6	Using an 8 mm socket, remove the remaining two screws that hold the cover plate on. See Figure 8-6. This gives access to the keyboard cable connections.
7	Raise the QWERTY keyboard. This exposes a grounding bolt on each side of the keyboard. It grounds the conductive strap that comes out of the keyboard. The conductive strap is hard to see in the low levels of light. Using an 8 mm socket, remove the two grounding bolts from under the keyboard tray.
8	Disconnect the keyboard interface cable from the keyboard. See Figure 8-7.
9	Disconnect the trackball cable from the keyboard.
10	To remove the keyboard, grasp the keyboard where it joins the station frame and lift it while pulling it towards yourself.

8. Integrated Keyboard (IKB) Operation and Service

8.3. Removing and Replacing Integrated Keyboard

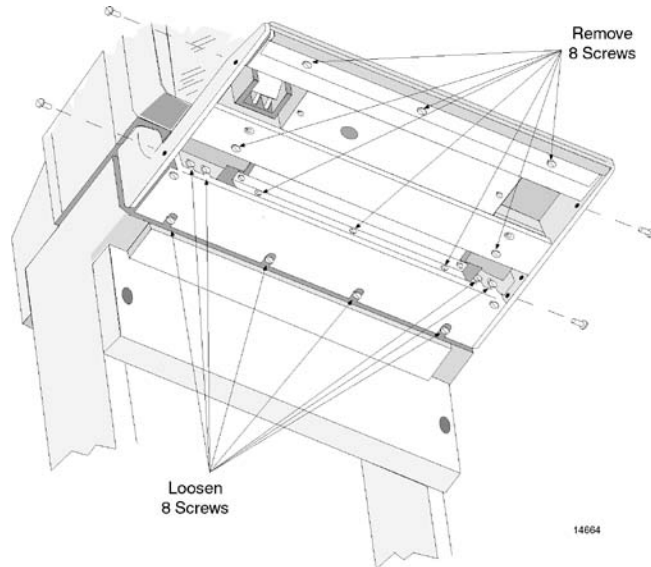


Figure 8-6 Location of Keyboard Screws

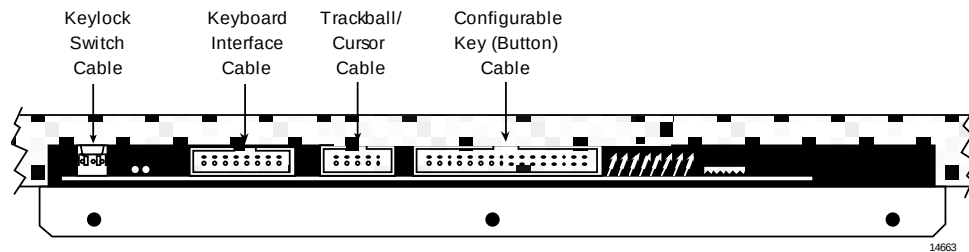


Figure 8-7 Keyboard Connectors from the Rear of the Keyboard

8. Integrated Keyboard (IKB) Operation and Service

8.3. Removing and Replacing Integrated Keyboard

Install the Integrated Keyboard assembly

Step	Action
1	Set the new keyboard on top of the support tray, but not in place.
2	Connect the keyboard interface cable and trackball cables to the keyboard connectors. See Figure 8-7.
3	Place the new keyboard in place on the tray, push it back against the station frame and down into place.
4	Replace the two grounding bolts (do not tighten), one on each side under the QWERTY keyboard panel. See Figure 8-8.
5	Lift the front edge of the keyboard and replace the rubber-edged bump strip between the keyboard and the frame. The tallest edge of the bumper should be on the top side.
6	Replace, but do not tighten, the eight screws under the keyboard shown as removed in Figure 8-6.
7	Replace the cover panel and the remaining two screws (do not tighten) shown in Figure 8-6. After all the screws are in place, tighten all 16 screws firmly.

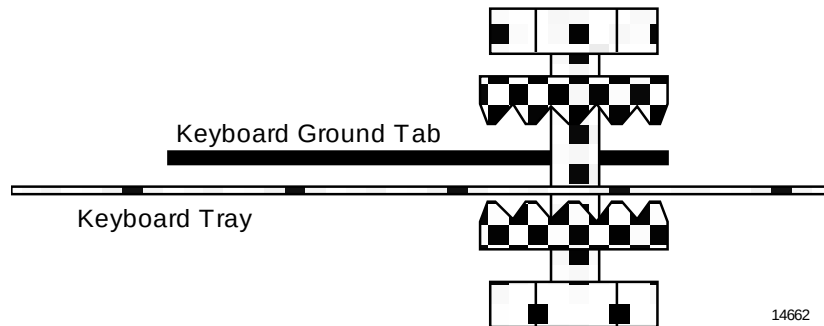


Figure 8-8 Keyboard Grounding Bolt Assembly

8. Integrated Keyboard (IKB) Operation and Service

8.3. Removing and Replacing Integrated Keyboard

Remove existing relegeable function panel

Step	Action
1	Shut down the Operating System for the node and go to another station and do a SHUTDOWN on this node before turning off the AC power. This prevents the LCN from ISOLATING the station.
2	Turn off the AC power to the Global User Station node. • If the GUS is the only node in the station, the main AC power switch on the rear of the footrest can be used. • If there is another node in the station, remove the upper-front cover panel and unplug the AC power cord from the GUS module.
3	Remove the screw on each side of the panel and from the ground straps, and slide the panel away from the swivel mechanism on each side.
4	Disconnect the interface ribbon cable from the center connector at the edge of the IKB Control board (located closest to the RFKP).

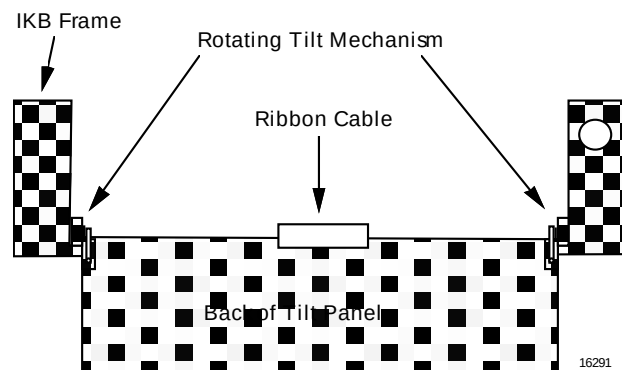


Figure 8-9 Removing/Installing Relegeable Function Key Panel

8. Integrated Keyboard (IKB) Operation and Service

8.3. Removing and Replacing Integrated Keyboard

Install releasable function panel

Step	Action
1	Connect the ribbon cable to the center connector on the new panel, as shown in Figure 8-9.
2	Rotate the left-panel mount mechanism counterclockwise and slip the panel mounting screw through the ground strap eyelet and into the slot of the mechanism.
3	Rotate the right-panel mount mechanism clockwise and slip the panel mounting screw through the ground strap eyelet and into the slot of the mechanism.
4	Tighten both screws after ensuring that the panel is fully inserted into position.
5	Release the panel and it should rotate into the closed position.

Remove the existing IKB controller board

The IKB Controller Board can be replaced by itself after the Integrated keyboard has been removed from the Global User Station (GUS.)

Step	Action
1	Shut down the Operating System for the node and go to another station and do a SHUTDOWN on this node before turning off the AC power. This prevents the LCN from ISOLATING the station.
2	Turn off the AC power to the Global User Station node. • If the GUS is the only node in the station, the main AC power switch on the rear of the footrest can be used. • If there is another node in the station, remove the upper-front cover panel and press the AC power switch on the GUS module.
3	Remove the IKB assembly from the station.
4	Remove the six screws from the IKB Control board, as shown in Figure 8-10.
5	Lift the board from the assembly.

8. Integrated Keyboard (IKB) Operation and Service

8.3. Removing and Replacing Integrated Keyboard

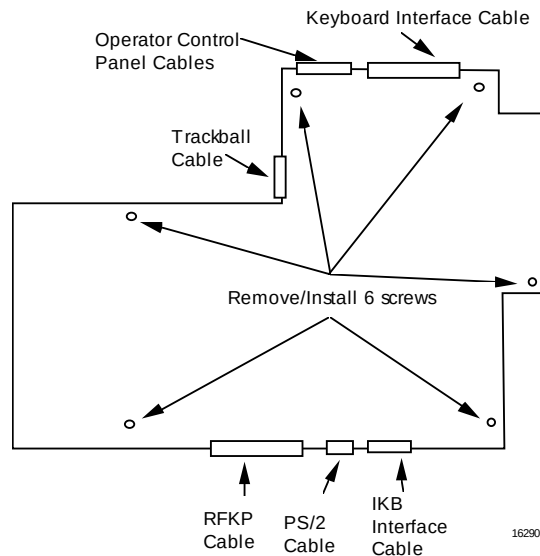


Figure 8-10 Removing/Installing the IKB Controller Board



ATTENTION

The controller board has been replaced with a newer design that only has four screw holes **that must match the four holes in the IKB frame**.

Install the new IKB controller board

Step	Action
1	Mount the board with the six screws (the surface mount version of the board only has four screws), as shown in Figure 8-10.
2	Connect the cables, as shown in Figure 8-10.
3	Install the IKB.
4	Press the AC power switch on the front of the platform. Ensure the LED next to the button is lit, indicating power is on.

8. Integrated Keyboard (IKB) Operation and Service

8.3. Removing and Replacing Integrated Keyboard

Remove full-travel keyboard

The Full-Travel Keyboard can be replaced by itself after the Integrated Keyboard assembly is removed from the Global User Station.

Step	Action
1	Shut down the Operating System for the node and go to another station and do a SHUTDOWN on this node before turning off the AC power. This prevents the LCN from ISOLATING the station.
2	Turn off the AC power to the Global User Station node. • If the GUS is the only node in the station, the main AC power switch on the rear of the footrest can be used. • If there is another node in the station, remove the upper-front cover panel and press the AC power switch on the GUS module.
3	Remove the IKB assembly from the station, as described in Section 8.3.
4	Turn the IKB assembly upside down. Locate and remove the screws in both the ground/retaining straps attached to the Full-Travel Keyboard. This strap on each side of the Full-Travel Keyboard is what prevents the keyboard from tilting up more than 4 inches.
5	Remove the IKB Controller Board.
6	Slide the Full-Travel Keyboard interface cable through the slot in the IKB Bezel that is shown in Figure 8-11.
7	Lift the keyboard gently away from the keyboard bezel, sliding the ground straps through the openings in the bezel.

8. Integrated Keyboard (IKB) Operation and Service
8.3. Removing and Replacing Integrated Keyboard

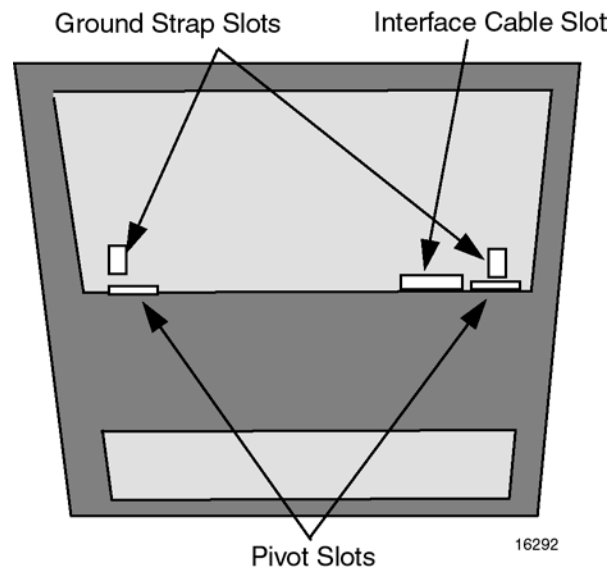


Figure 8-11 IKB Bezel Showing Cable Slots and Ground Strap Slots

8. Integrated Keyboard (IKB) Operation and Service

8.3. Removing and Replacing Integrated Keyboard

Install full-travel keyboard

Step	Action
1	Stand the IKB assembly on its left side with the front edge of the assembly nearest your body.
2	With the new keyboard held flat in front of you as if you were using it, grasp it at the rear edge and rotate it counterclockwise 90° (to the vertical position ready to place in the IKB bezel) and hold it in place, next to the keyboard bezel.
3	Insert the keyboard interface cable through the slot in the keyboard bezel, as shown in Figure 8-11.
4	Feed the ground straps through the two slots in the bezel and fit the two pivot tabs, located on the front edge of the keyboard, into the pivot slots in the bezel.
5	Install the two screws in the ground strap so that the straps hold the keyboard tightly into the bezel.
6	Press the keyboard into the bezel.
7	Replace the IKB Control Board and connect the cables, as shown in Figure 8-10 .
8	Remount the IKB Assembly to the station, as shown in Section 8.3.
9	Press the AC Power On/Off switch on the back of the platform to turn the AC power back on. Ensure the LED in the button is lit, indicating power is on.

8. Integrated Keyboard (IKB) Operation and Service

8.3. Removing and Replacing Integrated Keyboard

Remove the operator control panel

The Operator Control Panel can be replaced by itself after the integrated keyboard assembly is removed from the Global User Station.

Step	Action
1	Shut down the Operating System for the node and go to another station and do a SHUTDOWN on this node before turning off the AC power. This prevents the LCN from ISOLATING the station.
2	Turn off the AC power to the Global User Station node. If the GUS is the only node in the station, the main AC power switch on the rear of the footrest can be used. If there is another node in the station, remove the upper-front cover panel and press the AC power switch on the GUS module.
3	Remove the IKB assembly from the station, as described in Section 8.3.
4	Remove the IKB Controller board, as shown in Figure 8-10 .
5	Loosen the four nut and washer sets located near the front edge of the IKB assembly that retain the Operator Control Panel.
6	Remove the three screws and one retaining bracket that retain the other edge of the Operator Control Panel.
7	Lift the free edge of the board slightly to clear obstructions and pull forward until the board is free.
8	Transfer the relegendable insert to the new Operator Control Panel.
9	Transfer the four screw, washer, and nut sets from the edge of the old board to the new board.

8. Integrated Keyboard (IKB) Operation and Service

8.3. Removing and Replacing Integrated Keyboard

Install operator control panel

Step	Action
1	Insert the new panel into the IKB bezel, leaving the four nuts loose.
2	Replace the three screws and one bracket to retain the free edge of the board.
3	Firmly tighten the four nuts and three screws.
4	Replace the IKB Controller board.
5	Replace the bottom cover and screws.
6	Replace the IKB on the station, as shown in Section 8.3.
7	Press the AC Power On/Off switch on the back of the platform to turn the AC power back on. Ensure the LED in the button is lit, indicating that power is on.

8.4 Using the Adapter to Connect the IKB

Connecting the IKB to the computer platform

Use the procedure in this section to connect the IKB to the SVR2/WKS2 platform. The IKB is connected to the COM3 serial port on the SVR2/WKS2 platform.

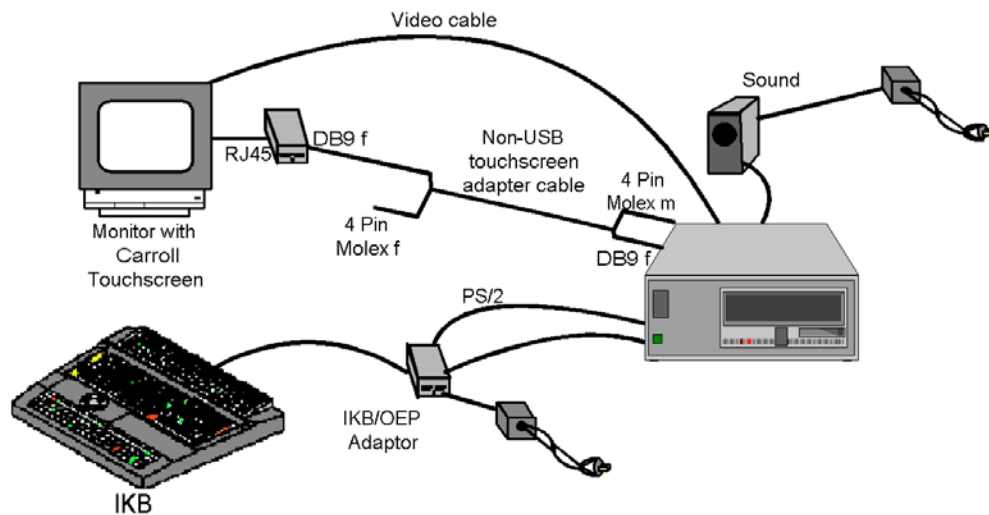
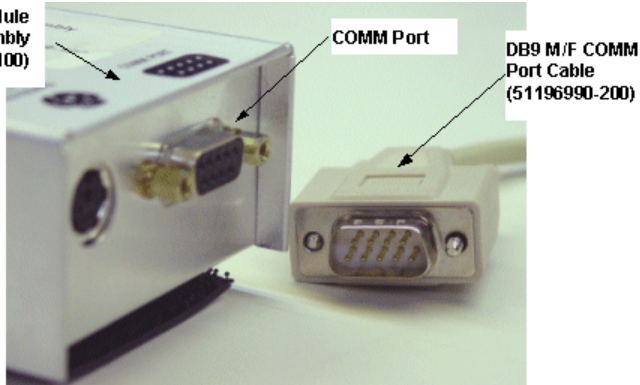
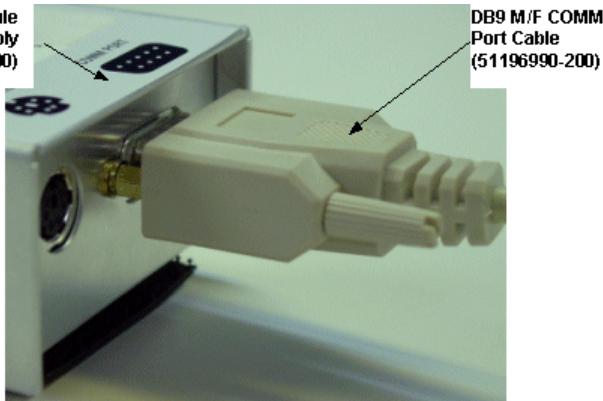


Figure 8-12 IKB Connected to SVR2/WKS2 Using Adapter

8. Integrated Keyboard (IKB) Operation and Service

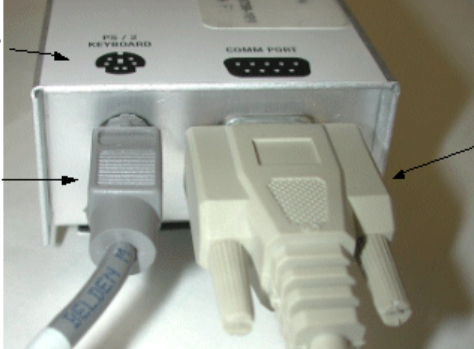
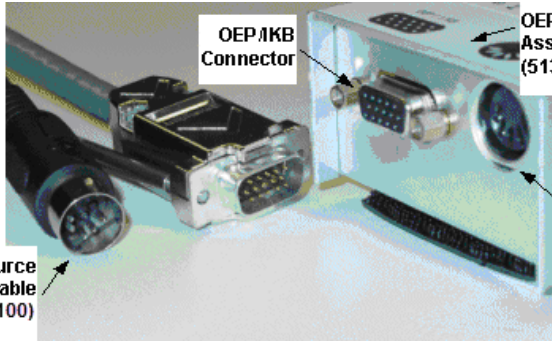
8.4. Using the Adapter to Connect the IKB

Connect the OEP/IKB adapter module to the device

Step	Action
1	Power down the computer and other peripheral devices. DO NOT connect power to the OEP/IKB Module Assembly until told to do so.
2	Connect the male end of the DB9 M/F Comm Port Cable (51196990-200) to the COMM PORT on the OEP/IKB Module Assembly (51305776-100) as shown.  The image shows a close-up of the OEP/IKB Module Assembly (51305776-100) with a DB9 M/F Comm Port Cable (51196990-200) connected to its COMM PORT. Labels with arrows point to the 'OEP/IKB Module Assembly (51305776-100)', the 'COMM Port', and the 'DB9 M/F COMM Port Cable (51196990-200)'.
3	Tighten the thumbscrews on the COMM Port cable to secure the cable to the module assembly.  The image shows a close-up of the DB9 M/F Comm Port Cable (51196990-200) being secured to the OEP/IKB Module Assembly (51305776-100) by tightening the thumbscrews. Labels with arrows point to the 'OEP/IKB Module Assembly (51305776-100)' and the 'DB9 M/F COMM Port Cable (51196990-200)'.

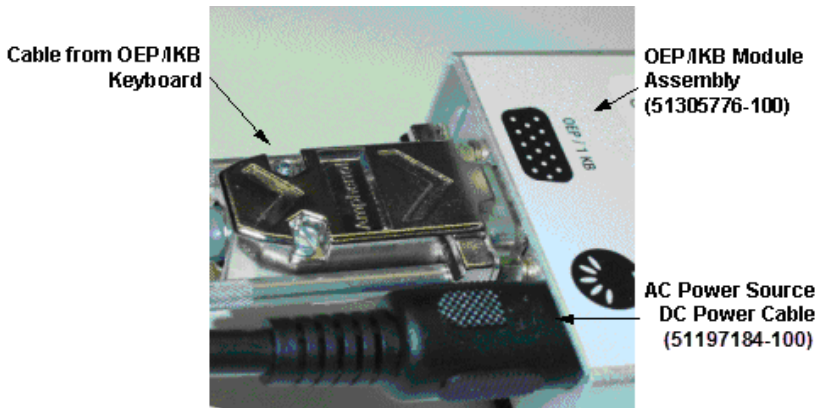
8. Integrated Keyboard (IKB) Operation and Service

8.4. Using the Adapter to Connect the IKB

Step	Action
4	If you are connecting an IKB, insert the round DIN plug of the IKB12 PS/2 cable (51305381) into the PS/2 Connector on the OEP/IKB Module Assembly (51305776-100).  OEP/IKB Module Assembly (51305776-100) IKB12 PS/2 KB Cable (51305381-300) DB9 M/F COMM Port Cable (51196990-200)
5	Insert the round DIN plug from the AC Power Source DC Power Cable (51197184) into the Power Connector on the OEP/IKB Module Assembly (51305776-100).  OEP/IKB Connector OEP/IKB Module Assembly (51305776-100) Power Connector AC Power Source DC Power Cable (51197184-100)

8. Integrated Keyboard (IKB) Operation and Service

8.4. Using the Adapter to Connect the IKB

Step	Action
6	Connect the proper end of the cable from the OEP/IKB Keyboard to the OEP/IKB Connector on the OEP/IKB Module Assembly (51305776-100) and tighten the thumbscrews on the OEP Keyboard cable to secure the cable to the module assembly.  Cable from OEP/IKB Keyboard OEP/IKB Module Assembly (51305776-100) AC Power Source DC Power Cable (51197184-100)
7	Connect the female end of the DB9 Comm Port Cable to COM Port 3.
8	Tighten the thumbscrews on the COMM Port cable to secure the cable to the computer.
9	If you are connecting an IKB, connect the other end of the IKB12 PS/2 cable (51305381) to the Keyboard PS/2 Port on the computer.
10	Connect the AC power cord (51305567) to the AC power source (51197184-100).
11	Plug the AC power cord in.
12	Power up the system.

9. Operator Entry Panel (OEP) Keyboard Operation and Service

9.1. Overview

PS2 Keyboard

This keyboard is the keyboard that comes with the platform. It connects to the Keyboard connector on the back of the platform.

Three-Button Optical Mouse

Table 9-1 Optical Mouse Features and Specifications

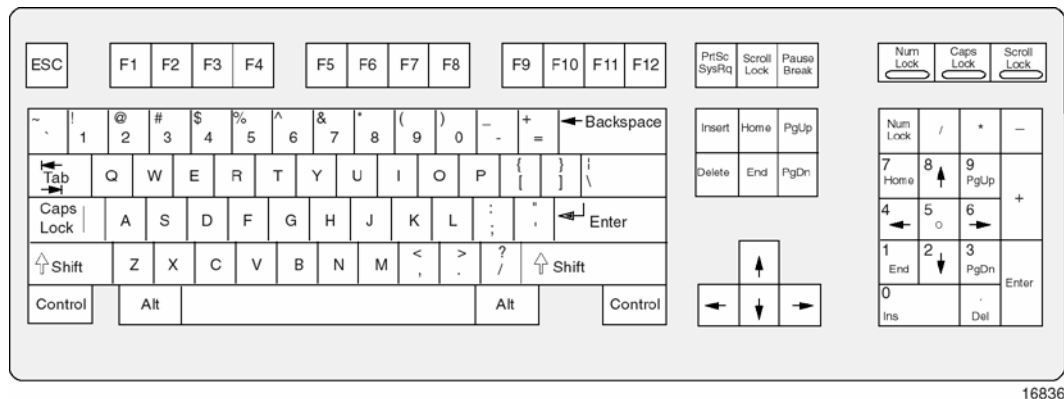
Description	Product Specifications
The three-button optical mouse functions with any PC-equipped with a PS/2 mouse port, and supports Windows 95, Windows 98, Windows 2000, Windows Millennium, or Windows XP. No software is required. Choose either the left or right button as your main click button. This mouse supports Figure 9-2 Three-Button Optical Mouse 	Features • FCC Class B certified • CE approved for use in Europe • C-Tick Mark for use in Australia • One Scrolling wheel • Color: Silver Grey • Includes: PS/2 adapter Cable Length: 1721 mm Dimensions: 107.95 mm (l) x 60.33 mm (w) x 38.1 mm (h) Weight: 130 g

9. Operator Entry Panel (OEP) Keyboard Operation and Service

9.1. Overview

Engineering Keyboard EC-PC101

This keyboard connects to the Keyboard connector on the back of the platform.



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Figure 9-3 Engineering Keyboard

9. Operator Entry Panel (OEP) Keyboard Operation and Service

9.2. Servicing the Operator's Entry Panel

9.2 Servicing the Operator's Entry Panel

Clean the keyboards

Both keyboards can be cleaned with a soft cloth dampened, not saturated, with a light, dishwashing soap solution.

Step	Action
1	Turn OFF the main power switch and disconnect the keyboard from the station.
2	Hold the keyboard in an upright position with the front edge of the keyboard resting on a stable surface.
3	Using a soft cloth dampened with a dishwashing soap in water, wash the surface of the keyboard, ensuring that no excess solution (there should not be any excess) runs into the keyboard assembly.
4	Rub the surfaces of the keyboard with a soft cloth to remove any remaining soap residue.
5	Reconnect the keyboard to the station

Test/Troubleshooting Keyboards

Refer to the *System Maintenance Guide* for further information.

In general, if one or a few keys don't work, the keyboard must be replaced.

- If none of the keys work, check the power LED on the electronics module (see Figure 9 1) and the cable connected to the OEP/IKB adapter.
- Check cabling between the module and the keyboard to ensure power is being supplied and signals are being passed.

9.3 Removing and Replacing the Operator's Entry Panel

Remove the operator's keyboard

Perform the procedure in the following table to remove the operator's keyboard.

Step	Action
1	To access the Operator's Entry Panel, located inside the keyboard frame, remove the six recessed screws from the keyboard top.
2	Lift and rotate the keyboard towards the monitor and disconnect the cables from J1, J2, J3, and J11 on the near edge. See Figure 9-4.
3	Disconnect the cables from J7 and J12, one at a time and reconnect them to the new panel. See Figure 9-4.
4	Reverse this procedure to install the new panel.

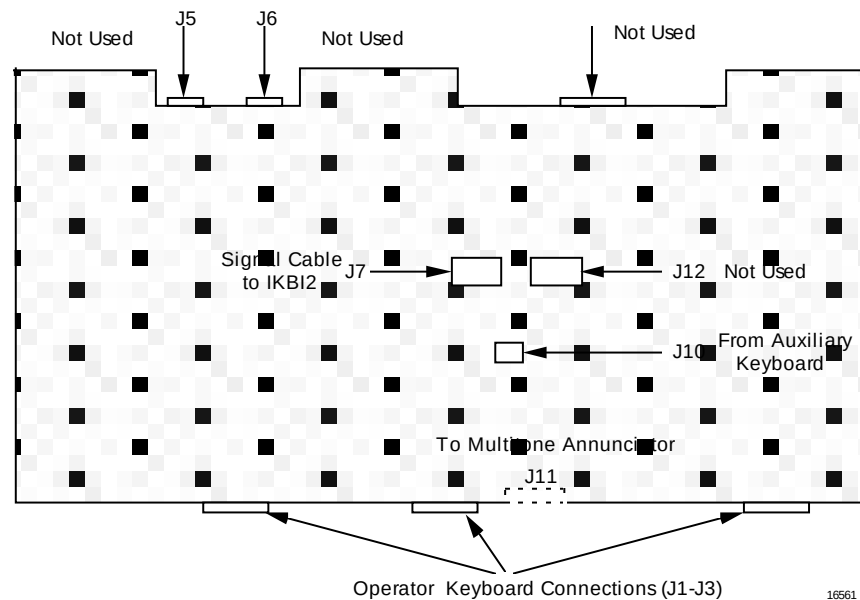


Figure 9-4 Operator Entry Panel Cable Connections

9. Operator Entry Panel (OEP) Keyboard Operation and Service

9.3. Removing and Replacing the Operator's Entry Panel

Remove the operator entry panel

Perform the procedure in the following table to remove the Operator Entry Panel (OEP) from the station.

Step	Action
1	Remove the access panel under the keyboard tray by removing the four screws.
2	Disconnect the keyboard cable from the manifold board and pull the cable out of the station.
3	Grasp the OEP on the side closest to the monitor.
4	Tilt the OEP up to the vertical position.
5	Disconnect the ground strap on each lower corner of the OEP.
6	Lift the OEP from the station and lay it face down on the extended keyboard shelf.

Install the operator entry panel

Perform the procedure in the following table to install an Operator Entry Panel (OEP) on the station.

Step	Action
1	Grasp the OEP, holding it in the vertical position with the space bar to the bottom.
2	Place the bottom edge of the OEP into the groove located at the front edge of the open area in the keyboard tray.
3	Reconnect the grounds straps to the OEP.
4	Tilt the OEP down flat in the keyboard tray.
5	Feed the keyboard cable through the hole in the keyboard tray and connect it to the manifold board.

9. Operator Entry Panel (OEP) Keyboard Operation and Service

9.4. Replacing the Manifold Board

Remove the operator's keyboard from console

To access the Operator's Keyboard, first remove the bezel.

Step	Action
1	Remove the four M5 screws from under the keyboard tray. Use an 8 mm spin-tight or wrench.
2	Lift the bezel from the tray.
3	Push the RESET button cover towards the monitor.
4	Remove the four M5 screws from under the keyboard tray. Use an 8 mm spin-tight or wrench.
5	Lift and rotate the keyboard towards the monitor and disconnect the cables from J1, J2, and J3 on the near edge. See Figure 9-5.
6	Rotate the keyboard back to the tray and disconnect the rest of the cables.

Install the operator's keyboard

To install the Operator's Entry Panel, reverse the previous procedure.

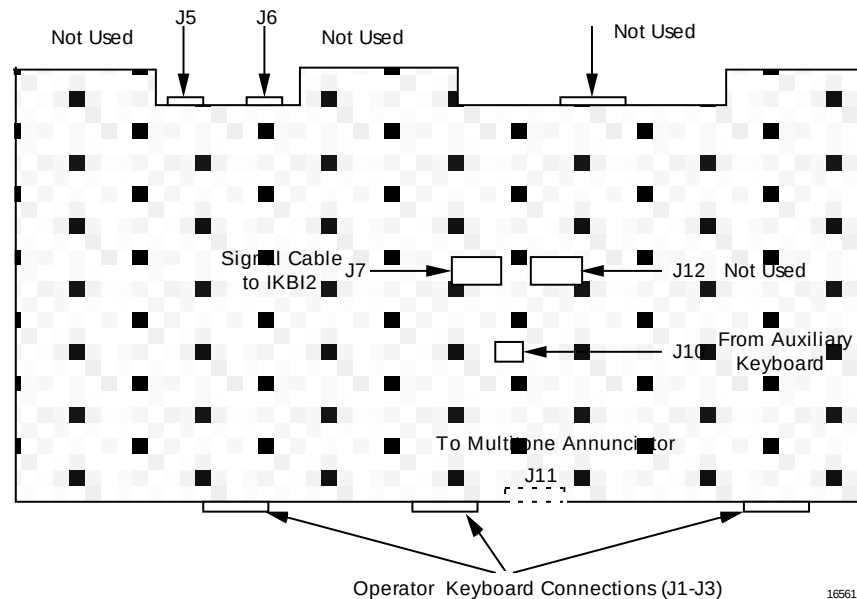


Figure 9-5 Operator's Entry Panel Cable Connections

9. Operator Entry Panel (OEP) Keyboard Operation and Service

9.4. Replacing the Manifold Board

Removing and Installing the EC-PC101 Keyboard

To remove or replace the Engineering Keyboard (EC-PC101), simply disconnect or reconnect the keyboard cable from the Keyboard connector on the back of the platform.

9.4 Replacing the Manifold Board

Manifold Board

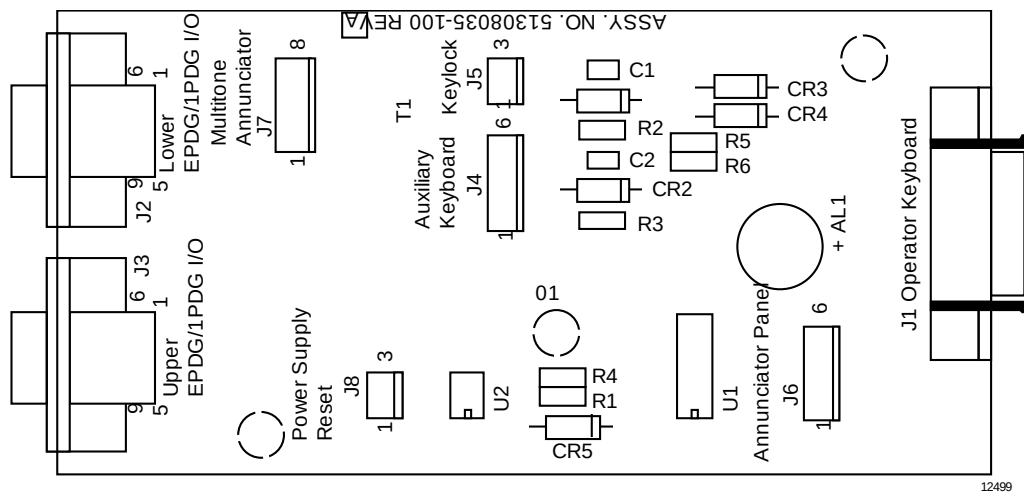


Figure 9-6 Manifold Board

9. Operator Entry Panel (OEP) Keyboard Operation and Service

9.4. Replacing the Manifold Board

Remove the manifold board

Step	Action
1	Remove the access cover from the bottom of the keyboard tray by removing the four hex-head screws (next to the module front cover).
2	Disconnect the remaining cables from the manifold board. • Operator Entry Panel (OEP) Assembly (J1) • GUS Right-hand Connector (J2) • Auxiliary Keyboard Power Cable (J4) • Keylock Cable (J5) • Annunciator Terminal Board • Multitone Annunciator Cable (J7) • Power Supply Reset Cable (J8)
3	Remove the three nuts from the three corners of the manifold board.
4	Remove the old manifold board and install the new manifold board using the three nuts removed in the previous step.
5	Reconnect the cables removed in Step 2.
6	Replace the access cover and the four hex-head screws.

9.5 Using the Adapter to Connect the OEP

Connecting the OEP to the computer platform

Use the procedure in this section to connect the OEP to the SVR2/WKS2 platform. The IKB is connected to the COM3 serial port on the SVR2/WKS2 platform.

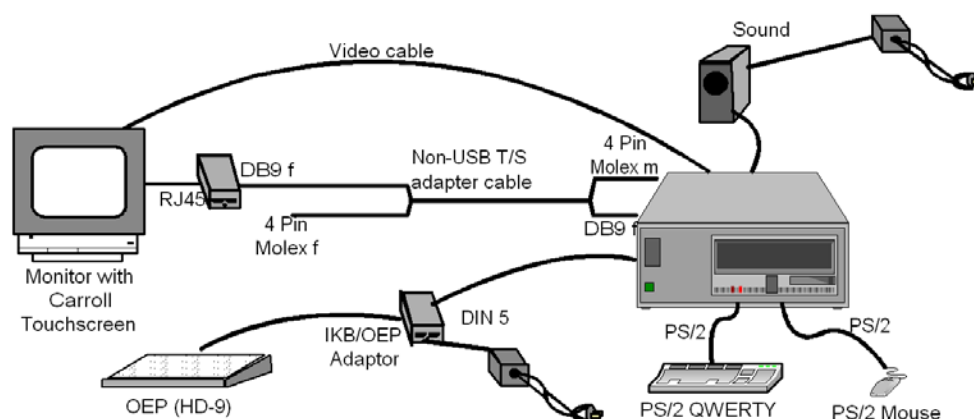
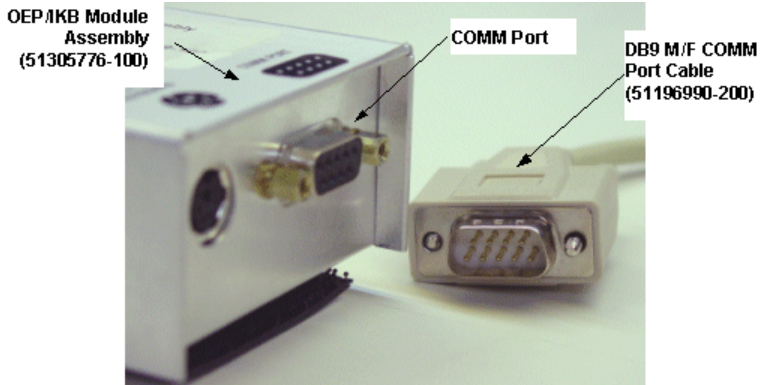


Figure 9-7 OEP Connected to SVR2/WKS2 Using Adapter

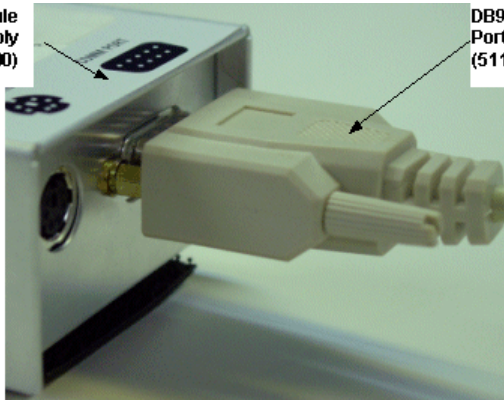
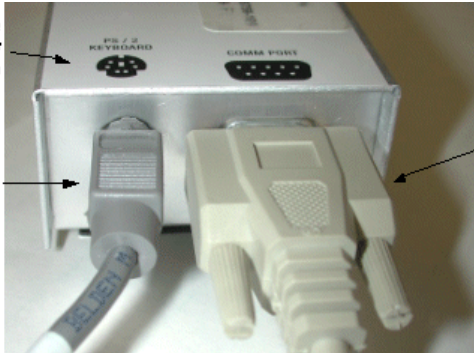
9. Operator Entry Panel (OEP) Keyboard Operation and Service
9.5. Using the Adapter to Connect the OEP

Connect the OEP/IKB adapter module to the device

Step	Action
1	Power down the computer and other peripheral devices. DO NOT connect power to the OEP/IKB Module Assembly until told to do so.
2	Connect the male end of the DB9 M/F Comm Port Cable (51196990-200) to the COMM PORT on the OEP/IKB Module Assembly (51305776-100) as shown.  The diagram illustrates the connection of a DB9 M/F Comm Port Cable to the COMM PORT on the OEP/IKB Module Assembly. The OEP/IKB Module Assembly (51305776-100) is shown on the left, with a label pointing to its COMM PORT. The DB9 M/F COMM Port Cable (51196990-200) is shown on the right, with a label pointing to its male end. The cable is being inserted into the COMM PORT of the module assembly.

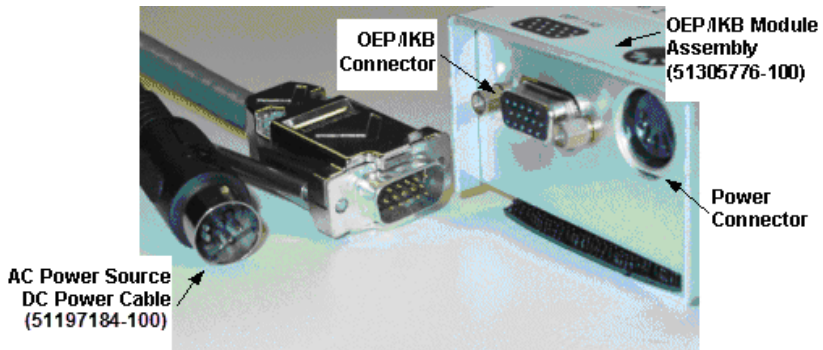
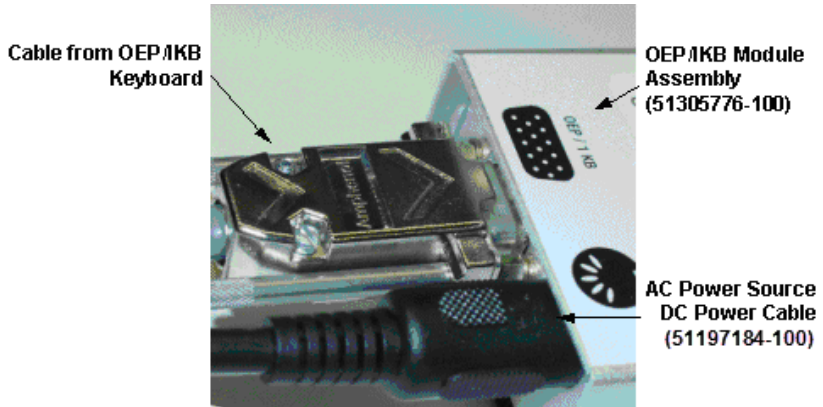
9. Operator Entry Panel (OEP) Keyboard Operation and Service

9.5. Using the Adapter to Connect the OEP

Step	Action
3	Tighten the thumbscrews on the COMM Port cable to secure the cable to the module assembly. OEP/IKB Module Assembly (51305776-100)  DB9 M/F COMM Port Cable (51196990-200)
4	If you are connecting an IKB, insert the round DIN plug of the IKB12 PS/2 cable (51305381) into the PS/2 Connector on the OEP/IKB Module Assembly (51305776-100). OEP/IKB Module Assembly (51305776-100)  IKB12 PS/2 KB Cable (51305381-300) DB9 M/F COMM Port Cable (51196990-200)

9. Operator Entry Panel (OEP) Keyboard Operation and Service

9.5. Using the Adapter to Connect the OEP

Step	Action
5	Insert the round DIN plug from the AC Power Source DC Power Cable (51197184) into the Power Connector on the OEP/IKB Module Assembly (51305776-100).  The diagram shows a close-up of the OEP/IKB Module Assembly. A round DIN plug from the AC Power Source DC Power Cable is being inserted into the Power Connector. The OEP/IKB Connector is also visible.
6	Connect the proper end of the cable from the OEP/IKB Keyboard to the OEP/IKB Connector on the OEP/IKB Module Assembly (51305776-100) and tighten the thumbscrews on the OEP Keyboard cable to secure the cable to the module assembly.  The diagram shows the OEP/IKB Module Assembly with the Cable from OEP/IKB Keyboard connected to the OEP/IKB Connector. The AC Power Source DC Power Cable is also shown.
7	Connect the female end of the DB9 Comm Port Cable to COM Port 3.
8	Tighten the thumbscrews on the COMM Port cable to secure the cable to the computer.

9. Operator Entry Panel (OEP) Keyboard Operation and Service

9.6. Replacing the Multitone Annunciator Panel

Step	Action
9	If you are connecting an IKB, connect the other end of the IKB12 PS/2 cable (51305381) to the Keyboard PS/2 Port on the computer.
10	Connect the AC power cord (51305567) to the AC power source (51197184-100).
11	Plug the AC power cord in.
12	Power up the system.

9.6 Replacing the Multitone Annunciator Panel

Purpose

The optional Multitone Annunciator Panel has a multitone annunciator that is used as an alternative to the standard alarm annunciator. It is located next to the manifold board under the keyboard tray.

Replace the Multitone Annunciator Panel

Follow the procedure in the following table to remove the multitone annunciator from the station.

Step	Action
1	Remove the access panel under the keyboard tray by removing the four screws.
2	Disconnect the cable from the Multitone Annunciator Panel and replace the board.
3	Reconnect the cable to the Multitone Annunciator Panel.
4	Replace the access panel removed in Step 1.

Installing the Wiring to the Annunciator Board

The Annunciator board has three relays that are activated when alarm conditions (that have been defined) are detected by one of the alarm parameters. The operator keyboard board causes the activation of the relays through the OEP keyboard.

Table 9-2 Annunciator Relay Specifications

Parameter	Specification
Input Turn-on Voltage Turn-off Voltage	3.0 to 6.0 VDC -3.0 to 1.0 VDC
Output Voltage Range Current Range Surge Current (1 second)	10 to 60 VDC 0.02 to 3.0 A

9. Operator Entry Panel (OEP) Keyboard Operation and Service

9.6. Replacing the Multitone Annunciator Panel

Annunciator Board in the Cabinet Base

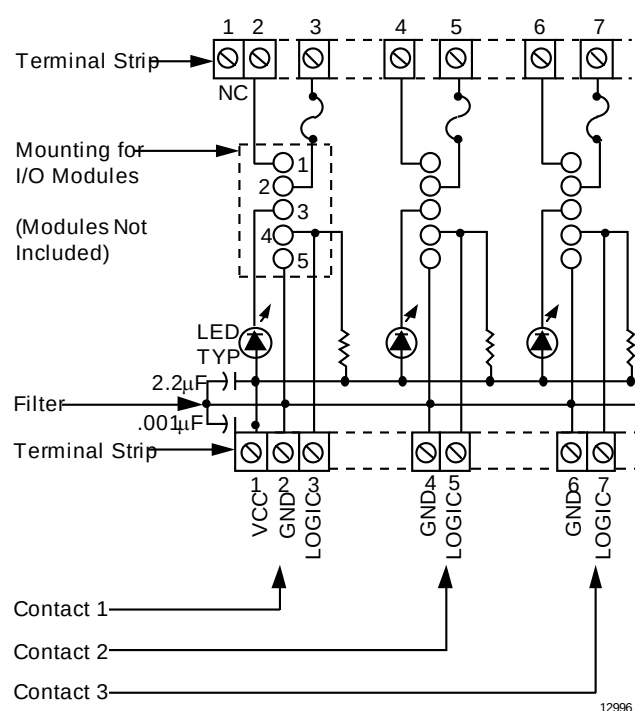


Figure 9-8 Diagram of the Annunciator Board in the Cabinet Base

9. Operator Entry Panel (OEP) Keyboard Operation and Service

9.6. Replacing the Multitone Annunciator Panel

Replace the annunciator board

Step	Action
1	Remove the right and left leg cover panels. Slide the footrest cover to the front, exposing a compartment.
2	Loosen the three screws at the back edge of the footrest cover.
3	Slide the footrest cover to the front, exposing a compartment.
4	Remove the four screws retaining the annunciator panel to the left rear of the compartment and replace the panel with the new one.
5	Remove the access panel covering the manifold board (under the keyboard, up close to the base of the station).
6	Pass the cable for the new assembly up the left leg of the station and connect it to J6 on the manifold board.
7	Replace the access panel, footrest cover, and leg covers.

9. Operator Entry Panel (OEP) Keyboard Operation and Service

9.6. Replacing the Multitone Annunciator Panel

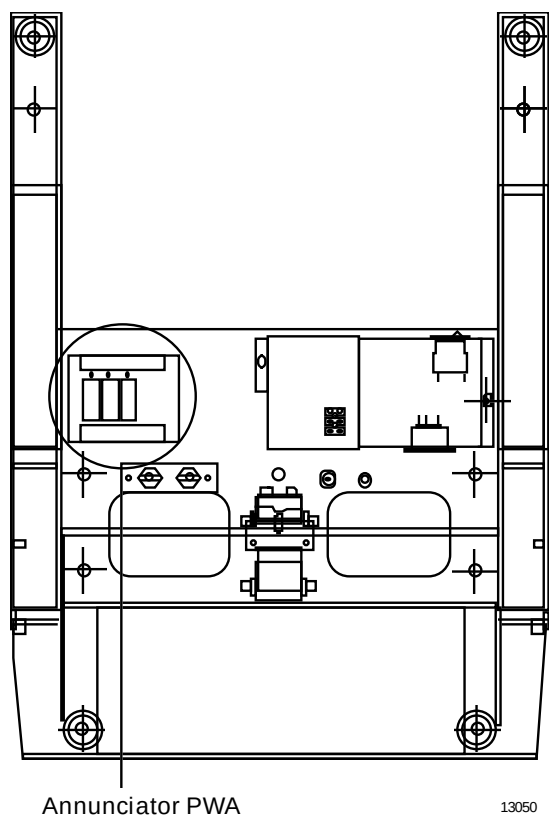


Figure 9-9 **Location of Annunciator Board in the Cabinet Base**

9.7 Engineering Keyboard EC-PC101

Description

The EC-PC101 Keyboard is a QWERTY key configuration that can be rested on the keyboard shelf that slides out from under the front of keyboard tray. This keyboard performs the Engineer's Keyboard function for Global User Stations with Operator Keyboards. The Auxiliary Keyboard Tray is shown in Figure 9-11.

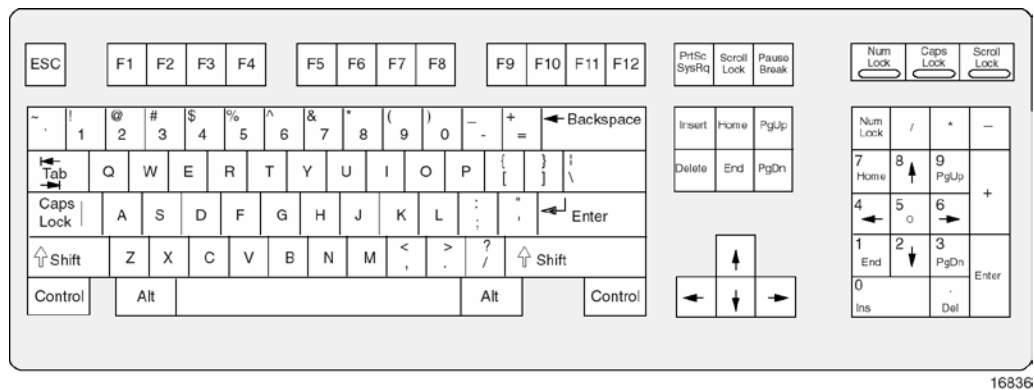


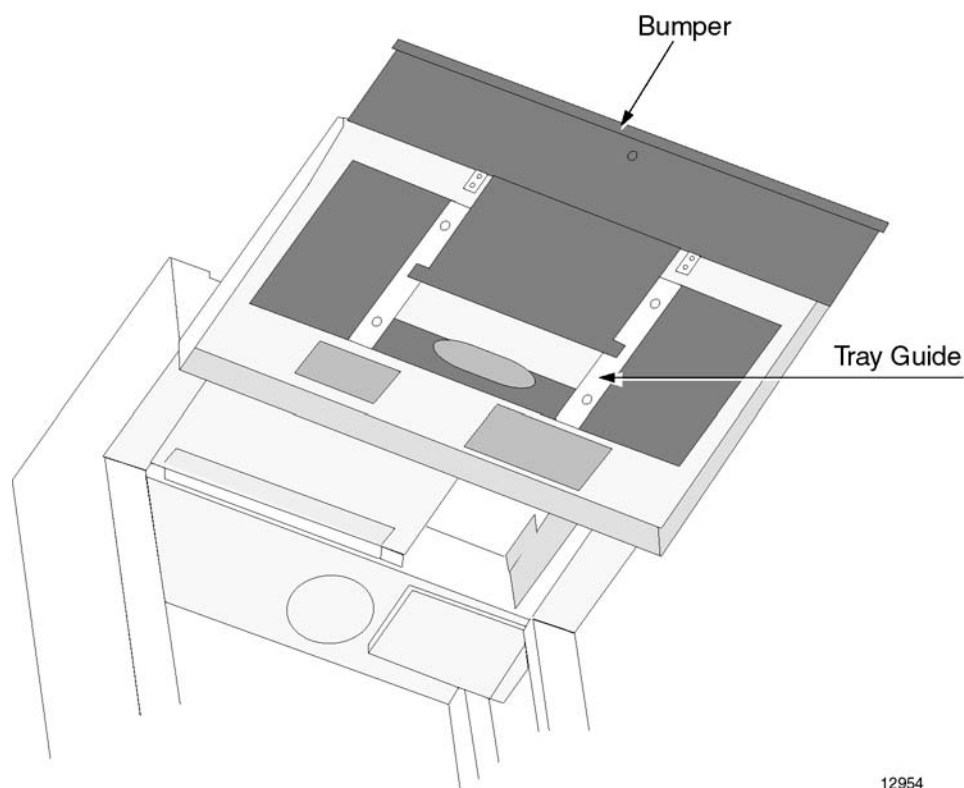
Figure 9-10 Engineering Keyboard

9. Operator Entry Panel (OEP) Keyboard Operation and Service

9.7. Engineering Keyboard EC-PC101

Engineer's Keyboard Tray

The Engineer's Keyboard rests on the keyboard slide tray located under the front edge of keyboard tray. The tray can be pulled forward to hold the engineer's keyboard when it is in use.



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Figure 9-11 Engineer's Keyboard and Keyboard Slide Tray

10. Classic Console Operation and Service

10.1 Classic Console Furniture

Overview

The following figures provide illustrations of Classic furniture that house a Global User Station and an optional single Five-Slot Module or a Dual Node Module.

Front view

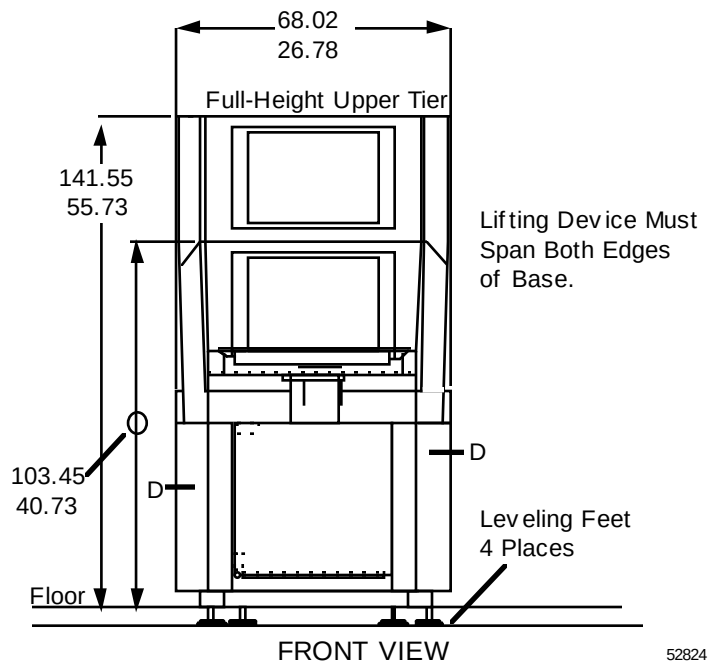


Figure 10-1 Classic Furniture Dimensions Front View

10. Classic Console Operation and Service

10.1. Classic Console Furniture

Top view

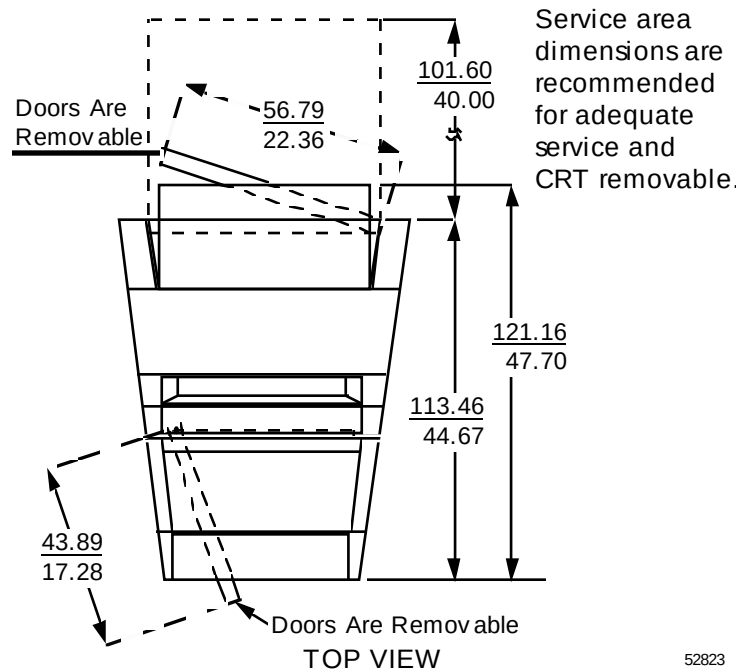


Figure 10-2 Classic Furniture Dimensions Top View

10.1. Classic Console Furniture

Technical drawing of a wheelchair showing dimensions for side view and floor clearance.

Side View Dimensions:

- Overall width: 120.95
- Distance from front wheel to back of seat: 47.62
- Distance from front wheel to back of footrest: 113.61
- Distance from front wheel to back of seat (lower section): 44.73
- Height of backrest: 141.71
- Height of seat: 55.80
- Height of backrest to top of seat: 125.81
- Height of seat to top of backrest: 49.53
- Height of footrest: 70.97
- Height of seat to top of footrest: 27.94

Floor Clearance:

- Leveling feet: 4 places
- Dimension may vary from 1.58 min clearance to 0.62 to 4.12 max clearance
- 1.62

52817

Figure 10-3 **Classic Furniture Dimensions Side View**

10. Classic Console Operation and Service

10.1. Classic Console Furniture

Wire and cable entry

Areas marked “open” are for vertical cable entry from cable raceways.

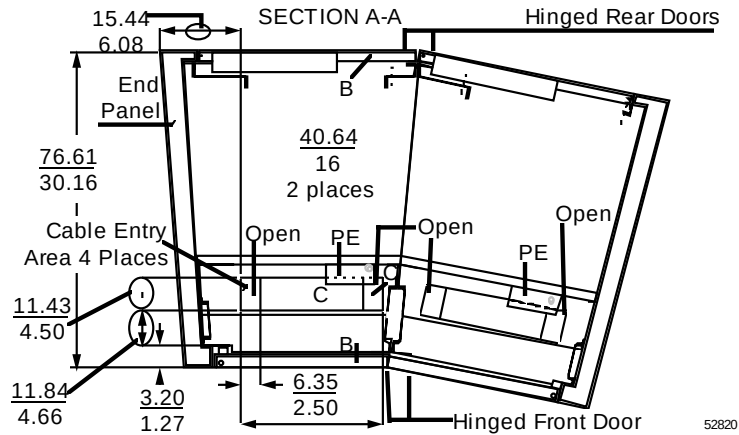


Figure 10-4 Classic Furniture Wire and Cable Entry

Wire and cable entry for complexed consoles

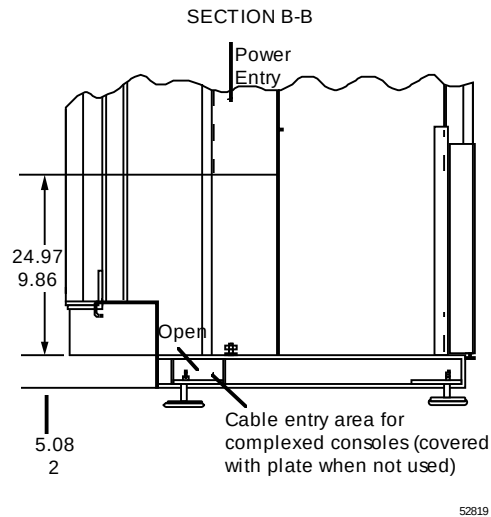


Figure 10-5 GUS Cable Entry for Complexed Console

10. Classic Console Operation and Service

10.1. Classic Console Furniture

Power Entry wiring - US

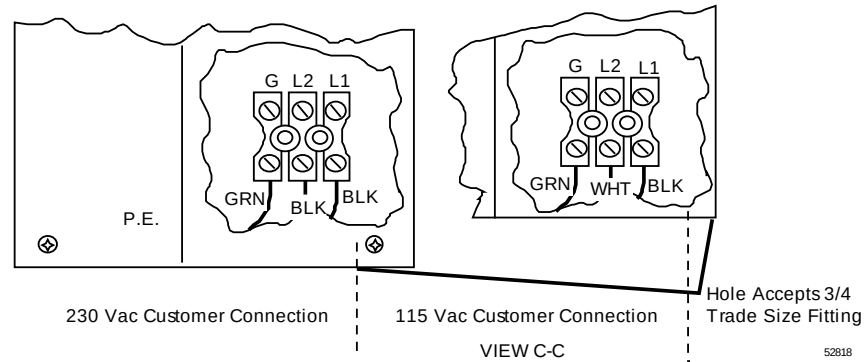


Figure 10-6 Classic Furniture AC Wiring (US Installations)

Power Entry wiring - Europe

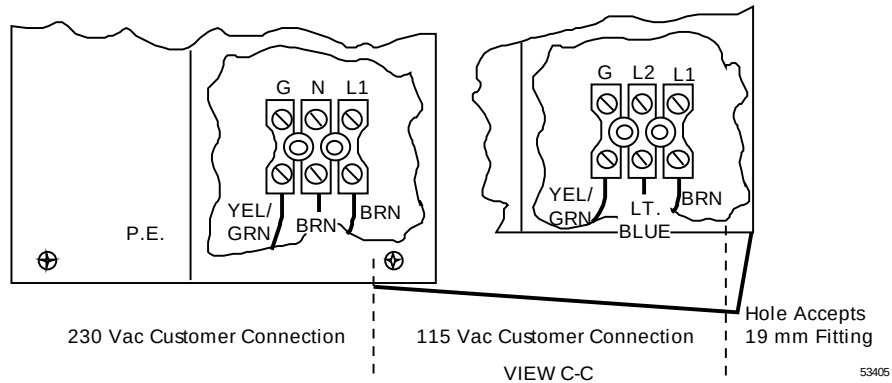


Figure 10-7 Classic Furniture AC Wiring (European Installations)

10. Classic Console Operation and Service

10.1. Classic Console Furniture

Console templates

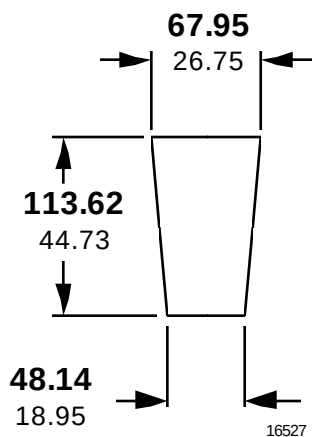


Figure 10-8 Classic Furniture Template for a Single Station

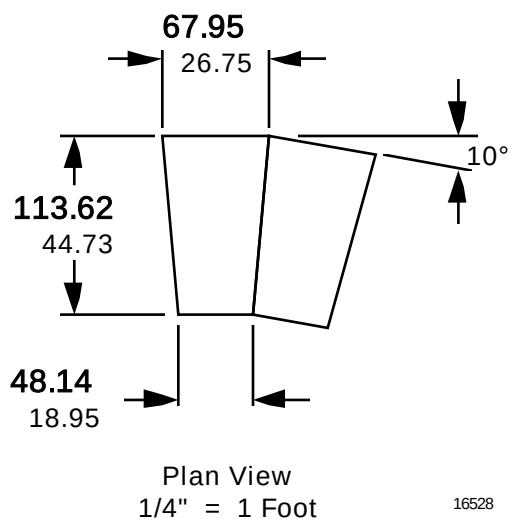


Figure 10-9 Classic Furniture Template for Double Station

TotalPlant Solution (TPS) system – console layout

The console-planning template shown here is available in printed pads of the template. Parts of the template are shown in the figures that follow.

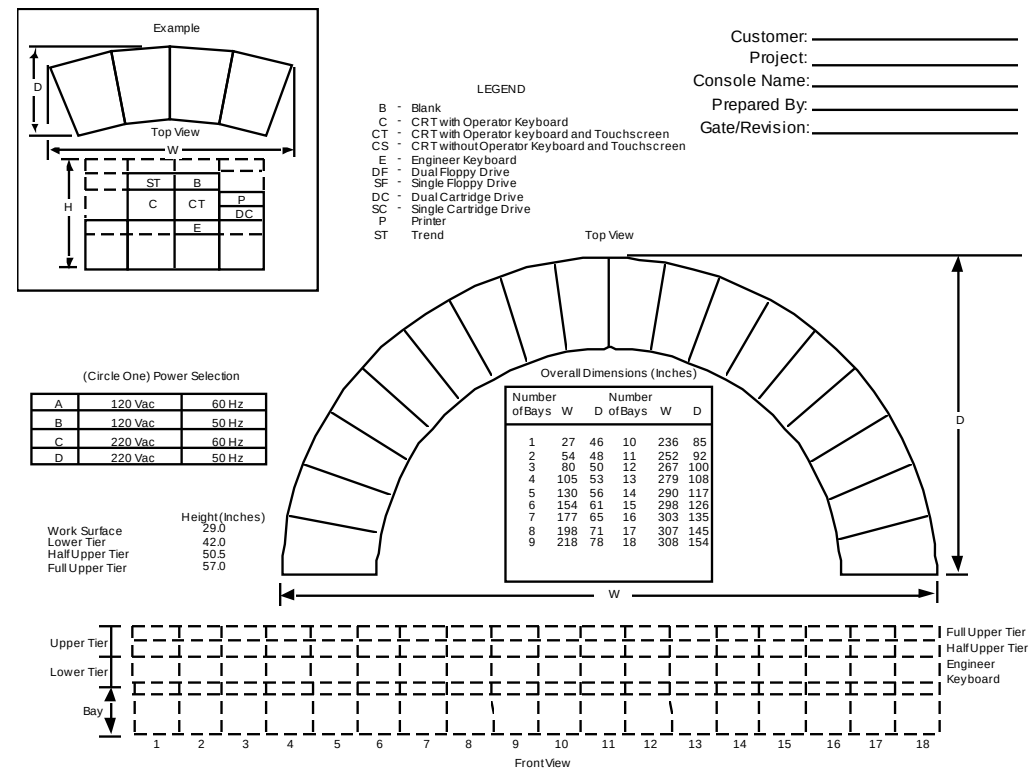


Figure 10-10 TPS System Console Layout for Classic Furniture

10. Classic Console Operation and Service

10.1. Classic Console Furniture

Console dimensions

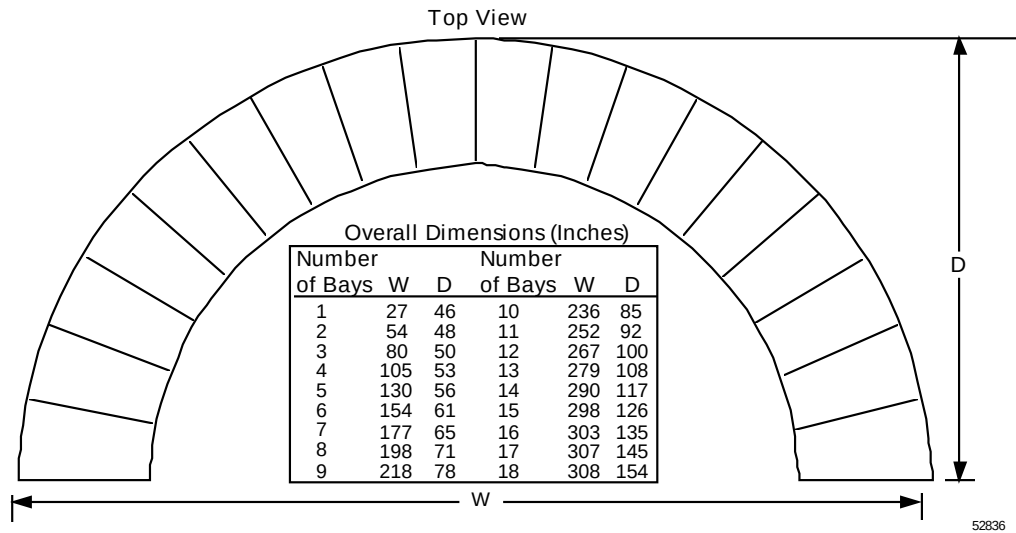


Figure 10-11 Classic Furniture Console Dimensions

10.2 Classic Console Operation

AC power

The AC circuit breaker switch, located on the center of the cabinet, behind the door under the keyboard is the master switch for the station. See the following figure.

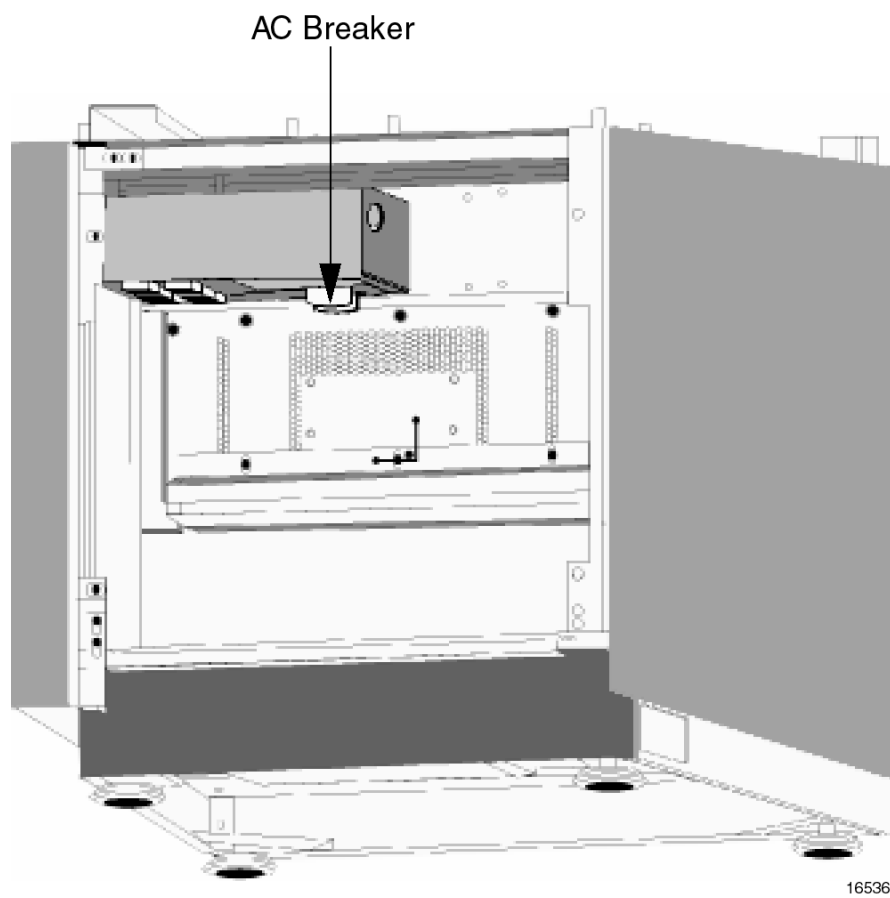


Figure 10-12 Location of AC Power Switch

10. Classic Console Operation and Service

10.2. Classic Console Operation

Station with IKB

The furniture allows for single or dual monitors, Integrated Keyboard (IKB), or the original Operator Entry Panel (OEP).

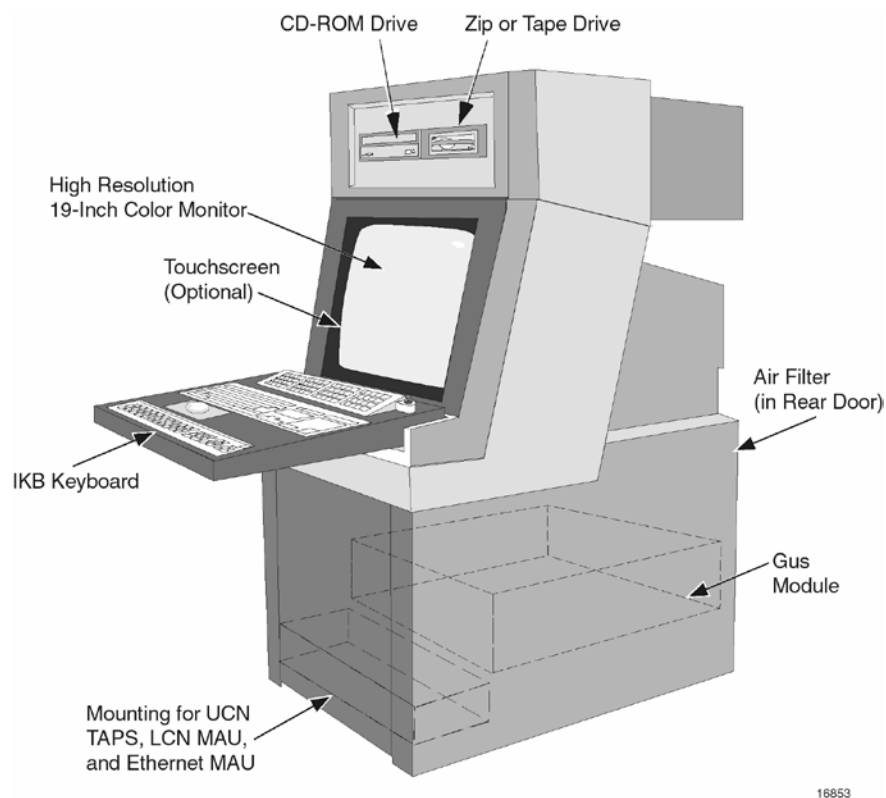


Figure 10-13 Features of the Station with IKB

Station with OEP

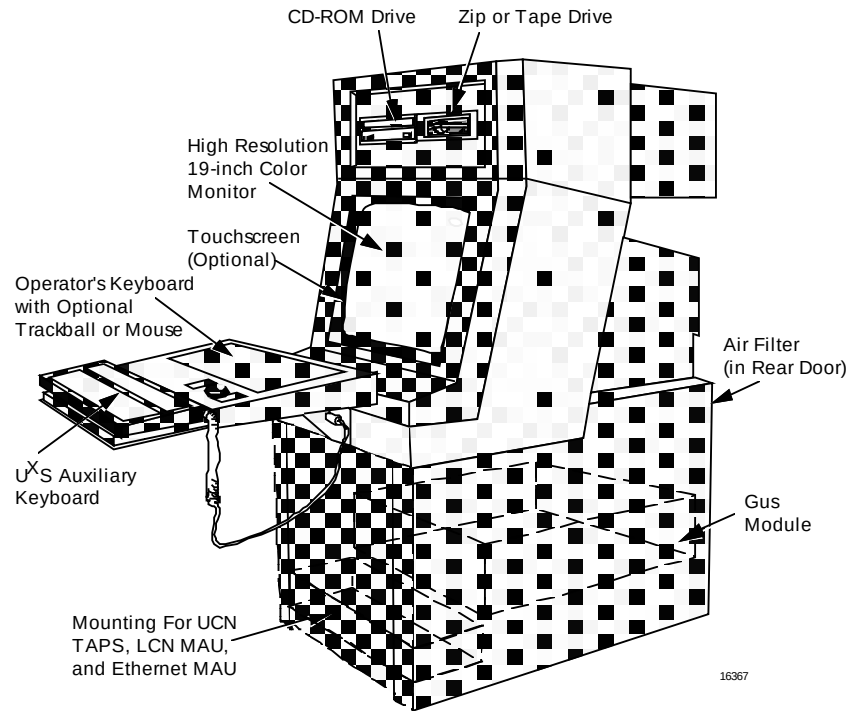


Figure 10-14 Features of the Station with OEP

10. Classic Console Operation and Service

10.3. Removing and Replacing the Flat Screen Open-Frame CRT in a Classic Console

Monitor controls

Monitor controls that control the following functions are located on the right-hand side of each monitor:

- Intensity—a thumbwheel control to adjust the intensity of the monitor
- Contrast—a thumbwheel control to adjust the contrast of the monitor
- Degauss button—press the round button below the thumbwheels until color purity is established.

10.3 Removing and Replacing the Flat Screen Open-Frame CRT in a Classic Console

CRT description

This 19-inch flat screen high-resolution color display monitor with digital multi-scan capability is an Optimum Replaceable Unit (ORU). The removal/replacement procedure and screen adjustments are given in this section.



Figure 10-15 Sony 19-inch Open-Frame CRT

Monitor assembly description

The monitor assembly is composed of the following:

- Monitor is a Flat Square Tube (FST) enclosed in an EMI shield.
- Monitor is virtually a flat screen [radius of curvature – 50,000 mm (164 ft)] that minimizes image distortion and reflective glare.
- Digital panel control and a full suite of on-screen display (OSD) programming/calibration menus.
- The lower tier monitor slides out of the Classic furniture on slide rails.
- The upper tier monitor is bolted to a CRT mounting bracket.
- A Control Panel connector is bracket-mounted on the side of the monitor mounting chassis for the cable plug from the Control Panel assembly on the bezel.

In general, the monitor:

- Is secured to the monitor mounting chassis and onto the slide tray with screws and PEM nuts.
- Can be replaced by removing the monitor mounting chassis from the slide tray. The replacement monitor is secured to the slide tray.
- Can be removed from the station without the disassembly of the touchscreen assembly or monitor control assembly.

10. Classic Console Operation and Service

10.3. Removing and Replacing the Flat Screen Open-Frame CRT in a Classic Console

Replace the lower and upper monitor

Step	Action
1	Shut down the operating system. Go to another station and do a second shutdown on this node number to prevent the LCN from isolating nodes.
2	Turn off the AC breaker, located in the front of the station, under the keyboard.
3	Remove the screws retaining the rear cover on the monitor and slide the rear cover off of the station.
4	Disconnect the AC power cable from the back of the monitor. See Figure 10-16.
5	Disconnect the video interface cable from the monitor.
6	Disconnect the Control Panel assembly cable from the monitor.
7	Lower Monitor: First, remove the two hex head screws securing the slide trays restraint bar to the console frame. Release the two detent buttons located on each side of the rear of the monitor-mounting tray. Remove the six hex-head screws from the bottom of the slide tray. NOTE: Once these 6 hex-head screws are removed, the monitor may slide off the slide tray. Carefully slide the monitor out of the cabinet, ensuring that the cable in the cabinet does not get caught or pinched. Upper Monitor: This monitor is mounted directly to the support tray. Go to the next step.

10. Classic Console Operation and Service

10.3. Removing and Replacing the Flat Screen Open-Frame CRT in a Classic Console

Step	Action
8	UPPER MONITOR ONLY: - Loosen the four vertical oriented hex-head screws on one side of the monitor. - Remove the three horizontally oriented hex-head screws from each side of the monitor-mounting bracket. - Lift the monitor from the mounting tray. - Place the new monitor on the support tray between the two mounting brackets. - Replace the three horizontally mounted hex-head screws on each side of the monitor. - Tighten the four vertically oriented hex-head screws that were previously loosened
9	LOWER MONITOR ONLY: - Place the new monitor assembly on the slide tray. - Replace the six screws and washers in the slide tray/monitor. - Install the six screws through the bottom of the slide tray and into the PEM nuts mounted on the monitor assembly. Align the monitor with the bezel before tightening the screws.
10	Reconnect the monitor Control Panel cable to the connector on the left side of the monitor. Reconnect the video interface cable and AC power cord.
11	Press the button on each side of the slide mechanism and push the monitor forward until it contacts the bezel and the side mechanism clicks into the detent.
12	Install the monitor support channel assembly, 51201879-100, by inserting a screw and grounding washer through a wide flat washer and spacer, and into the slide rail assembly.
13	Reinstall the lower pod rear cover and retaining screws.

10. Classic Console Operation and Service

10.3. Removing and Replacing the Flat Screen Open-Frame CRT in a Classic Console

Monitor cabling

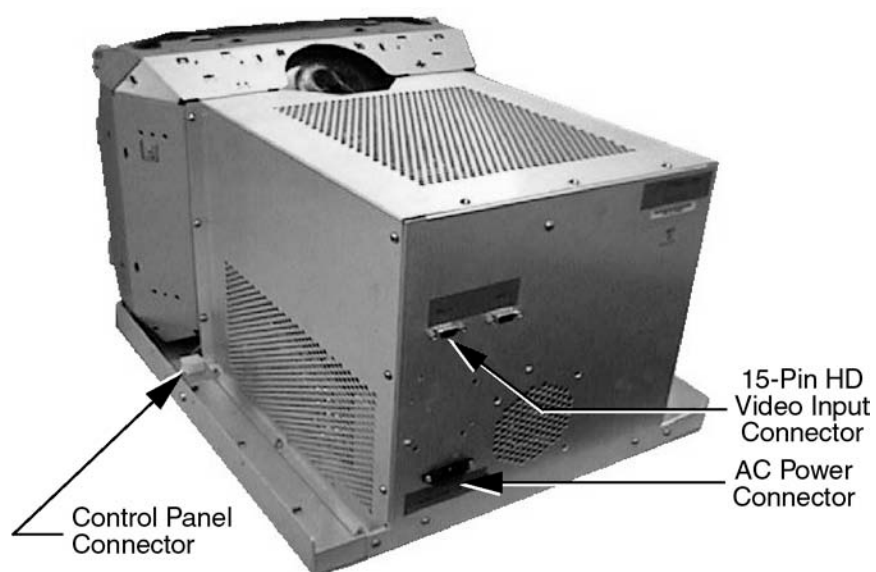


Figure 10-16 Monitor Cabling

Replace the monitor control assembly

To replace the monitor control panel assembly, the rear cover and the monitor must be removed first. Refer to the “Replace the lower and upper monitor” procedures in this section.

**ATTENTION**

If the monitor contains a touchscreen, the bezel/control panel assembly must be removed first before the Control Panel can be accessed and removed.

Step	Action
1	Remove the monitor, as detailed in this section.
2	Remove the screws holding the monitor control assembly into the touchscreen bezel. See Figure 10-17.
3	Remove the monitor Control Panel assembly from the bezel.
4	Install the new monitor control panel assembly on the bezel. See Figure 10-17. Orient assembly so switches appear at front of bezel as shown in Figure 6-6.
5	Install the four screws to hold the assembly to the bezel.
6	Replace the monitor, as detailed in this section.

10. Classic Console Operation and Service

10.4. Operating Media Drives

Control Panel Assembly for 19-Inch Monitor

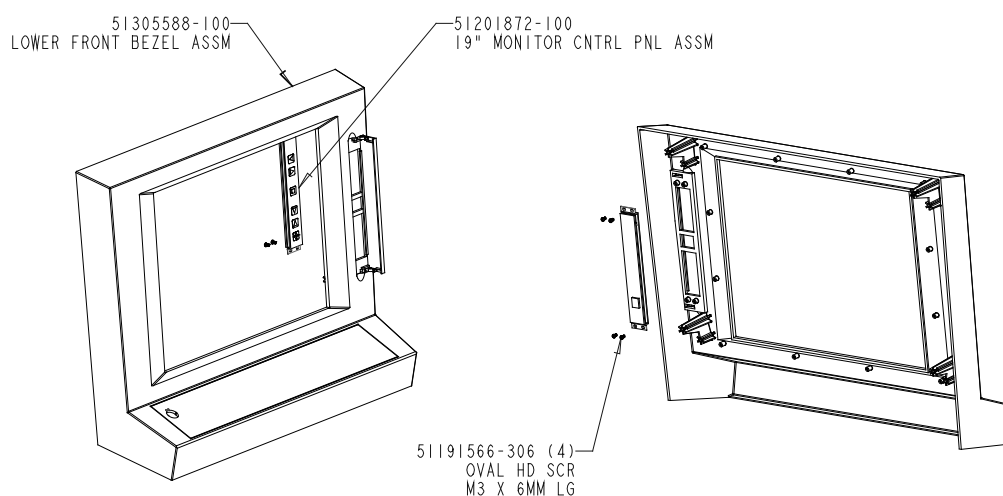


Figure 10-17 Control Panel Assembly for 19-Inch Monitor

10.4 Operating Media Drives

Load/unload CD-RW

Function	Operation
Load	1. Push the button—the loader tray will extend. 2. Insert the CD-RW into the tray. 3. Push the loader tray into the drive.
Unload	1. Push the button—the loader tray will extend. 2. Remove the CD-RW from the tray. 3. Push the loader tray into the drive.

Zip drive operation

Function	Operation
Load	1. Insert the cartridge in the drive—the LED will flash until the drive has spun up to speed.
Unload	1. Press the unload button—the LED will flash until the drive has unloaded. 2. Remove the cartridge by pulling the cartridge out of the drive.
Access	The LED glows when the cartridge is electronically accessed.

Floppy drive operation

Function	Operation
Load	1. Insert the floppy into the slot in the drive. 2. Press the floppy into the slot until the load mechanism pulls the floppy all the way in and seats it in the drive.
Unload	1. Press the button on the front of the drive. 2. Remove the floppy after it ejects.

10.5 Servicing Media Drives

Remove the drive enclosure facade

The rear cover of the half-upper pod must be removed to access the drive enclosure.

No.	Step
1	Remove the four M5 screws and washers from the four corners of the rear cover of the half-upper tier and slide the cover back off the mounting tray.
2	Remove the four screws on each side of the enclosure.
3	Remove the rear half of the top cover on the drive enclosure.
4	Disconnect the dc power cable from the back of the drive to be replaced.
5	Disconnect the SCSI interface cable from the back of the drive.
6	Remove the four vertically mounted screws from both sides of the enclosure.
7	Remove the four vertically mounted screws from the mounting bracket of the drive to be replaced.
8	Remove the two horizontally mounted screws from each side of the mounting bracket.

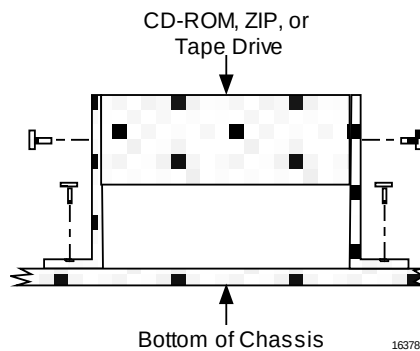


Figure 10-18 Drive Enclosure Box

Replace a drive in the enclosure

Step	Action
1	Attach the two mounting brackets to the drive with two screws on each side.
2	Place the drive assembly in the drive enclosure by installing the four nuts on the four vertically positioned screws.
3	Reconnect the SCSI interface cable to the Zip drive. Refer to Figure 10-20.

Assemble the drive enclosure

Step	Action
1	Replace the cover panel and all of the retaining screws.
2	Place the drive enclosure on the half-upper, tier-mounting tray.
3	Install the four vertically positioned screws in the drive enclosure-mounting bracket. Ensure that the faceplate on the front of the drive is firmly against the front bezel before tightening the screws.
4	Reconnect the SCSI interface cable to the rear of the drive enclosure.
5	Reconnect the AC power cable to the rear of the drive enclosure.
6	Replace the half-upper tier rear cover by sliding it on the internal mounting guide.
7	Replace the four M5 screw and washer sets in the four corners of the cover.

Troubleshooting an Inoperative CD-ROM Drive

In general, if only one drive fails, the disk drive is most likely at fault. If both drives are inoperative, check the power supply in the electronics module.

Isolation procedure

Table 10-1 Troubleshooting a CD-ROM Problem

Step	Action
1	If both drives fail, disconnect the power cable coming out of the rear of the electronics module; +5 VDC and +12 VDC must be present.
2	Check for power at the input to the drive enclosure and at the input to the drive.
3	If no problem is found in Step 1, check that all cable connections are seated.
4	Replace the drive with a spare.
5	If all of the above failed to locate the problem, check the SCSI cable inside the electronics module.
6	The last resort is to replace the motherboard inside the electronics module.

10.6 Servicing Remote Peripherals

Overview

CD-ROM, tape, and Zip drives are used to load and store data as required by the operation of the system.

Description

The CD-ROM, tape, and Zip drives are located in the drive enclosure box, located in the half upper tier above the monitor. The CD-ROM drive can share the enclosure with either the optional Zip drive or the optional tape drive.

CE Compliant enclosure

The CD-ROM drive is enclosed in a universal metal box with an interface connector and one power connector. The SCSI connector interfaces the SCSI drives inside the box.

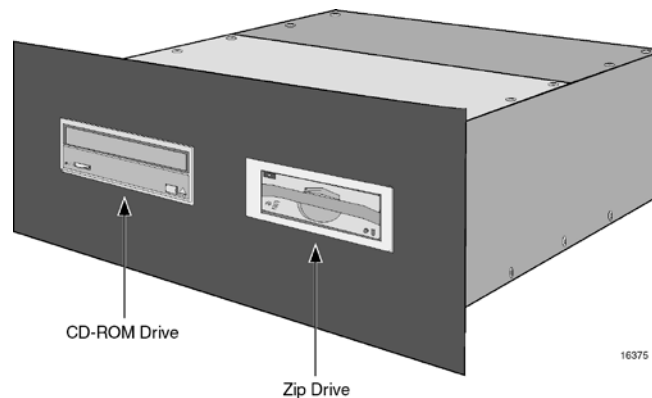


Figure 10-19 CE Compliant CD-ROM Drive Enclosure

The enclosure can have a one or two-piece cover. The rear cover of a two-piece set can be removed for inspection purposes. To replace a drive or power supply, the enclosure must be removed from the station.

10. Classic Console Operation and Service

10.6. Servicing Remote Peripherals

Cable connections

The SCSI connector interface to the SCSI.

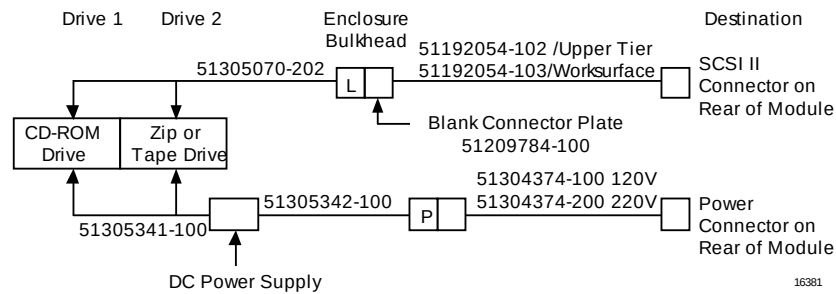


Figure 10-20 CD-ROM Drive Cabling Construction

Construction

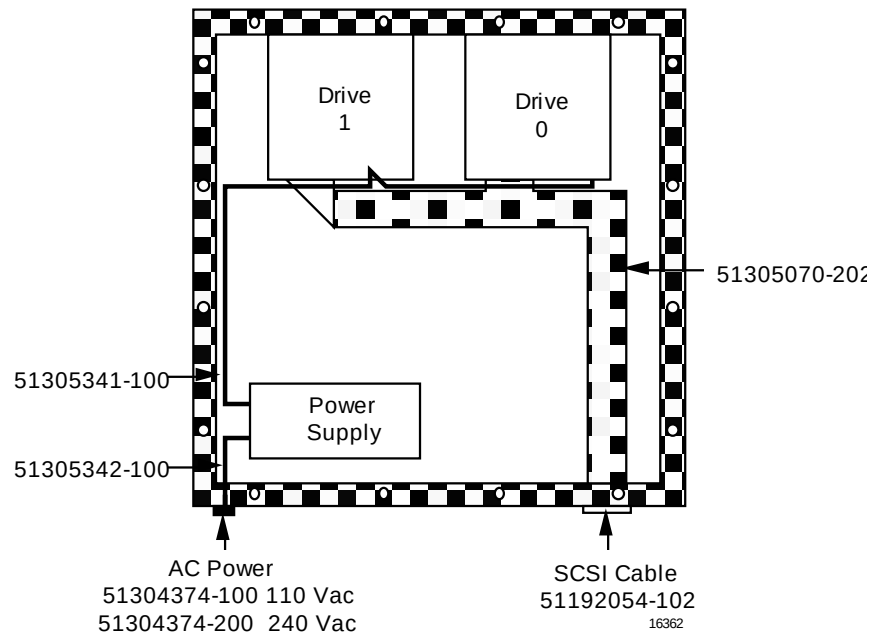


Figure 10-21 Cabling for a CE Compliant Drive Enclosure

10. Classic Console Operation and Service

10.6. Servicing Remote Peripherals

11. Z Console Operation and Service

11.1 Z-Console Planning

Cabling

The dc cables are routed up the left leg cavity and AC cables are routed up the right leg cavity, as seen from the front of the station.

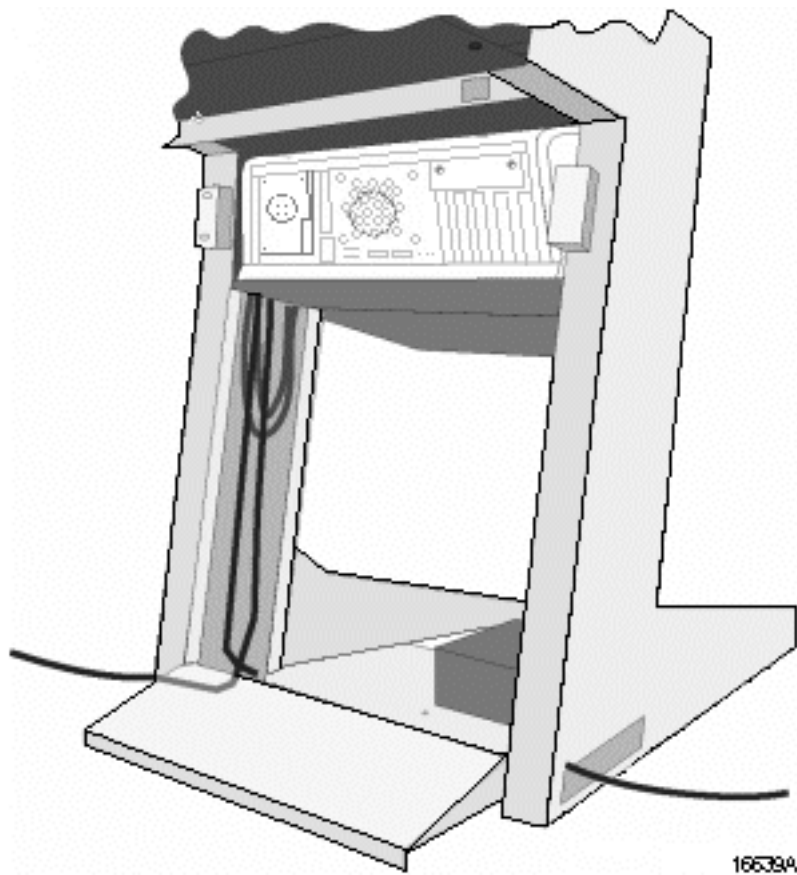


Figure 11-1 LCN Cabling

11. Z Console Operation and Service

11.1. Z-Console Planning

UCN Tap location

UCN trunk cable taps can be mounted in the bottom of the footrest compartment when a NIM module is located below the Global User Station (GUS).

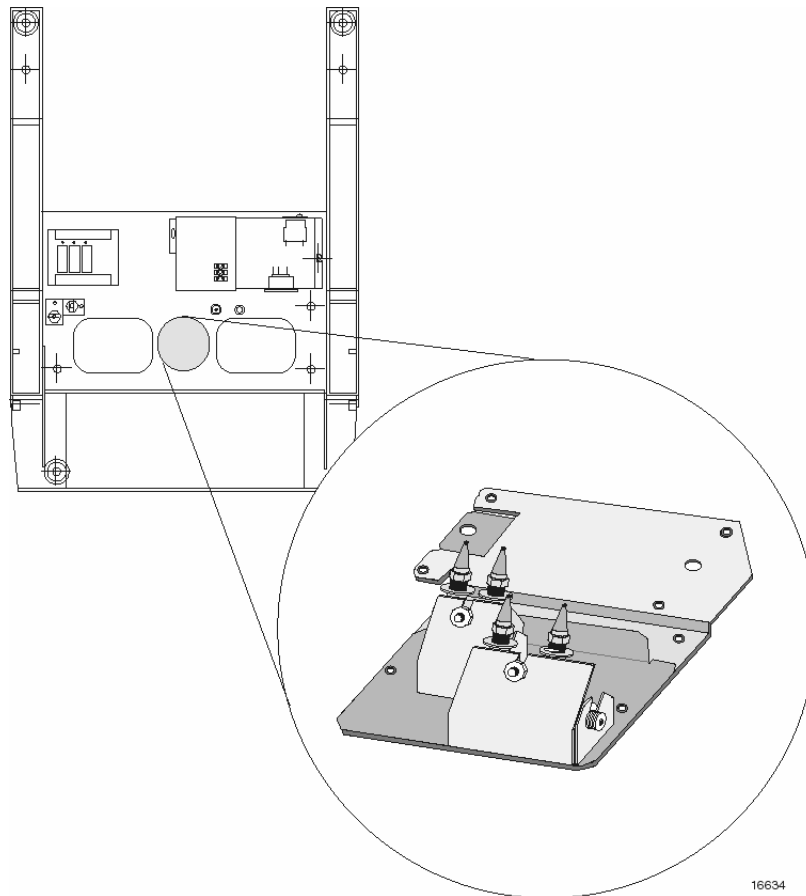


Figure 11-2 UCN Trunk Cable Mounting

LCN MAU mounting

The Global User Station has an LCN Media Adapter Unit (MAU) mounted in the bottom of the footrest compartment next to the optional UCN Tap.

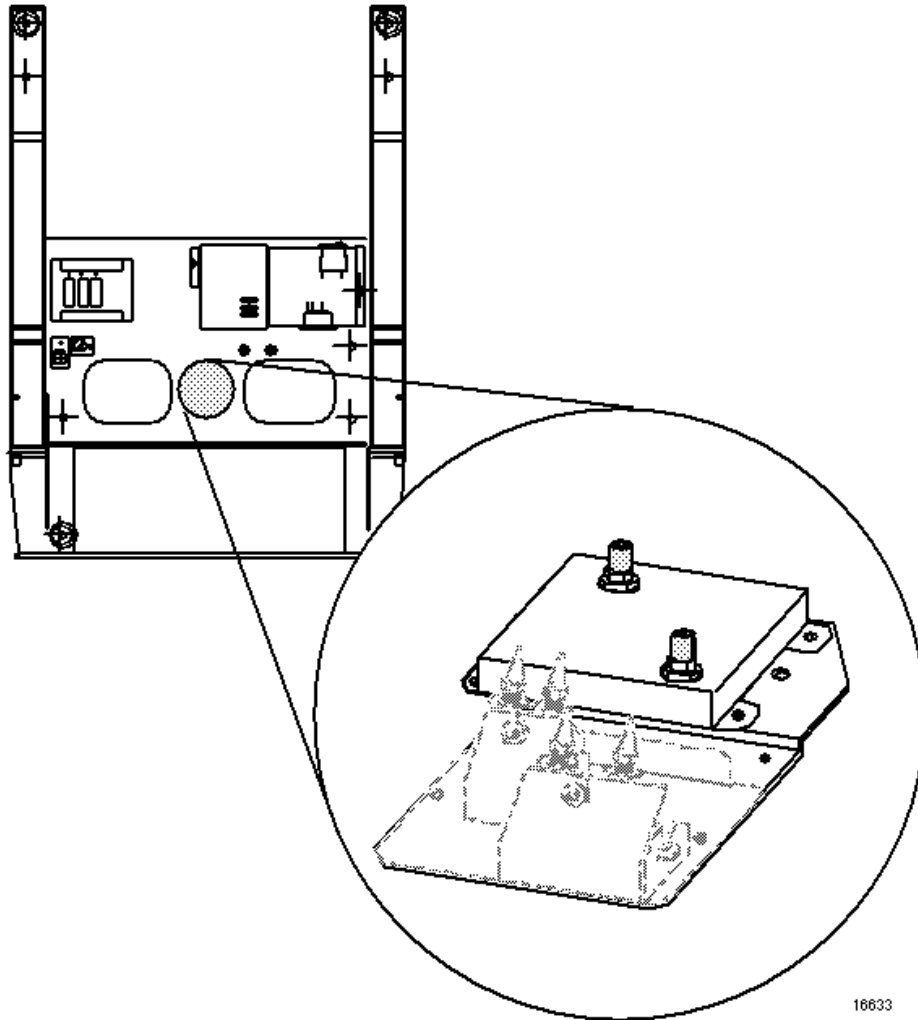


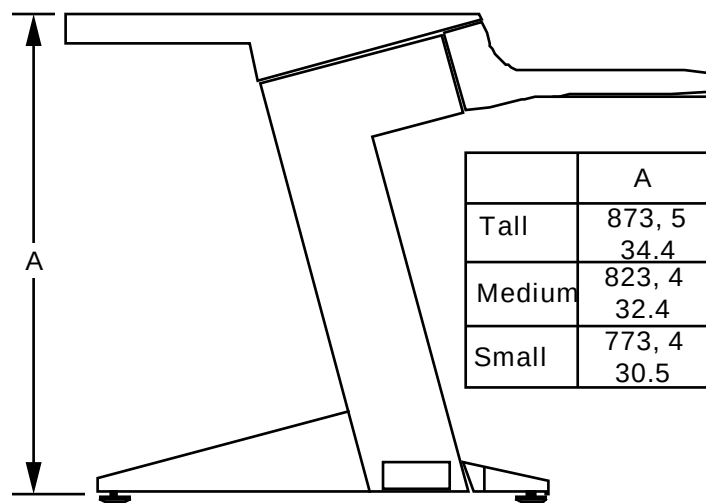
Figure 11-3 LCN MAU Mounted with UCN Tap

11. Z Console Operation and Service

11.1. Z-Console Planning

Work surface side view

Work surface cabinets match the new furniture base footprint.



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Figure 11-4 **New Furniture Work Surface Side View**

Work surface top view

Work surface cabinets have a blank plate instead of a keyboard.

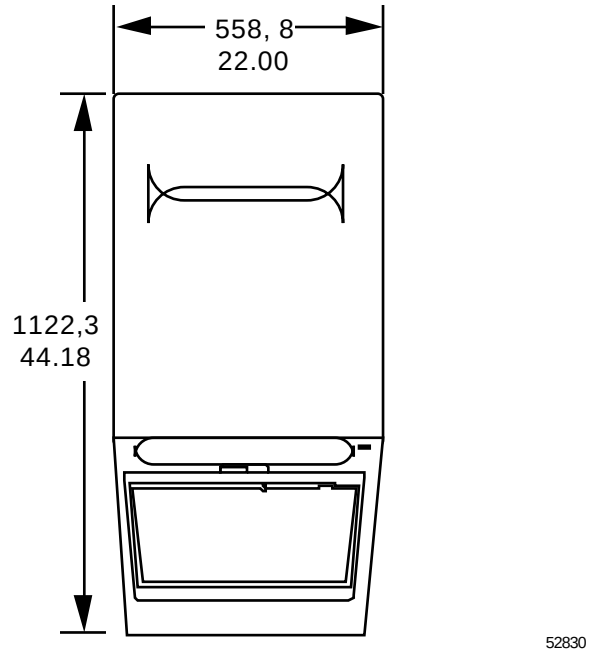


Figure 11-5 Work Surface Top View

11. Z Console Operation and Service

11.1. Z-Console Planning

Template

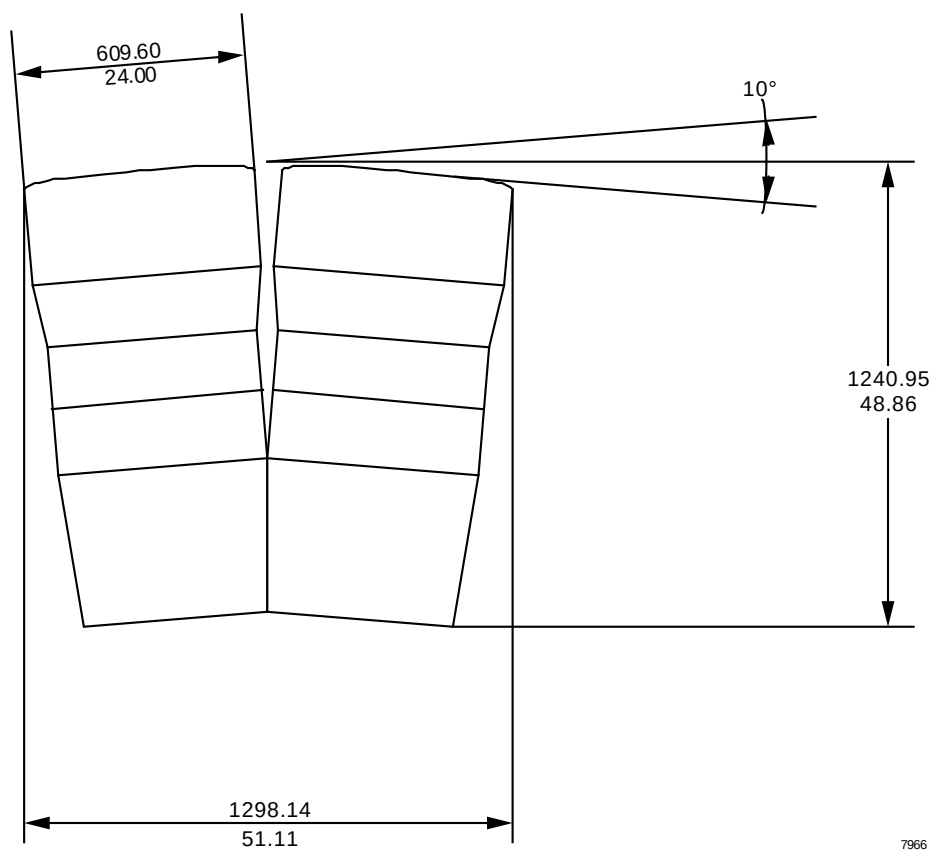


Figure 11-6 Furniture Template

TPS system – arc console layout

Part of this planning template is enlarged in the next figure.

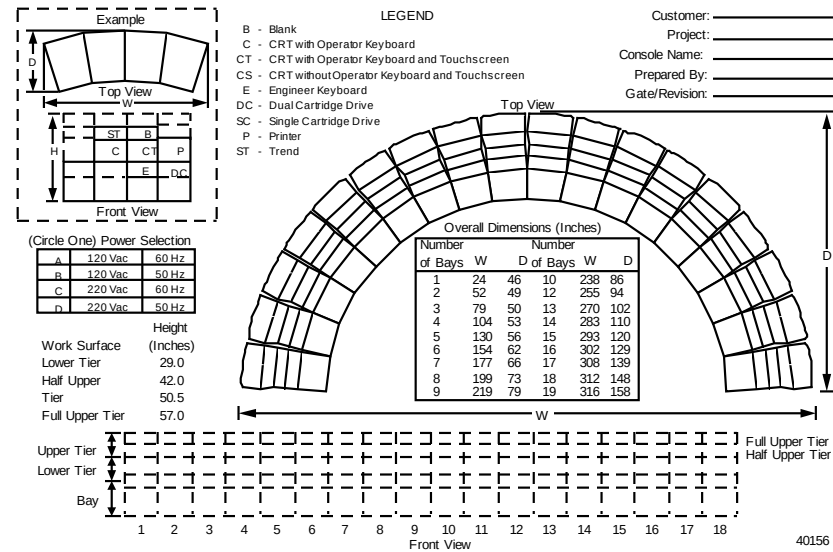


Figure 11-7 TPS System New Furniture Console Layout

11. Z Console Operation and Service

11.1. Z-Console Planning

Console dimensions

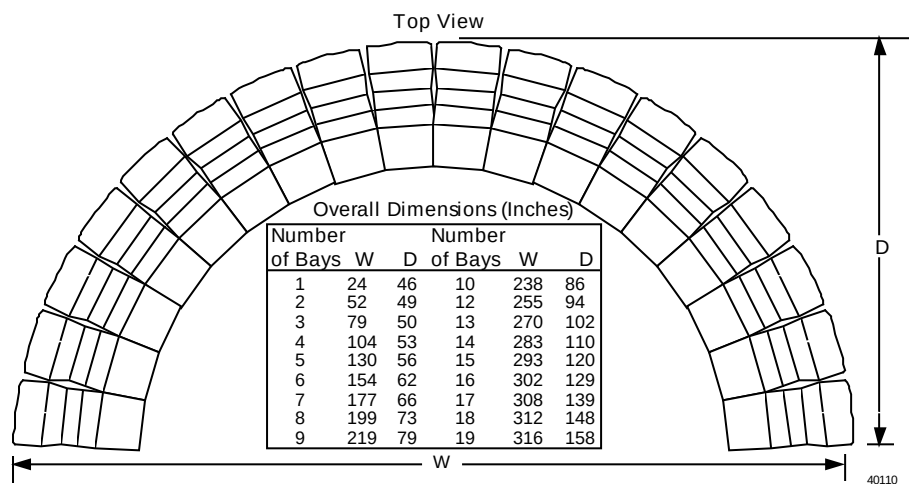


Figure 11-8 Console Dimensions Planning Template

11.2 Z Console Furniture

Overview

Figure 11-9 through Figure 11-1 provide detailed illustrations of ergonomic furniture that houses either single or multiple monitors for Global User Stations or a single Global User Station. A Global User Station can have one other module mounted in the station.

The furniture is available in three heights referred to as 650 mm (small), 700 mm (medium), and 750 mm (tall). This terminology refers to the height of the keyboard above the floor. Refer to Figure 11-9 and Figure 11-10.

The work-surface furniture has the same footprint dimensions as the Global User Station.

Front view

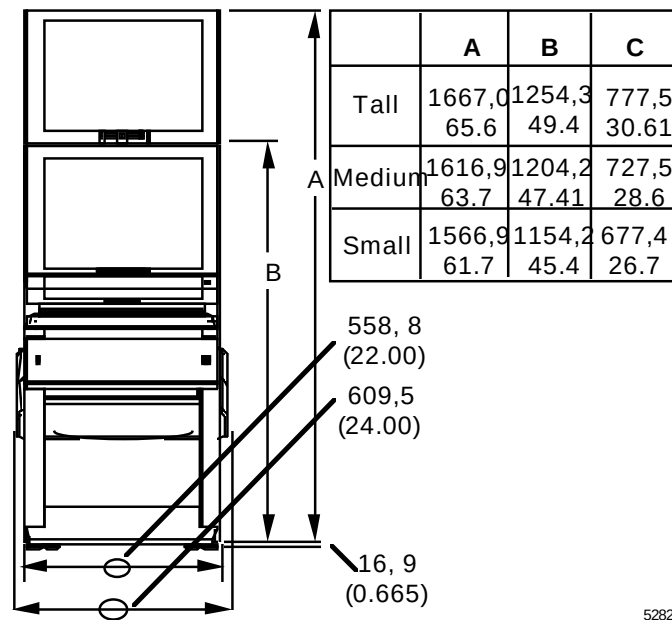


Figure 11-9 New Furniture Front View

11. Z Console Operation and Service

11.2. Z Console Furniture

Side view

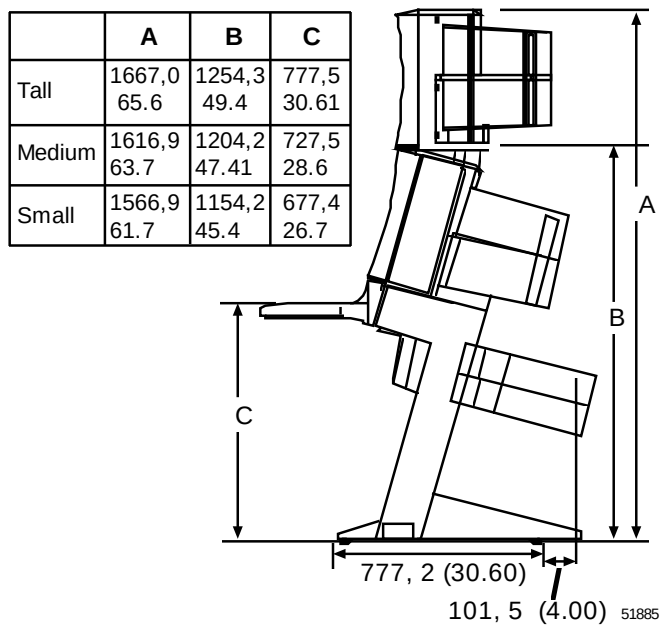


Figure 11-10 New Furniture Side View

Top view

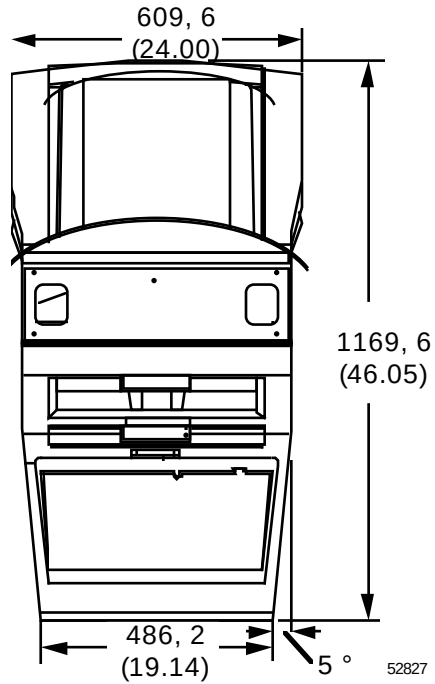


Figure 11-11 New Furniture Top View

11. Z Console Operation and Service

11.2. Z Console Furniture

Single monitor side view

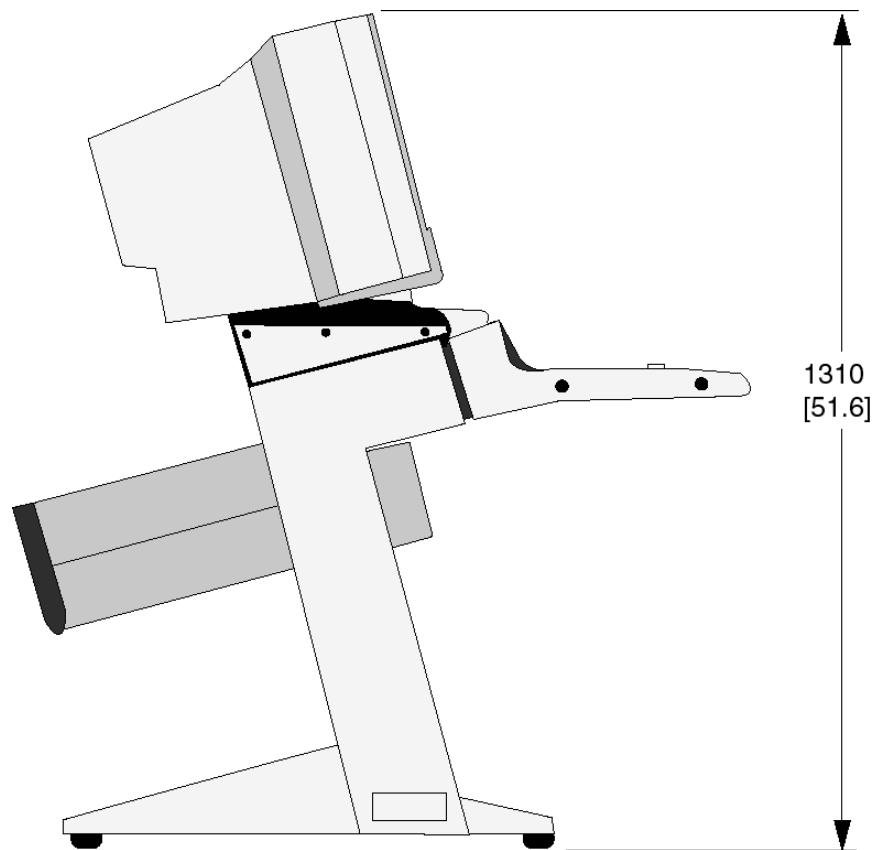


Figure 11-12 Single Monitor Z-Furniture Side View

Dual monitor side view

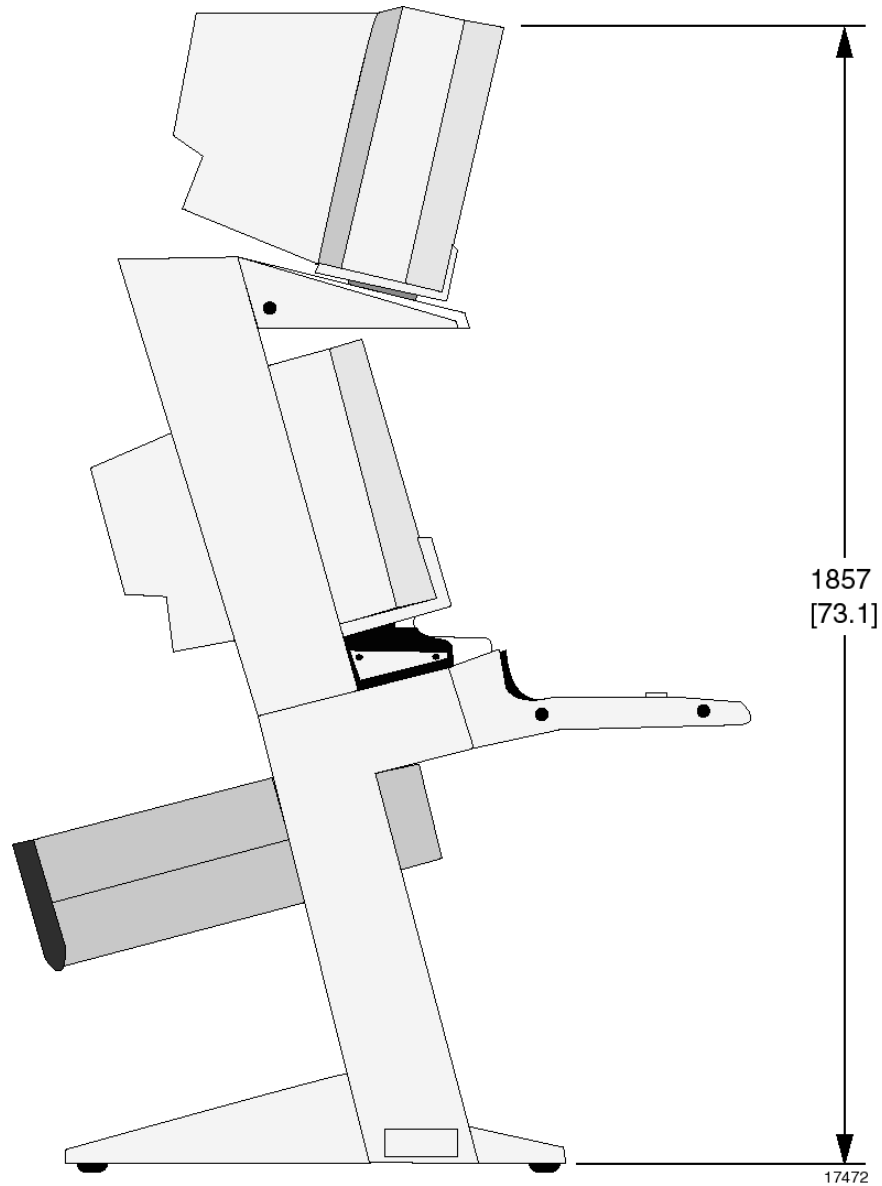


Figure 11-13 Dual Monitor Z-Furniture Side View

11. Z Console Operation and Service

11.2. Z Console Furniture

Multi-purpose enclosure

The multi-purpose enclosure can be used as an enclosure for customer install device such as communications equipment, alarm panels, etc. The enclosure is a standard NEMA 19-inch rack that can be used to mount equipment or a flat panel used to mount equipment.

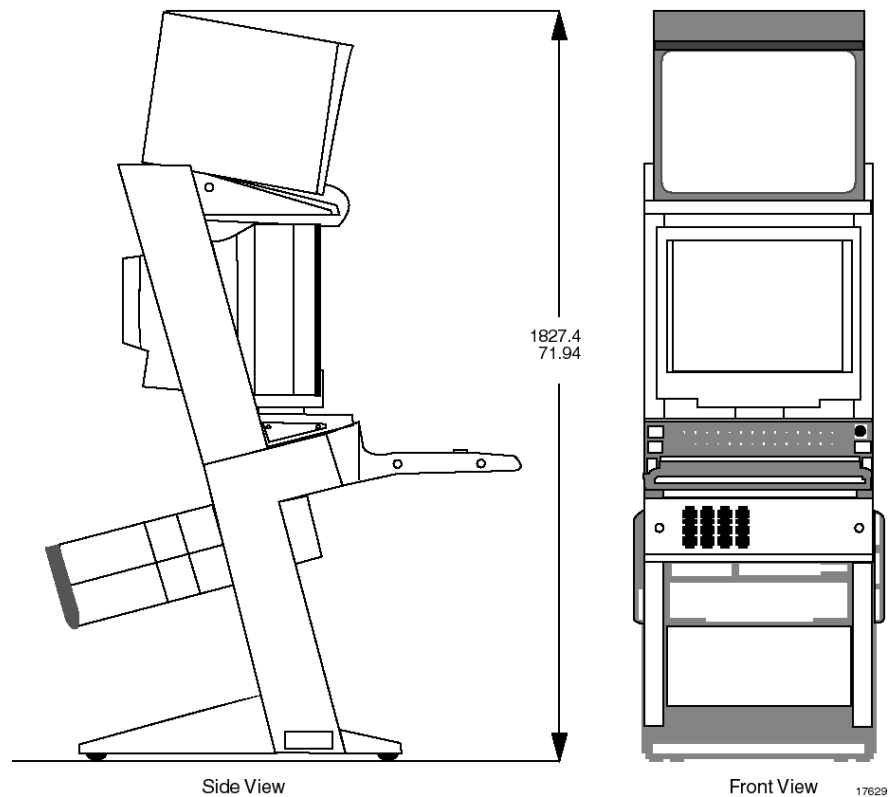


Figure 11-14 Z-Console Multipurpose Enclosure

Enclosure dimensions

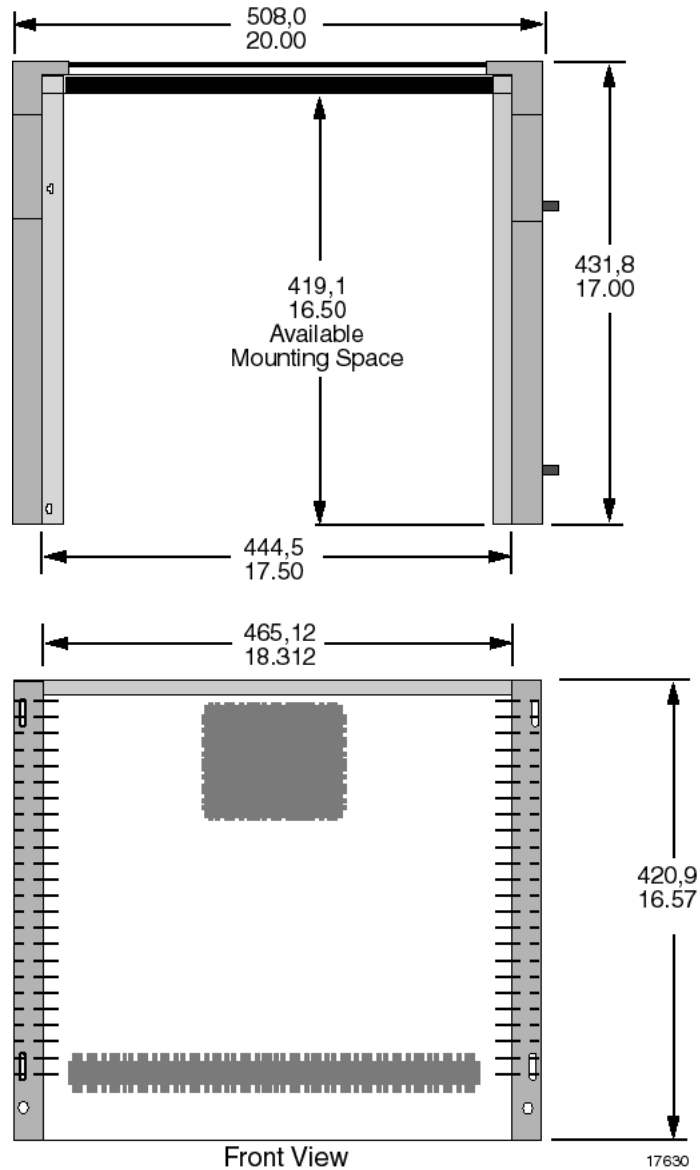


Figure 11-15 Enclosure Dimensions

11. Z Console Operation and Service

11.2. Z Console Furniture

Bezel dimensions

The bezel is used when a panel is placed on the front of the enclosure to allow various objects to be mounted on it. The bezel dimensions define the available area that can be used.

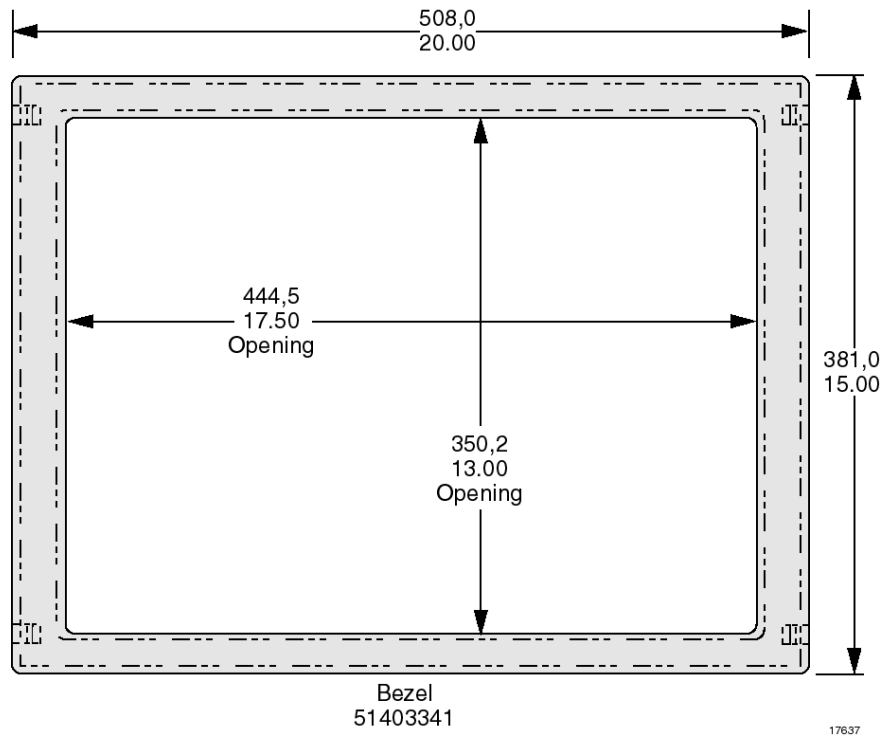


Figure 11-16 Z-Console Multipurpose Enclosure Bezel Dimensions

Z-Console Operation

The rocker switch located on the rear of the footrest is a master switch for the station. This switch controls the power to the entire station. See Figure 11-17.

AC power switch

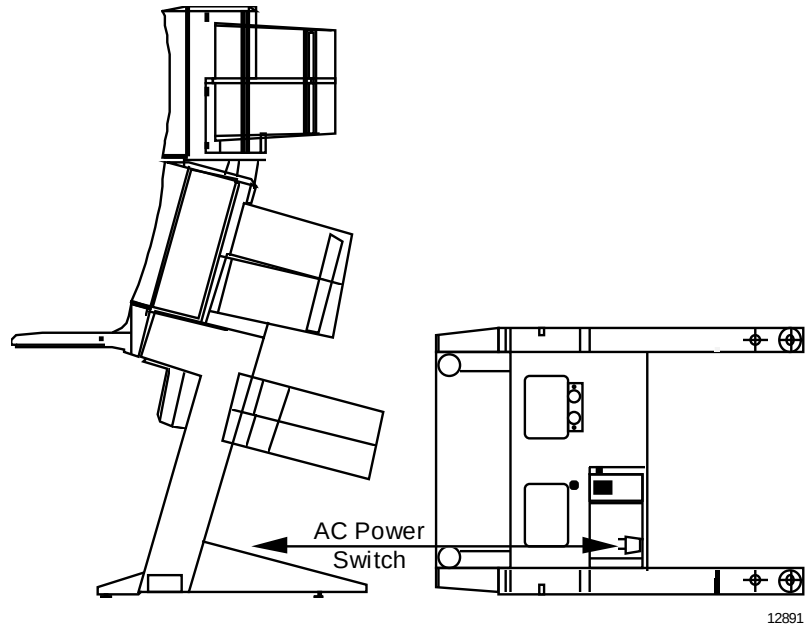


Figure 11-17 Location of AC Power Switch

11. Z Console Operation and Service

11.3. Accessing and Removing a Monitor in a Z Console

Station features

The furniture allows for dual monitors and/or dual equipment pods with front bezels and rear covers. Both monitor rear cover and a larger equipment rear cover are available.

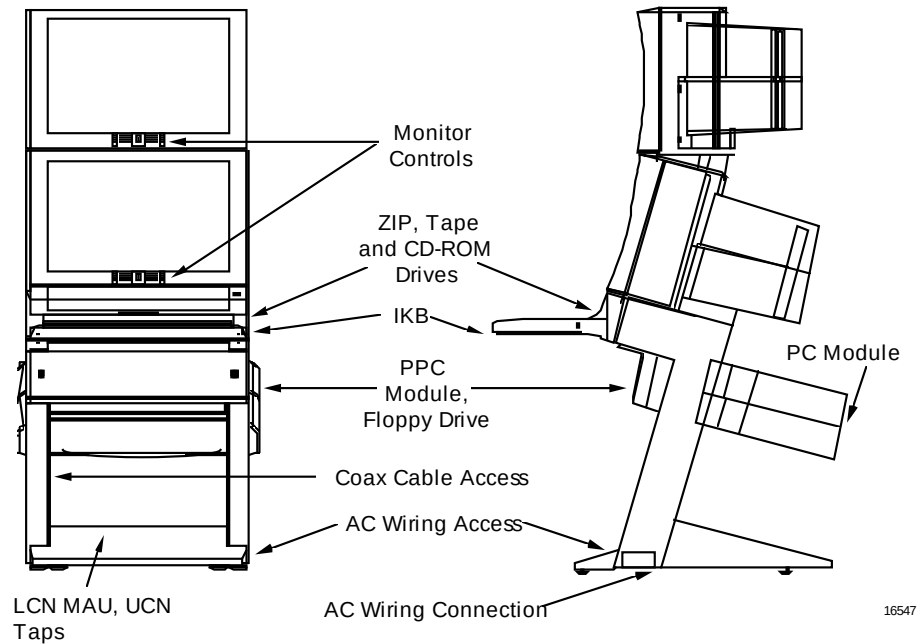


Figure 11-18 Features of the Station

Monitor controls

Monitor controls for the following functions are located behind the HONEYWELL logo panel at the bottom of each monitor:

- Intensity—a sliding control to adjust the intensity of the monitor
- Contrast—a sliding control to adjust the contrast of the monitor
- Degauss button—press the square button in the center until color purity is established.

Integrated keyboard

The IKB is the optional keyboard on the new furniture Global User Station units. This version of the IKB has a button panel that tilts forward.

There is an optional trackball integrated in IKB surface.

11.3 Accessing and Removing a Monitor in a Z Console

Access the monitor

To access the monitor, remove the rear monitor cover as follows.

Step	Action
1	Remove the four hex-head screws (two on each side) from the cover using an 8 mm socket.
2	Slide the cover off to the rear.

Monitor cable connections—external

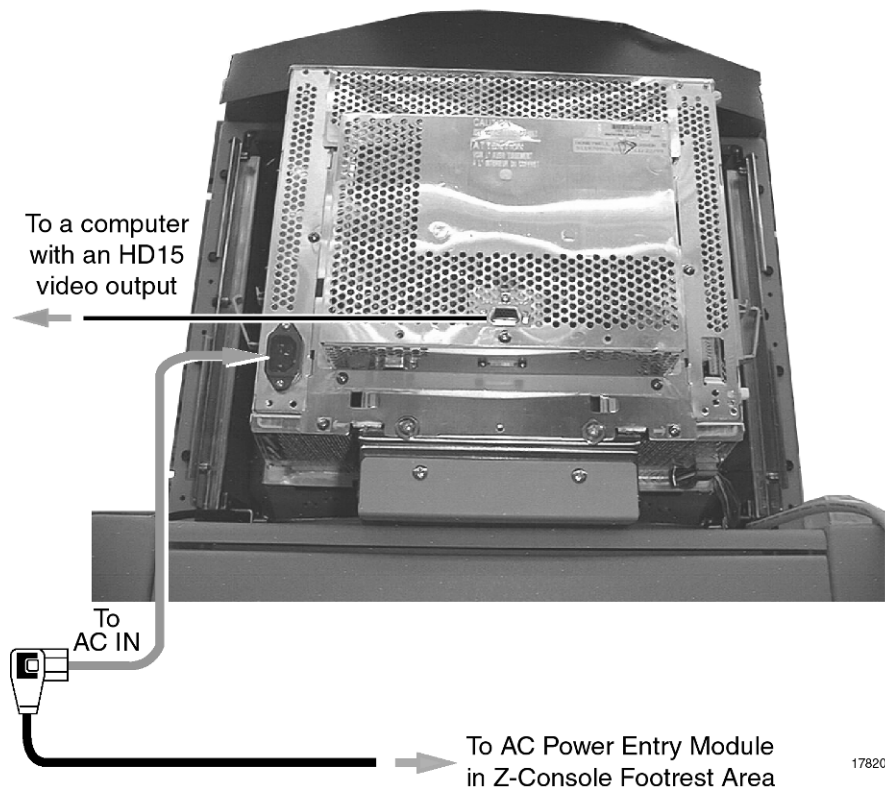


Figure 11-19 Monitor Cable Connections—External

11. Z Console Operation and Service

11.3. Accessing and Removing a Monitor in a Z Console

Remove the monitor

Perform the steps below to remove the monitor in the console.



CAUTION

The monitor assembly weighs in excess of 74 pounds. Whether removing the lower or upper monitor, once the screws have been loosened and/or removed, the monitor assembly is free to slide forward or backward onto the floor. For safety purposes, it is essential that two (2) individuals participate in the safe removal of the monitor from the console to prevent personal injury or equipment damage.

Step	Action
1	Disconnect the AC power cable from the back of the monitor. See Figure 11-19.
2	Disconnect the HD15 video signal cable from the connector at the rear of the monitor. See Figure 11-19.
3	The monitor control panel assembly remains mounted to the bezel. The interface connection must be disconnected. • On older units you may have to slide the monitor out several inches and disconnect from the front. • Disconnect the monitor control panel cable plug from the panel-mounted plug on the cable from the monitor electronics.
4	Remove the three hex-head screws on each side of the monitor-mounting bracket. See Figure 11-21. Hint: Loosen four bolts on each side for easier removal.
5	ATTENTION Remove the two screws (and spacers if no touchscreen) from the L-bracket at the bottom rear of the monitor. The monitor is ready for removal.
6	ATTENTION If this is a lower monitor, it will slide backward onto the floor if you let go of it. Slide the monitor to the rear of the cabinet as far as it will go.
7	ATTENTION Using two people, lift the monitor from the monitor-mounting shroud. The monitor chassis weighs in excess of 74 pounds.

Monitor and mounting chassis

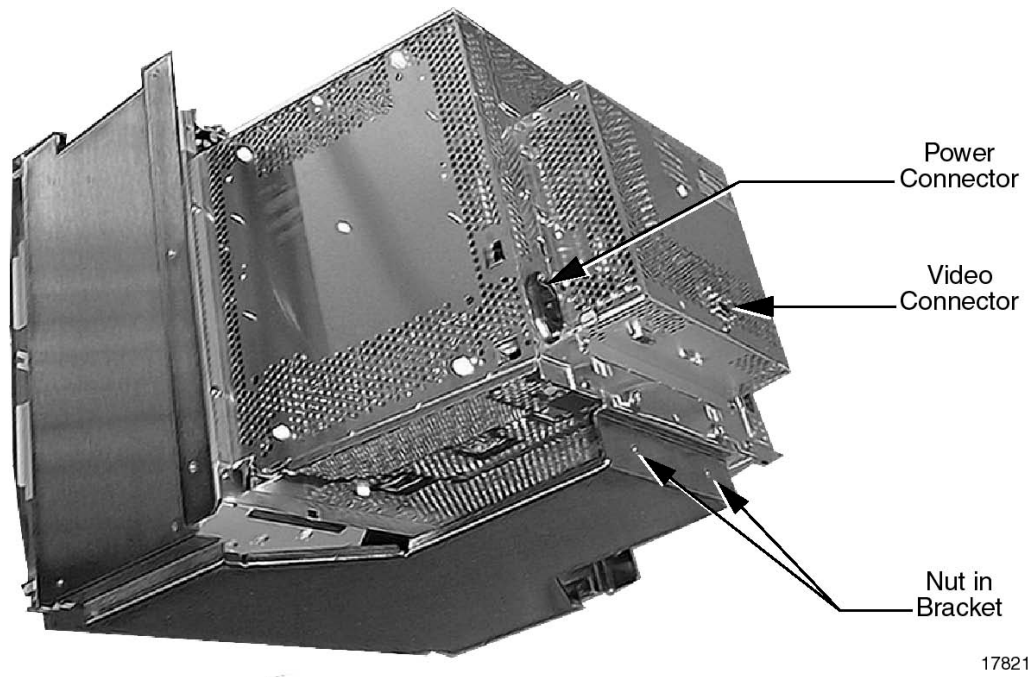


Figure 11-20 Monitor and Mounting Chassis

11. Z Console Operation and Service

11.3. Accessing and Removing a Monitor in a Z Console

Location of monitor mounting shroud assembly bolts

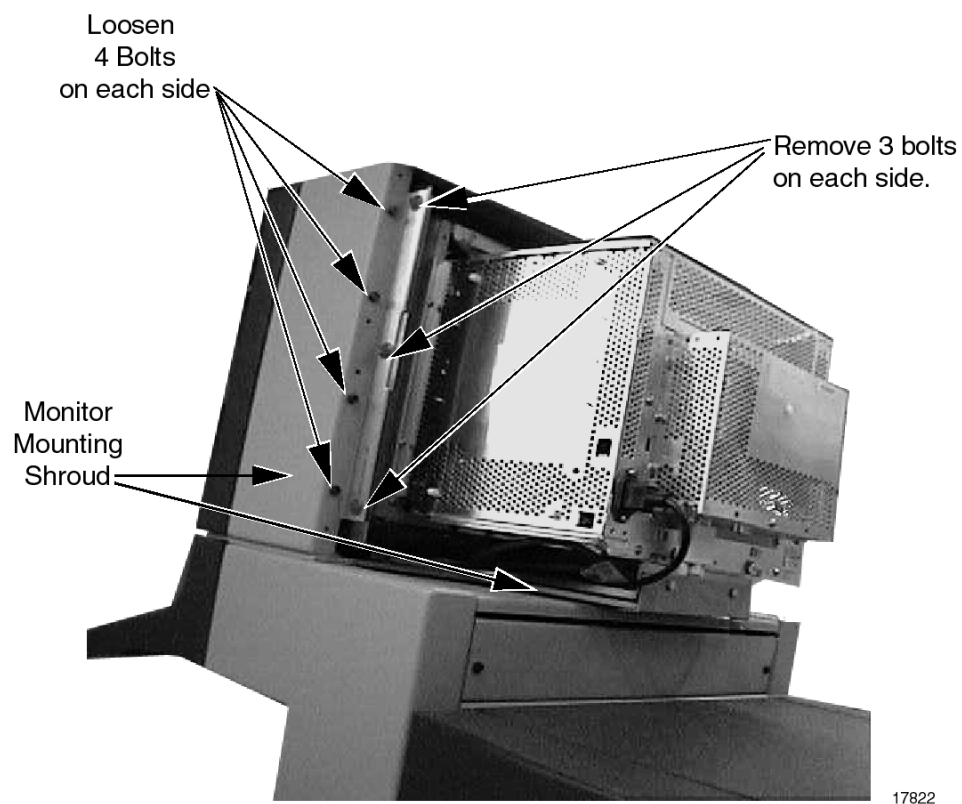


Figure 11-21 Location of Monitor Mounting Shroud Assembly Bolts

11.4 Installing and Replacing a Monitor in a Z Console

Install the monitor

Perform the steps below to install (replace with new) the monitor in the console.



CAUTION

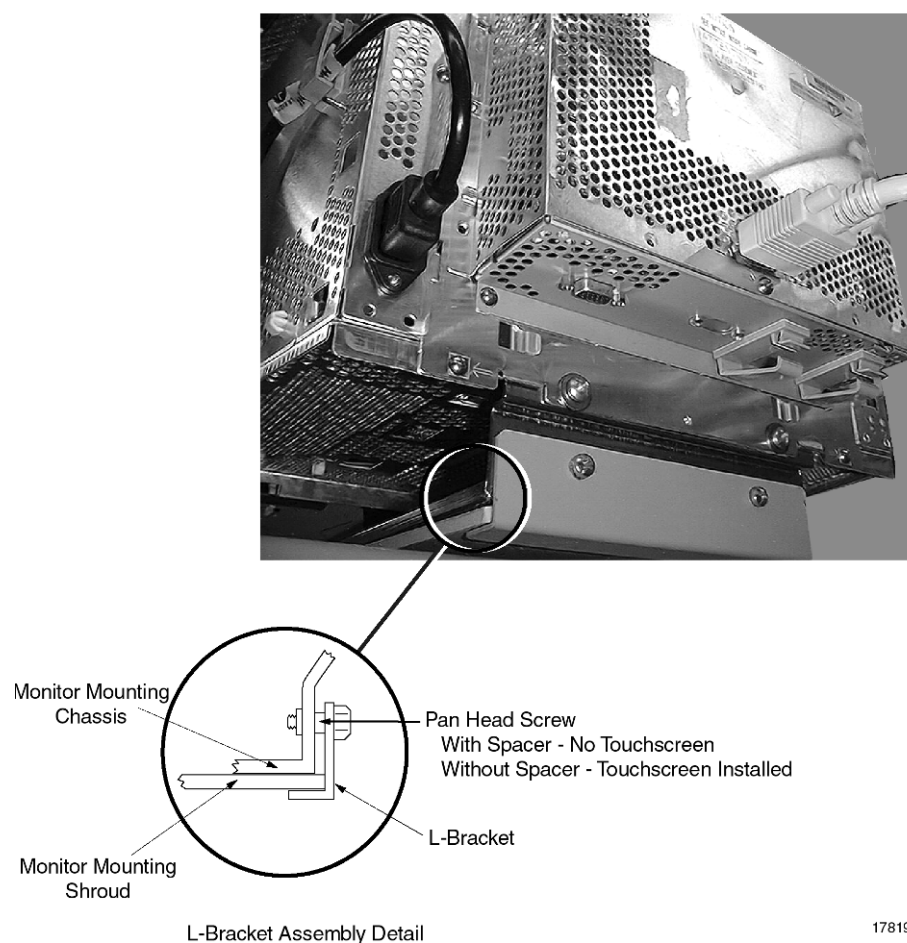
The monitor assembly weighs in excess of 74 pounds. Whether removing the lower or upper monitor, once the screws have been loosened and/or removed, the monitor assembly is free to slide forward or backward onto the floor. For safety purposes, it is essential that two (2) individuals participate in the safe removal of the monitor from the console to prevent personal injury or equipment damage.

Step	Action
1	ATTENTION Use two people to lift the monitor and set it on the monitor-mounting shroud. The monitor assembly weighs in excess of 74 pounds.
2	Carefully slide the monitor forward against the front bezel. Ensure that the Smartframe touchscreen interface cable and the monitor control panel cable are not pinched and that the monitor lines up in the bezel opening correctly. Connect the monitor control panel cable plug to the plug on the cable from the monitor electronics.
3	Replace the three hex-head screws on each side of the monitor-mounting bracket. Tighten the four bolts that were loosened. See Figure 11-21.
4	ATTENTION Remove the two screws (and spacers if no touchscreen) from the L-bracket at the bottom rear of the monitor. The monitor is ready for removal.
5	Connect the video signal cable to the single video-input connector (HD15) at the rear of the monitor.
6	Connect the AC power cord to the monitor.

11. Z Console Operation and Service

11.4. Installing and Replacing a Monitor in a Z Console

Location of CRT Mounting Bracket



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Figure 11-22 Location of CRT Mounting Bracket

Replace the monitor control assembly

To replace the monitor control panel assembly, the rear cover and the monitor must be removed first. Refer to Section 11.3



ATTENTION

If the monitor contains a touchscreen then the bezel/control panel assembly must be removed first before the control panel can be accessed and removed.

Step	Action
1	Remove the rear cover, as detailed in Section 11.3.
2	Remove the monitor, as detailed in Section 11.3
3	Remove the screws holding the monitor control assembly into the touchscreen bezel. See Figure 11-23.
4	Remove the monitor control panel assembly from the bezel. See the ATTENTION block below.
5	Install the new monitor control panel assembly on the bezel. See Figure 11-23.
6	Install the three screws to hold the assembly to the bezel.
7	Replace the monitor, as detailed in Figure 11-20.
8	Replace the rear cover over the monitor.



ATTENTION

The early version of the monitor control panel assembly required it to be permanently attached to the bezel using contact tape. If you have this version then you must replace the entire bezel assembly with a later version that contains a removable monitor control panel assembly installed onto it. Latest version part number is 51403530.

11. Z Console Operation and Service

11.4. Installing and Replacing a Monitor in a Z Console

Monitor control panel assembly

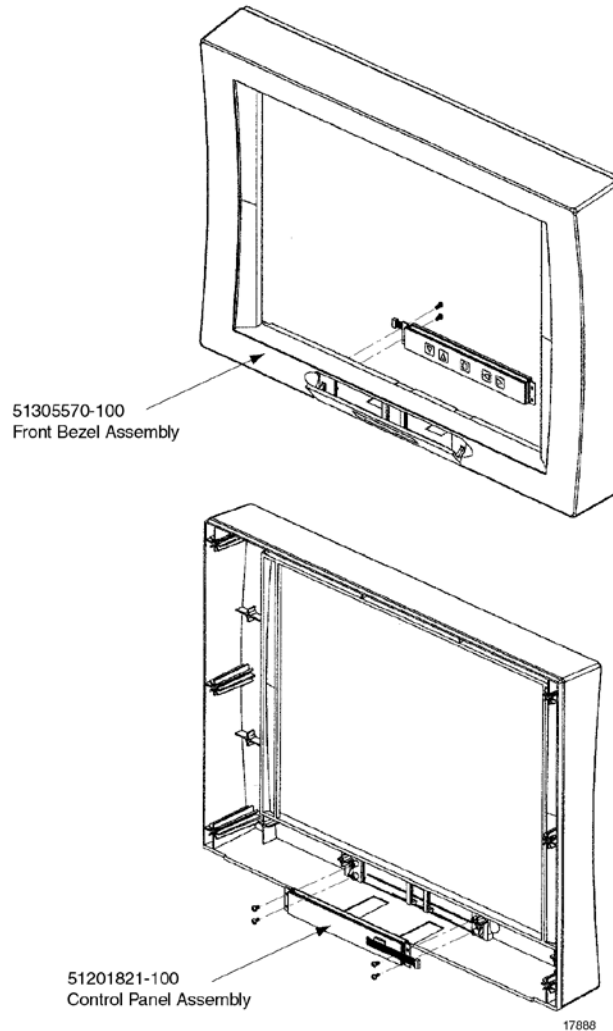


Figure 11-23 Monitor Control Panel Assembly

Cabling Connections for Global User Station

The AC IN and the video input connector(s) (HD15-1 and HD15-2) are shown here. Only the video input connector on the back facing the rear (HD15-1) is used.

11. Z Console Operation and Service
11.4. Installing and Replacing a Monitor in a Z Console

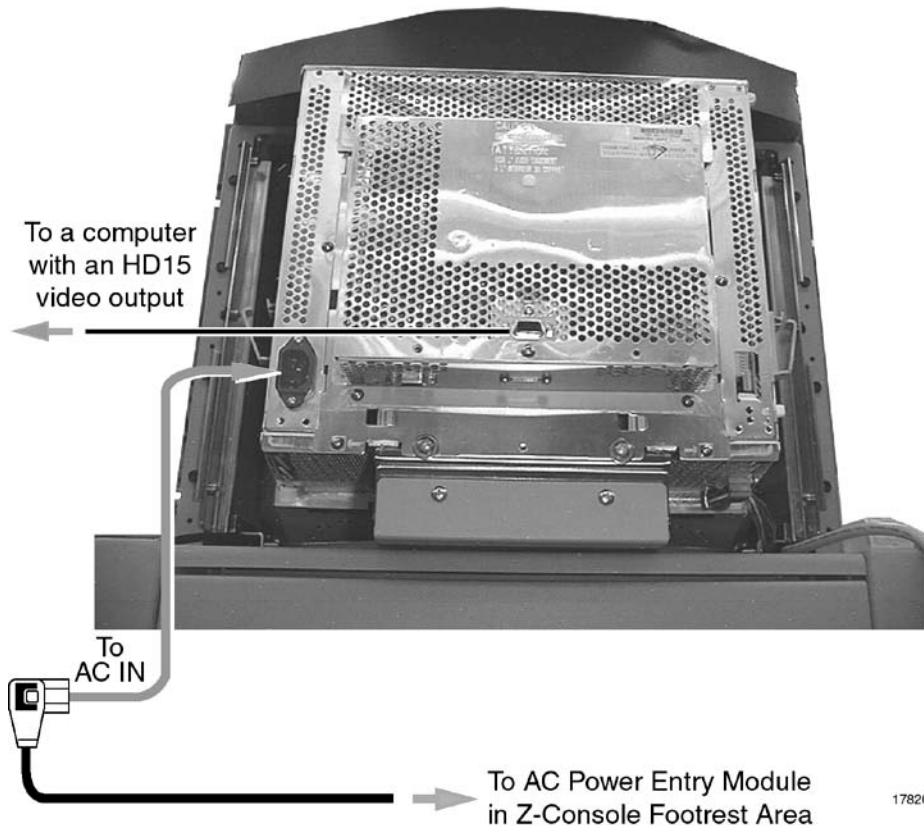


Figure 11-24 Cabling Connections for Global User Station

11. Z Console Operation and Service

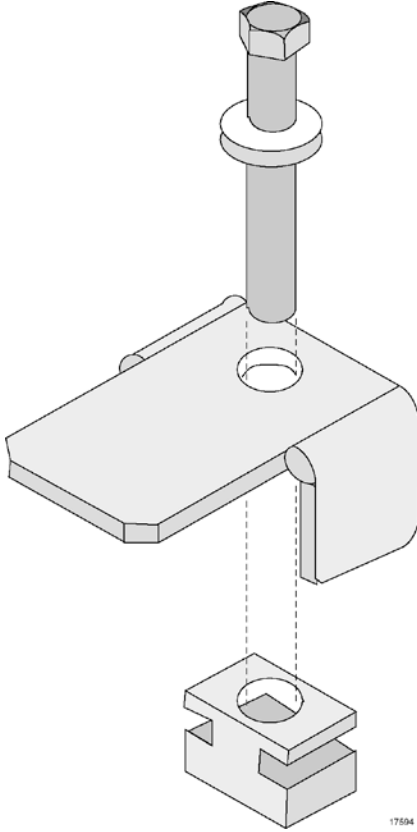
11.4. Installing and Replacing a Monitor in a Z Console

EZ-Console Monitor Installation

Step	Action
Mounting the Lower Monitor	
1	Remove the two screws from the rear edge of the rear top cover and set the cover aside.  Remove 2 screws 17504
2	Install the lower monitor's AC power cord in the left leg of the station (as seen from the rear of the station). Plug the cord into the AC distribution box.

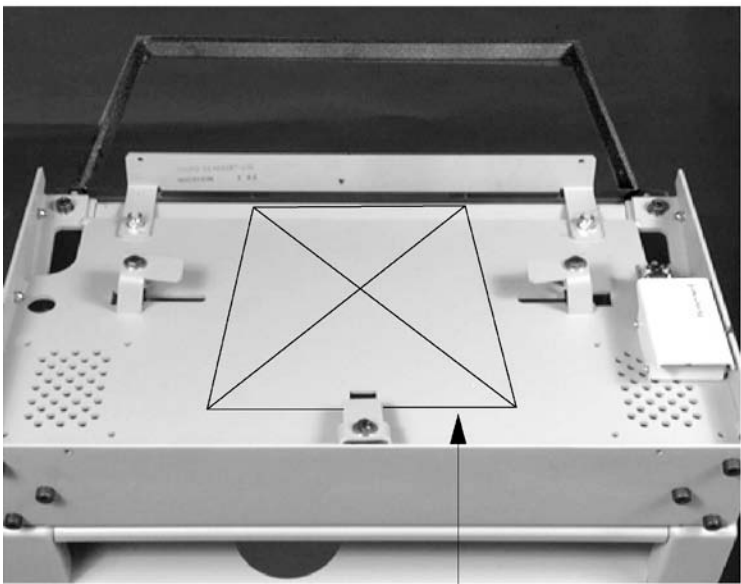
11. Z Console Operation and Service

11.4. Installing and Replacing a Monitor in a Z Console

Step	Action
3	There are three bolt-heads located over slots in the monitor shelf surface that must be loosened and slid out of the slot. Disassemble each bolt and nut from the foam piece and reassemble as shown, with the clamp piece from the plastic bag.  Slide the three clamp assemblies into the slots so that they will not interfere with the mounting of the monitor.

11. Z Console Operation and Service

11.4. Installing and Replacing a Monitor in a Z Console

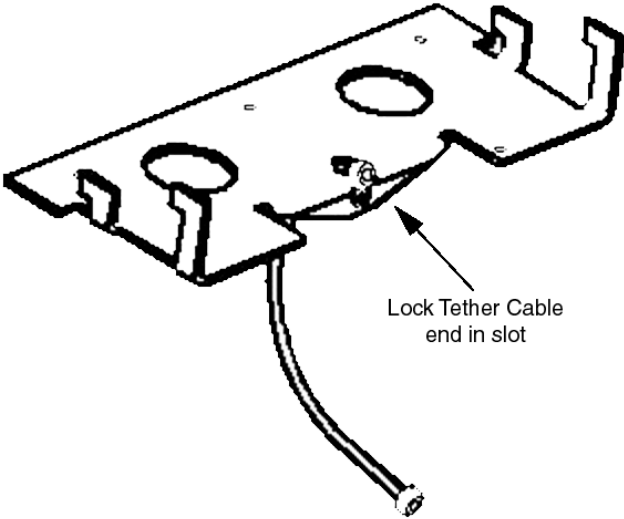
Step	Action
4	Close the tilt panel so it will not interfere with mounting the monitor. Remove the bracket from the front center to allow the monitor to be inserted from the front. Move to clear the two side-monitor mounting clamps and the rear-monitor mounting clamps to allow the monitor to be set in place. Do not slide the clamps over the square hole and remove the clamps—the nuts will fall into an area impossible to reach.  Monitor Footprint 17503

11.4. Installing and Replacing a Monitor in a Z Console

Locking Brackets

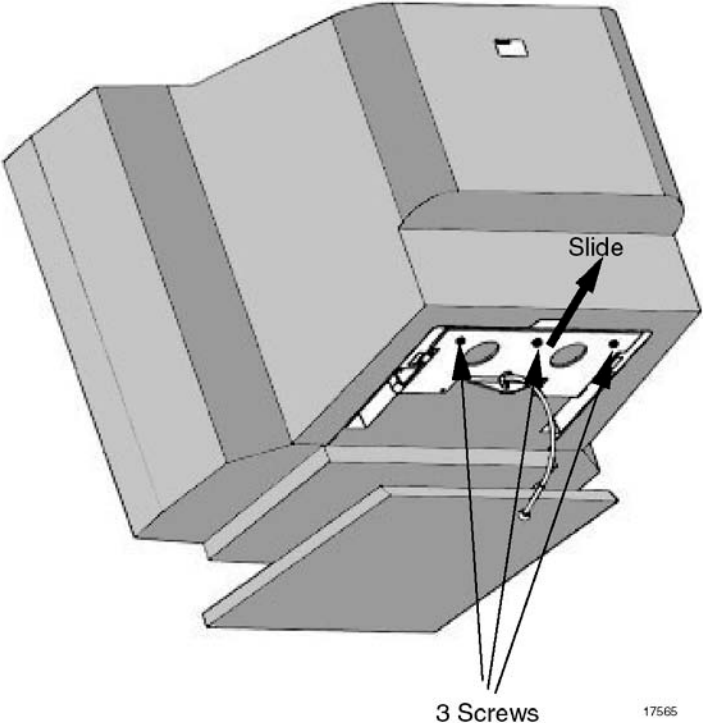
11. Z Console Operation and Service

11.4. Installing and Replacing a Monitor in a Z Console

Step	Action
12	Mount a cable-tie down to the rear of the monitor, just under the rear edge. Gather the power cords and interface cables in the tie down. Ensure the monitor can swivel in all four directions.
13	Reinstall the top rear cover removed in Step 1 and replace the two screws in the cover.
14	Configure the monitor sync switch on the rear of the monitor for VGA.
15	If this station is a single monitor station, the installation of the monitor and touchscreen is complete. If this is a dual monitor station, continue the procedure.
Mounting the Upper Monitor	
16	Remove the two screws in the rear cover and remove the cover.
17	Remove the two screws in the rear of the front cover and remove the cover.
18	Follow the same basic procedure to install the upper monitor.
19	While holding the mounting plate, insert the one end of cable through the slot and down into the locked position.  17564

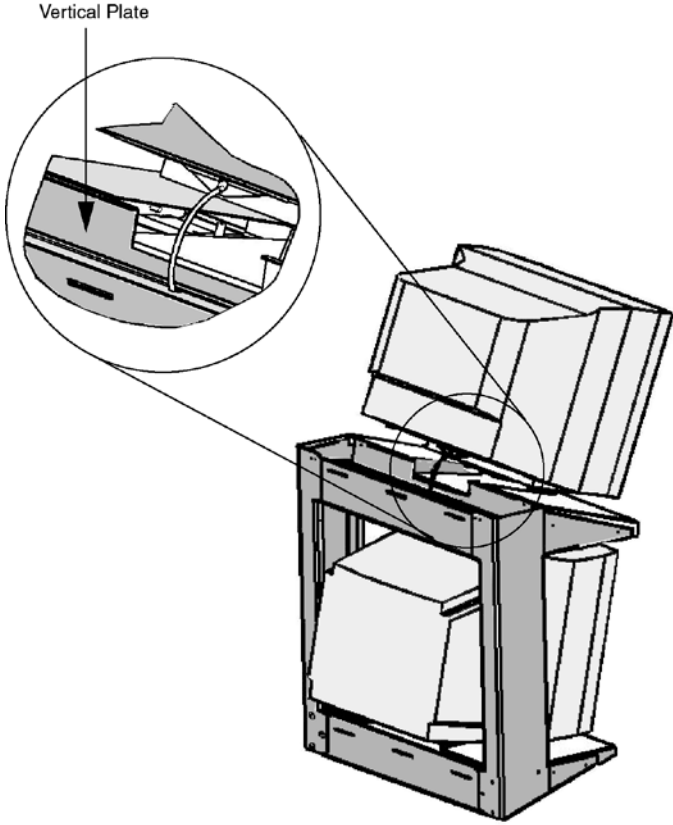
11. Z Console Operation and Service

11.4. Installing and Replacing a Monitor in a Z Console

Step	Action
20	Remove the center screw shown below. Position the mounting plate as show with the claw-like tabs inserted into the monitor and slide the mounting plate to the rear of the monitor until the holes in the plate line up with the screw holes in the monitor. Insert the longer of the three screws into the center hole and each of the other screws in the other screw holes in the plate. Attention: Tighten the screws without stripping them. If the monitor swivel were to break, this tether will keep the monitor from landing on personnel.  Slide 3 Screws 17565

11. Z Console Operation and Service

11.4. Installing and Replacing a Monitor in a Z Console

Step	Action
21	Remove the screw from each end of the vertical angle plate and insert the second end of the tether cable into the slot shown and replace the vertical plate and two screws, trapping the tether cable. Vertical Plate  17520

11.5 Removing and Replacing the 21-inch Flat Screen CRT in Z Console

Description of monitor assembly

The monitor assembly is composed of the following:

- Monitor is a Flat Square Tube (FST) enclosed in an EMI shield.
- Monitor is virtually a flat screen [radius of curvature = 50,000 mm (164 ft)] that minimizes image distortion and reflective glare.
- Monitor is a super fine pitch FD Trinitron picture tube using magnetic deflection, electrostatic focus precision in-line gun, and aperture grille technologies.
- Digital front panel control and a full suite of on-screen programming/calibration menus.

General removal tips

- Is secured to the monitor's mounting shroud by three screws on each side and by an L-bracket with screws (and spacers if no touchscreen) to the back of the monitor's mounting frame.
- Can be replaced by removing the monitor mounting chassis from the monitor mounting shroud and CRT mounting bracket. The replacement monitor is secured to the mounting shroud and the CRT mounting bracket.
- Can be removed from the station without the disassembly of the touchscreen assembly or monitor control assembly.



CAUTION

The monitor assembly weighs in excess of 74 pounds. Whether removing the lower or upper monitor, once the screws have been loosened and/or removed, the monitor assembly is free to slide forward or backward onto the floor. For safety purposes, it is essential that two (2) individuals participate in the safe removal of the monitor from the console to prevent personal injury or equipment damage.

11. Z Console Operation and Service

11.5. Removing and Replacing the 21-inch Flat Screen CRT in Z Console

Removing the 21-inch Flat Screen CRT monitor in a Z console

Follow the procedure to remove the Sony 21" monitor subassembly.



ATTENTION

The following procedures may vary, based on the model of monitor being replaced.

Because the monitor weighs in excess of 60 pounds, two people are required to safely remove it from the console.

Step	Action
1	Remove the rear cover from the monitor assembly.
2	Remove the two screws that secure the L-bracket to the bottom rear of the monitor. The L-shaped angle bracket secures the back of the monitor to the mounting ring. 
3	Disconnect the ac power cord from the back of the monitor.
4	Disconnect the monitor's front control panel cable from the lower-right side of the monitor.
5	Disconnect the 15-pin video connector cable from the back of the monitor.
6	Remove the three hex-head bolts on each side of the monitor-mounting bracket.

11. Z Console Operation and Service

11.5. Removing and Replacing the 21-inch Flat Screen CRT in Z Console

Step	Action
	CAUTION If this is a lower-tier monitor, it may slide backward. It is unrestrained once the four hex-head bolts on each side are removed. Be prepared to restrain the monitor after removing the first screw. HINT: To help the monitor slide smoothly out of place, loosen four bolts on each side of the monitor.
7	Tilt the back of the monitor up slightly to aid removal. 
7	Carefully slide the monitor back, away from the front bezel. Using two people, lift the monitor off the monitor-enclosure shroud, and place the monitor on a table.
	ATTENTION When sliding the monitor out of the monitor enclosure shroud, tilt the back of the monitor up to allow the top portion of the monitor chassis to clear the edges of the Z-Console monitor enclosure shroud.

Removal is now complete.

11. Z Console Operation and Service

11.5. Removing and Replacing the 21-inch Flat Screen CRT in Z Console

Installing the 21-inch Flat Screen CRT monitor in a Z console

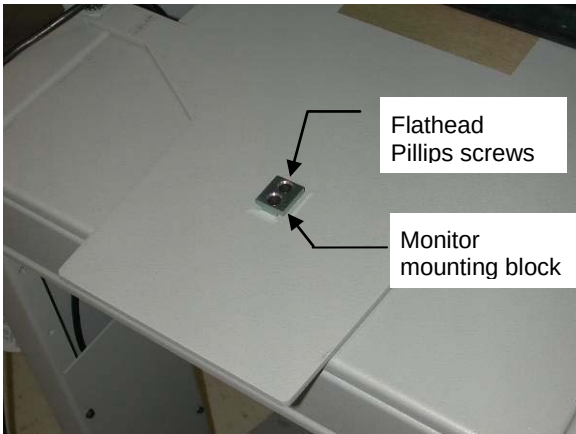
Follow the procedure to install or replace the Sony 21" monitor subassembly.



ATTENTION

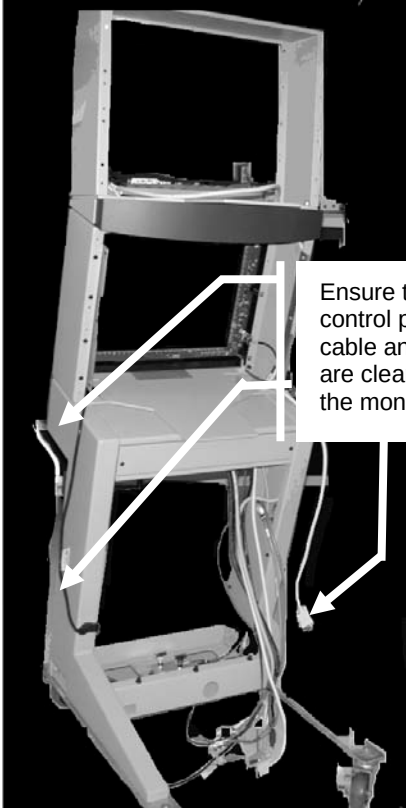
The following procedures may vary, based on the model of monitor being replaced.

Follow the steps outlined in this section to install (replace) the monitor in the console:

Step	Action
1	If upgrading from a predecessor monitor, remove the two flathead Phillips screws that secure the mounting block to the mounting ring. Discard the screws and the mounting block.  Flathead Phillips screws Monitor mounting block

11. Z Console Operation and Service

11.5. Removing and Replacing the 21-inch Flat Screen CRT in Z Console

Step	Action
2	 Ensure that the monitor control panel cable, video cable and AC power cord are clear of the area where the monitor will be installed.

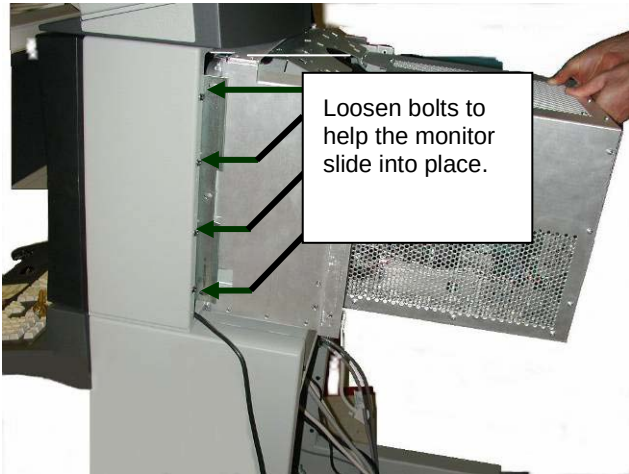
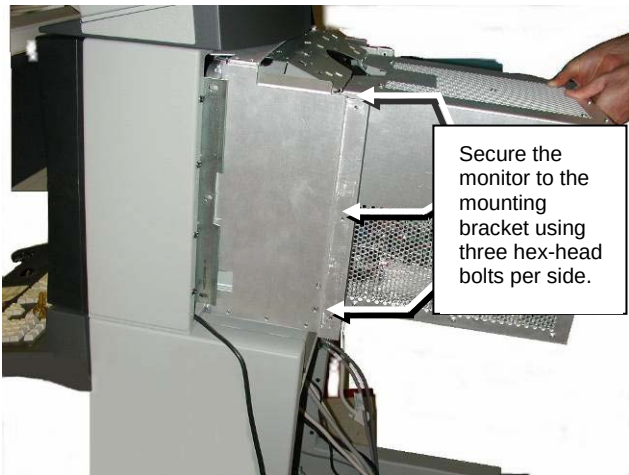
11. Z Console Operation and Service

11.5. Removing and Replacing the 21-inch Flat Screen CRT in Z Console

Step	Action
3	ATTENTION The monitor assembly weighs in excess of 60 pounds. We recommend using two people to lift the monitor and set it on the rear surface of the mounting ring. 

11. Z Console Operation and Service

11.5. Removing and Replacing the 21-inch Flat Screen CRT in Z Console

Step	Action
4	Carefully slide the monitor forward against the front bezel. HINT: Ensure that the four bolts shown below are loosened on each side, to help the monitor slide smoothly into place.  Loosen bolts to help the monitor slide into place.
5	When sliding the replacement monitor into the mounting ring, tilt the back of the monitor up to allow the top portion of the monitor chassis to clear the edges of the monitor mounting ring return flanges.  Secure the monitor to the mounting bracket using three hex-head bolts per side.

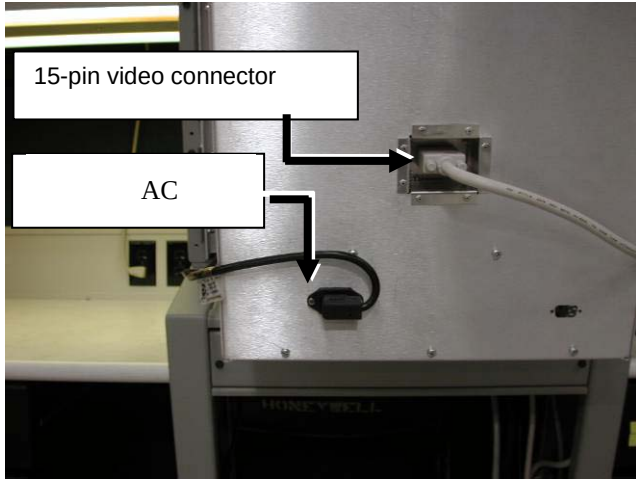
11. Z Console Operation and Service

11.5. Removing and Replacing the 21-inch Flat Screen CRT in Z Console

Step	Action
6	Secure the monitor to the monitor mounting bracket using six head bolts (three per side) as shown in Step 5.
7	Re-tighten the eight hex-head bolts (four per side) that secure the monitor mounting brackets to the monitor-mounting ring.
8	Connect the control panel interface cable to the monitor control panel bulkhead connector. 

11. Z Console Operation and Service

11.5. Removing and Replacing the 21-inch Flat Screen CRT in Z Console

Step	Action
9	Connect the 15-pin video connector cable to the video connector located on the back of the monitor. 
10	Connect the AC power cord to the monitor.
11	Install the L-shaped angle bracket that secures the back of the monitor to the mounting ring (the angle bracket comes packaged in a plastic bag with two screws). 

11. Z Console Operation and Service

11.5. Removing and Replacing the 21-inch Flat Screen CRT in Z Console

Step	Action
12	Replace the monitor cover. Installation is now complete. 

11.6 Removing and Replacing a Touchscreen in a Z-Console

Removing and Replacing the Touchscreen

The touchscreen assembly is mounted to the inside of the front bezel. The monitor must be removed from the station before the touchscreen assembly can be removed. See the procedures in Section 0 for instructions on accessing and removing the monitor.

Remove the touchscreen from a monitor

Perform the following procedure to remove a touchscreen assembly from the front bezel.

Step	Action
1	ATTENTION When this step is complete, the touchscreen assembly is loose and can be damaged if allowed to fall. Use ESD  Wrist Strap 16310 Remove the seven screws that secure the touchscreen PWA along the top and bottom onto four plastic mounting blocks on the inside of the bezel.
2	Remove the four screws and washers that secure the touchscreen PWA to the four plastic mounting studs along the sides of the bezel.
3	Lift the touchscreen assembly out of the bezel.
4	Lay the touchscreen assembly on a flat surface.
5	Remove the dust cover by lifting it up and off the window bezel.
6	Disconnect the touchscreen frame interface cable from the window bezel. See Figure 11-25.
7	Lift the touchscreen frame by two opposite sides. CAUTION Do not flex the touchscreen frame—handle carefully.

11. Z Console Operation and Service

11.6. Removing and Replacing a Touchscreen in a Z-Console

Touchscreen to IKB Cable Interconnection

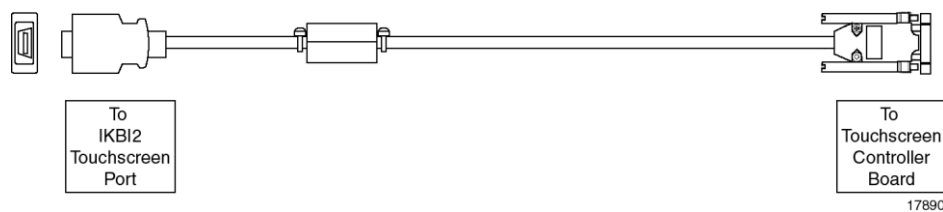
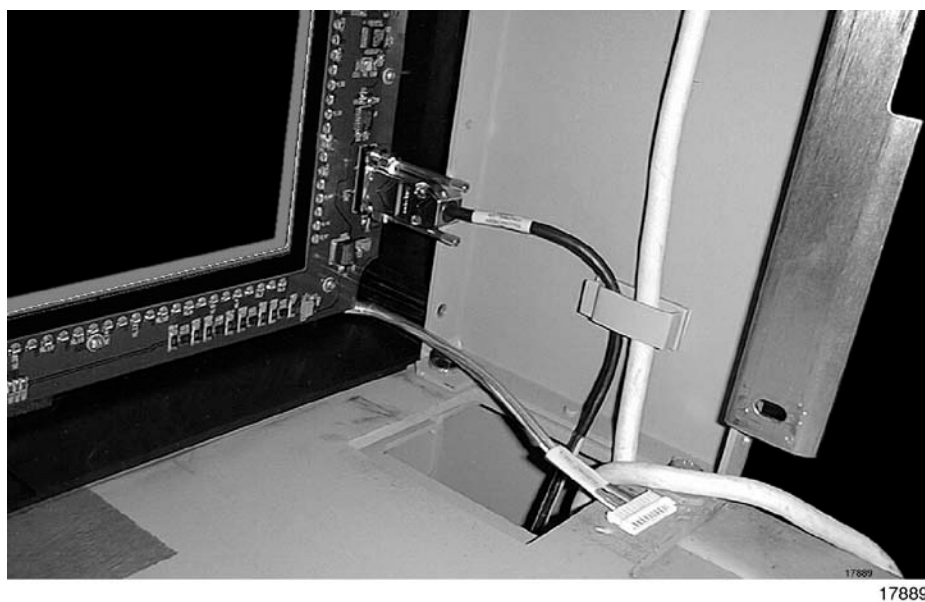


Figure 11-25 Touchscreen to IKB Cable Interconnection

Replace a touchscreen on a monitor

The touchscreen PWA is assembled into the plastic front bezel as follows:

Step	Action
1	Lay the front bezel on a flat surface with the cavity up.
2	Use ESD  Wrist Strap 16310 Place the touchscreen PWA inside the cavity with the opto-electronic devices facing up. CAUTION Do not flex the touchscreen frame—handle carefully.
3	Replace the seven screws that secure the touchscreen PWA along the top and bottom onto four plastic mounting blocks on the inside of the bezel.
4	Replace the four screws and washers that secure the touchscreen PWA to the four plastic mounting studs along the sides of the bezel.
5	Connect the touchscreen frame interface cable from the window bezel. See Figure 11-25.

11. Z Console Operation and Service

11.6. Removing and Replacing a Touchscreen in a Z-Console

Calibrate touchscreen

After a touchscreen is replaced, it should be calibrated to establish registration with the monitor. The following calibration procedure applies to the Carroll Touchscreen, for systems with Windows 2000 Operating Systems.

Step	Action
1	Click on Start > Run > Browse and select the path, C:\GUS\bin\CTPanel icon. Click OK. The calibration program is started. ATTENTION If there is a problem communicating with the touchscreen, an error message will be displayed. If this happens, reboot the system to enable the driver and start the calibration again.
2	Select the CT Panel item on the menu bar and then the Calibration... item. The calibration screen will appear, containing the following instructions: • "If the touch system does not respond to your touch after a few seconds, exit the calibration screen by pressing ESC, then select the Touch Event Settings ... menu item." If the check box in the Enable Mouse/Touch section reads "No Touch," select the Communication Settings ... menu item and make sure that the communication settings match those of your touch system.
3	Press ESC to abort calibration.
4	Touch the center of the upper-left calibration target using only one finger, inserting it straight into the touchscreen. Note: Even if beeps do <u>not</u> occur in the steps below, continue holding finger in place until next target appears. Hold your finger in place until you hear a beep or target changes, and then remove your finger.

11. Z Console Operation and Service

11.6. Removing and Replacing a Touchscreen in a Z-Console

Step	Action
5	Follow the above instructions and after the beep or target changes, the following new instructions appear: "Touch the center of the lower-right calibration target using only one finger, inserting it straight into the touchscreen. Hold your finger in place until you hear a beep, then remove your finger."
6	After you hear the second beep or target changes, keep the finger in place until the screen returns to the "Carroll Touch Control Panel" screen clearing the second target.
7	Close the window to exit the Carroll Touch Control Panel program.

The touchscreen is now calibrated.

11. Z Console Operation and Service

11.7. Media Drive Operation

11.7 Media Drive Operation

Load/unload CD-RW/DVD-RW(+/-)

Function	Operation
Load	1. Push the button—the loader tray will extend. 2. Insert the CD-RW into the tray. 3. Push the loader tray into the drive.
Unload	1. Push the button—the loader tray will extend. 2. Remove the CD-RW from the tray. 3. Push the loader tray into the drive.

Zip drive operation

Function	Operation
Load	1. Insert the cartridge in the drive—the LED will flash until the drive has spun up to speed.
Unload	1. Press the unload button—the LED will flash until the drive has unloaded. 2. Remove the cartridge by pulling the cartridge out of the drive.
Access	The LED will light when the cartridge is electronically accessed.

Floppy drive operation

Function	Operation
Load	1. Insert the floppy into the slot in the drive. 2. Press the floppy into the slot until the load mechanism pulls the floppy all the way in and seats it in the drive.
Unload	1. Press the button on the front of the drive. 2. Remove the floppy after it ejects.

11.8 Remote Peripheral Service

Overview

CD-RW/DVD-ROM, tape and Zip drives are used to load and store data as required by the operation of the system.

Description

The CD-RW/DVD-ROM drive is located in the drive enclosure box, located behind the tilt panel on the IKB. The drive can share the enclosure with either the optional Zip drive or the optional tape drive.

CE Compliant Drive Enclosure

The enclosure has a 2-piece shell with a shelf inside that the drives mount on. The cables pass under the mounting shelf and up behind the connectors on the drive(s). The enclosure has screws that pass through the enclosure bottom and into the mounting shelf inside the enclosure. One of the two cables located beside the right-hand drive is not used. This common assembly has other uses where the cable is used.

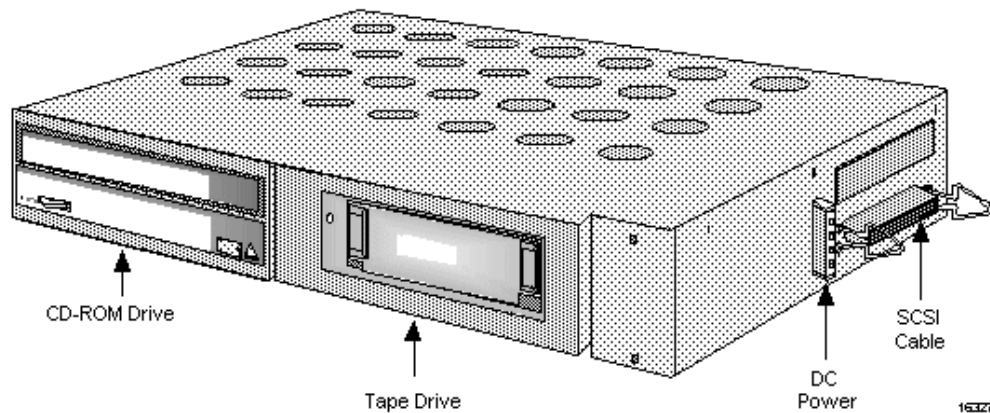


Figure 11-26 CE Compliant CD-RW/DVD-ROM/Tape Drive Enclosure

11. Z Console Operation and Service

11.8. Remote Peripheral Service

Cable connections

The SCSI connector on the outside of the enclosure is part of a cable that is connected to the CD-RW/DVD-ROM drive and tape or Zip drive. The interconnecting cable connected to this connector is connected to the SCSI II connector on the remote peripheral assembly located in the top slot on the rear of the electronics module.

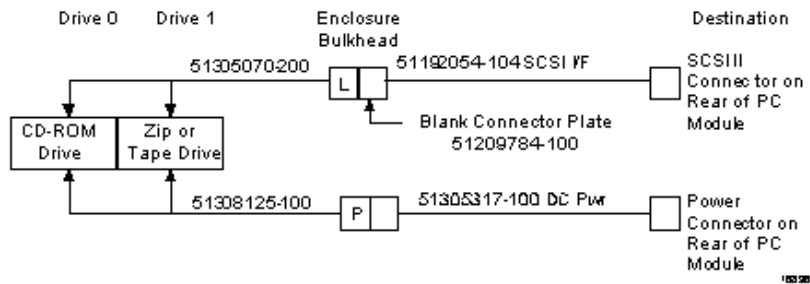


Figure 11-27 Remote Peripheral Cabling

Construction

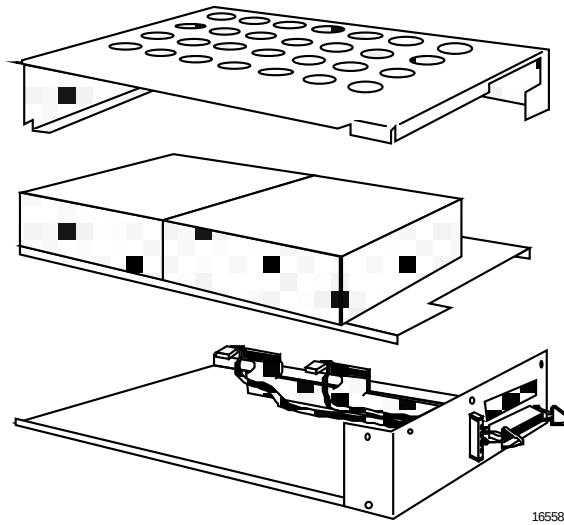


Figure 11-28 Exploded View of CE Compliant Drive Enclosure

12. Spare Parts

12.1 Overview

Scope

The parts numbers listed in this section are replacement parts for various options available with the SVR2/WKS2 platform. The Parts Online section at the <http://www.acs.honeywell.com/> website also contains up to date information on specific part and model numbers.

12. Spare Parts

12.2. Common Spare Parts

12.2 Common Spare Parts

SVR2/WKS2 spare parts

These are spare parts that are common to the SVR2/WKS2 enclosure.

Table 12-1 Spare Parts for SVR2/WKS2 Platform

DESCRIPTION	PART NO.
1.44 MB Floppy Drive	51199581-901
10/100 Base Tx NIC	51199171-909
2.4 GHz CPU, Pentium 4,	51199581-906
2-channel Ultra ATA/133 raid controller, PCI	51199581-902
512 MB SDRAM, DDR 333, PC2700, ECC, Unbuff, 64mx72, 2 banks, 184 pin Note: This is the only available replacement memory and it is compatible in a 256/512 environment.	51199581-910
61.4 GB ATA-100, 7200 RPM Hard Drive or equivalent	51199581-903
Air filter	51199171-914
Air filter	51308003-200
Air filter housing	51403585-100
ATX chassis	51403349-100
ATX motherboard	51199581-905
Cable restraint bracket	51305719-100
CD-ROM, R/W, 40x12x48x IDE	51199171-924
CD-ROM-R/W/DVD-R/W (+,-) IDE drive	51199581-911
Clamp bracket	51305599-100
Clamp bracket	51403551-100
Drive Bay Filler Bracket*	51199171-926
Fan assembly	51199171-913
Fan, fan/heatsink, socket 478	51199581-907

Table 12-1 Spare Parts for SVR2/WKS2 Platform

DESCRIPTION	PART NO.
Fan/card guide bracket	51403353-100
Hard disk hotswap base	51403575-100
Hard disk hotswap top	51403576-100
Hard disk restraint bracket	51403573-100
Internal to external power cable	51199171-928
Label, front panel switch	51201886-100
Left-side mounting bracket, chassis	51201883-200
Optical Mouse	51197073-200
Optical Mouse (Commercial Three-Button)	51197073-100
Optical Mouse (European Three-Button)	51197073-200
Power supply adapter plate	51305740-100
Power supply for WKS2, 200-250 watt	51199171-923
Power supply for SVR2, 300 watt dual	51199171-902
Remote power cable	51305472-100
Right-side mounting bracket, chassis	51201883-100
Roxio v6.1 oem software for 51199581-911	51199581-912
Switch guard	51201894-100
SXGA Dual head 16 MB AGP video card	51199171-905
Top cover assembly	51403572-100
ROXIO v5.3 OEM Software for 51199171-924	51199467-301

12. Spare Parts

12.3. Console/Desktop Spare Parts

12.3 Console/Desktop Spare Parts

These spare parts are used in consoles and desktops,

Table 12-2 Console/Desktop Spare Parts

21-inch Monitor (Sony) – Upper/Lower	*51197171-922
21-Inch Monitor (Sony) – Upper/Lower	*51197097-100
21-Inch Touchscreen (for Sony) – Lower	*51191573-100
20.1-Inch Flat Panel Display Monitor (for NEC) for desktop options only	51197178-100
Annunciator Assembly	*51303642-600
Carroll HBC Touchscreen Controller Board	*51196930-100
External Touchscreen Controller	*51197009-100
Mincartridge Tape Drive	*51196996-400
WIN95 Space Saver Quiet Key, 104 Key Keyboard	*51198718-901
Zip Drive	*51196929-135

12.4 21-Inch Sony CE-Compliant Monitor Spare Parts

Table 12-3 Spare parts for Sony 21" monitor

Description	Part Number
21-inch Monitor (Sony) – Upper/Lower	*51197178-100
Control Panel Subassembly	51197181-100
Monitor AC Power Cable (for lower CRT)	51305542-100
Monitor AC Power Cable (for upper CRT)	51305542-200

*= ORU

12.5 OEP/IKB Related Spare Parts

IKB

These spare parts are common to all versions of the IKB.

Table 12-4 IKB Spare Parts

Description	Part Number
Desktop Integrated Keyboard w/o Trackball	51196694-100
Desktop Integrated Keyboard with Trackball	51196694-300
IKB Controller Board (through-hole version)	51196694-903
IKB Full-Travel Keyboard Subassembly	51196694-904
Operator Control Panel	51196694-907
Relegendable Function Key Panel	51196694-908
IKB Relegendable Insert Set (3 Inserts)	51196694-911
IKB Controller Board—SMT version	51196694-923
Trackball Cursor Control Unit	51196694-901
Full-Travel Keyboard Legended KeyCap Set	51196694-905
Full-Travel Keyboard Unlegended KeyCap Set	51196694-906
Relegendable Function Keyboard for Z-Console	51196694-909
Operator Keyswitch with Keys	51196694-910
Key, Supervisor (M0053)	51190884-201
Key, Engineer (2h210)	51190884-202

12. Spare Parts

12.5. OEP/IKB Related Spare Parts

OEP/IKB Adapter parts

Table 12-5 OEP/IKB Adapter Spare Parts

Description	Part Number
OEP/IKB Module Assembly	51305776-100
DB9 M/F Comm Port Cable (10 feet)	51196990-200
AC Power Cord	51305567-100 (120 V)
	51305557-100 (220 V)
AC Power Source	51197184-100

12.6 Z-Console Spare Parts

Scope

The spare parts are unique to the Z-Console.

Table 12-6 Z-Console Unique Spare Parts

Description	Part Number
Z-Console Integrated Keyboard w/o Trackball	*51196694-150
Z-Console Integrated Keyboard with Trackball	*51196694-350
Operator Entry Panel Assembly (CE Spare)	*51401577-100
Keyboard-2 Assembly,	*51401952-200
Keyboard Membrane	*51401560-100

12.7 Classic Console Spare Parts

Scope

These spare parts are unique to the Classic Console.

Table 12-7 Classic Console Unique Spare Parts

Description	Part Number
Classic Console Integrated Keyboard w/o Trackball	*51196694-110
Classic Console Integrated Keyboard with Trackball	*51196694-310
19-Inch Monitor (Multisync)	*51197017-100
Touchscreen Assembly	*51400988-200
Operator Entry Panel Assembly (CE Spare) complete keyboard	*51401577-300
Keyboard Membrane	*51401560-100
Keyboard-2 Assembly	*51401952-200

12. Spare Parts

12.8. Printer Spare Parts

12.8 Printer Spare Parts

HP B/W LaserJet

Table 12-8 HP B/W LaserJet Spare Parts

Description	Part Number
HP LaserJet B/W 115 Vac	*51196317-100
HP LaserJet B/W 220 Vac	*51196317-101
Replacement Cartridge	*51196317-700
JetDirect Ethernet Card, BNC 10Base2	*51196317-400
JetDirect Ethernet Card, 10/100 Base T	*51196317-500

HP Color Printer

Table 12-9 HP Color LaserJet Spare Parts

Description	Part Number
HP LaserJet Color 115 Vac	*51196318-100
HP LaserJet B/W 220 Vac	*51196318-101
Toner, Cyan	*51196318-208
Toner, Yellow	*51196318-209
Toner, Magenta	*51196318-210
Toner, Black	*51196318-211
JetDirect Ethernet Card, BNC 10Base2	*51196317-400
JetDirect Ethernet Card, 10/100 Base T	*51196317-500

12.9 Periodic Replaceable Items

Periodic Replaceable Items

Table 12-10 Periodic Replaceable Items

Description	Part Number
Relegendable Insert Set for Integrated Keyboard	51196694-911
Printer Ribbon, Black—Signum ASPI 2043	51196690-401
Printer Ribbon, Color—Signum ASPI 2043	51196690-501
Blank Zip Cartridge, 100 MB, One Pack	51196929-150
Blank Zip Cartridge, 100 MB, 10 Pack	51196929-160
Minicartridge TR-5 tape media, One Pack	51196966-500
Tape Cleaning Cartridge	51196966-600
Battery (Panasonic CR2032)	

12. Spare Parts

12.9. Periodic Replaceable Items

13. Glossary

13.1 Acronyms and Abbreviations

Acronyms and abbreviations used in this document include:

AGP — Accelerated Graphics Port

AMR — Audio Modem Riser

APP Node— Application Processing Platform (MS Windows-based user station)

DIMM — Dual Inline Memory Modules

ECC RAM —Error Checking and Correction Random Access Memory

EIDE — Enhanced Integrated Drive Electronics

ESD — Electro-Static Discharge

FPD — Flat Panel Display

GUS — Global User Station (MS Windows-based window to process)

IR — Infrared

ISA — Industry standard Architecture

LAN — Local Area Network, in this case the Ethernet

LCN — Local Control Network (Also referred to as TPN)

LCNP4 — PCI Card providing K4LCN (TDC Processor functionality)

MAU — Media Access Unit (provides LCN I/O functions)

PCI — Peripheral Component Interconnect

SCSI — Small Computer System Interface

SDRAM — Synchronous Dynamic Random Access Memory

RAM — Random Access Memory

SVGA — Super Video Graphics Array

TPN — **TotalPlant** Process Network (formerly known as the LCN)

TPS system — **TotalPlant** Solutions system

13.2 Terminology

Specialized terminology used in this document includes:

LCNP4 — Local Control Network Processor board—Controls the communications between the **TotalPlant** Solutions (TPS) System nodes and the **TotalPlant** Process Network (TPN).

Moiré — A type of natural interference that produces soft, wavy lines on your screen. It may appear due to interference between the pattern of the picture on the screen and the phosphor pitch pattern of the monitor.

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